SPEAR3 RF System

RF Physics, Control Theory and Physical Plant

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This document states the physics of the SPEAR3 RF plant and the control problem that follows from it: what disturbs the cavity field, over what frequency range, and which loop is responsible for each band. It is *implementation-independent* — the content does not change when the VXI hardware is replaced by the LLRF9 controller.

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Revision History

Revision	Date	Scope of Change	Summary
R0	2026-10-22	Initial draft	Initial draft of the Tier 1 physics and plant reference for the SPEAR3 RF system: plant models, the LLRF signal chain, disturbance analysis, control architecture and loop-by-loop design, transient matching, and synchrotron damping and the noise-driven motion of the beam, with a procedure for setting the direct loop. Operating basis: 500 mA, 2.85 MV, $U_0 = 1.09$ MeV (measured; dipoles and insertion devices), legacy loop delay $\approx$ 625 ns. Legacy-station values come from normal-operation data of 7 October 2026, the closed-loop transfer-function measurements from 2 October 2026 and the synchrotron-line measurements from 8 October 2026, with station records from 2003, 2013, 2020 and 2025 for comparison. Open measurements are listed in §8.10.

Document Scope and Provenance

This document is the **Tier 1 physics and plant reference** for the SPEAR3 RF system: the physics of the RF plant, the disturbances that act on the cavity field, and the feedback loops that answer them. The physics does not change when the VXI controller is replaced by the LLRF9; where the legacy implementation matters via loop settings, delays, calibration and other hardware-specific details, it is stated explicitly in the text.

Each feedback loop exists because a class of disturbance threatens the cavity field. Its bandwidth follows from the frequency content of that disturbance; its gain follows from the field-stability requirement and from what the beam tolerates (§8).

All technical content is cited to original source documents; every numbered [Rn] tag in the text links to its entry in Appendix A. Where this document derives a value rather than quoting one, the derivation is given as a numbered equation.

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1 System Overview and Field Stability Requirements

1.1 The Control Problem in One Sentence

The SPEAR3 RF system must deliver a 476.3 MHz accelerating voltage of ~ 2.85 MV across four cavities with amplitude stability $< 0.1\%$ RMS and phase stability $< 0.1^\circ$ RMS, while the beam changes with every fill and top-off.

1.2 Storage Ring Parameters

Parameter	Symbol	Value	Unit	Source
Beam energy	E_0	3.0	GeV	[R1]
Beam current (top-off)	I_b	500	mA	[R1]
Circumference	C	234.14	m	[R2]
Revolution frequency	f_{rev}	1.2804	MHz	c/C
Revolution period	T_{rev}	0.781	μs	$1/f_{\text{rev}}$
Harmonic number	h	372	—	[R1]
RF frequency	f_{RF}	476.3	MHz	[R4]
Momentum compaction	α_c	1.18×10^{-3}	—	[R2]
Energy loss per turn (dipoles + IDs)	U_0	1.09	MeV	Measured (§8.8); dipoles alone 0.912 [R1]
Total accelerating voltage (design)	V_{RF}	3.2	MV	[R1]
Total accelerating voltage (operational)	V_{RF}	~2.85	MV	[R4], [R5]
Synchrotron frequency	f_s	~10.0	kHz	Equation (2.7)
Synchrotron tune	ν_s	~0.008	—	Equation (2.7); [R1]
Longitudinal radiation damping time (amplitude)	τ_s	2.36	ms	With IDs; 2.87 bare lattice [R1]
Transverse radiation damping times	$\tau_{x,y}$	3.65, 4.30	ms	With IDs; 4.24, 5.14 bare lattice [R1]

Note on f_{RF}: The nominal RF frequency is 476.3 MHz. The precise frequency varies slightly as the ring circumference changes (D9); the master-oscillator readback `SPEAR:RFFreq` gave $476.303\,964$ MHz on 7 October 2026. For consistency, 476.3 MHz is used throughout this document as nominal.

Note on U_0 and the damping times: $U_0 = 1.09$ MeV is measured, from the power balance of the four cavities, $\sum_k (P_{\text{fwd}} - P_{\text{refl}} - P_{\text{wall}})_k / I_b$, on 7 October 2026 at 499 mA (§8.8). It is the radiation loss of the dipoles (0.912 MeV, the bare lattice) plus the present insertion devices. The damping times of [R1] ($\tau_s = 2.87$ ms, $\tau_{x,y} = 4.24, 5.14$ ms) belong to the bare lattice; the insertion-device radiation, emitted where the dispersion is small, damps with partition numbers $J_s \approx 2$, $J_x \approx J_y \approx 1$, which with $J_s = 1.79$ for the dipoles gives $J_s = 1.82$ overall and the shorter times in the table, $\tau_i = 2E_0 T_{\text{rev}}/(J_i U_0)$.

Note on V_{RF}: The **design** value of 3.2 MV (800 kV/cavity) is from the original SPEAR3 RF system specification [R1]. The **operational** value of ~ 2.85 MV (712 kV/cavity) is from LLRF9 test measurements [R4], and is also the current nominal operating point at 500 mA. It is used throughout the beam loading calculations in this document.

Note on α_c: Published values range from 0.0011 (SSRL parameter page [R3]) to 0.00118 (used in this document) depending on the specific lattice optics. Differences of $\sim 7\%$ in α_c produce $\sim 3\%$ differences in f_s.

1.3 RF System Configuration

The RF station consists of: a single 1.5 MW CW klystron (Marconi/CPI K3512S) → waveguide circulator → 3× Magic-Tee power splitter → four PEP-II-type single-cell HOM-damped copper cavities [R1], [R6], as seen in Figure 1. A single I/Q modulator at the klystron input is the primary actuator for all fast feedback loops.

1.4 Field Stability Requirements

The requirement column is the specification [R5]; the achieved column is from LLRF9 commissioning [R4].

Parameter	Requirement	Typical Achieved	Rationale
Amplitude stability	$< 0.1\%$ RMS	$< 0.05\%$ RMS	Photon beam position, energy spread
Phase stability	$< 0.1^\circ$ RMS	$< 0.05^\circ$ RMS	Bunch timing, longitudinal emittance
HVPS voltage regulation	$< \pm 0.5\%$	$< \pm 0.3\%$	Klystron gain/phase stability
HVPS ripple	$< 1\%$ P-P	$< 0.5\%$ P-P	RF phase modulation budget

What the beam responds to. The beam does not respond to the integrated rms of the field error. Its centroid is a driven, damped oscillator (§8.1): it follows phase and amplitude errors below its coherent synchrotron frequency f_{coh} (8.8 kHz at 500 mA), amplifies those near f_{coh} by up to $Q\,(f_s/f_{\text{coh}})^2$, about 33 today, and ignores those far above. A field specification that protects the beam is therefore a limit on the spectral density of the error near f_{coh} and on the line harmonics below it, not an rms value alone (§8.7). On 8 October 2026 the beam's phase moved by 0.39–0.58° rms near f_{coh}, most of it driven by two HVPS harmonics beside the line (§8.8).

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2 Plant Physics Model

This section models the plant that the LLRF controller acts on: the cavities and their beam loading, the klystron, its high-voltage supply, the cavity tuners, and the delay around the loop.

2.1 High-Level System Block Diagram

Figure 1 shows the RF plant as a control system: four cavities driven by one klystron, regulated by loops that span seven decades of frequency.

Key architectural features:

- **Single klystron, four cavities**: One I/Q modulator drives all four cavities simultaneously. Field regulation uses the vector sum of all four cavity probes.
- **Bandwidth hierarchy**: direct loop (≈ 70 kHz) $\gg$ ripple loop (lines ≈ 1 Hz wide at 60 Hz–2.9 kHz) $\gg$ EPICS loops (HVPS, DAC and gain tracking $\lesssim 0.1$ Hz; tuners 0.01–0.05 Hz).
- **Analog fast path**: The legacy direct loop operates entirely in analog baseband; its measured total loop delay is about 625 ns, most of it in the klystron and the cables to and from the tunnel (§2.6).

2.2 RF Cavity as a Narrowband Resonator

2.2.1 Equivalent Circuit

Near its fundamental mode at $\omega_0 = 2\pi \times 476.3$ MHz, each cavity is modeled as a parallel RLC circuit. Shunt impedances are in the circuit convention $R_s = V^2/2P$.

Parameter	Symbol	Value	Unit	Source
Resonant frequency	f_0	≈ 476.3	MHz	[R6]
Shunt impedance	R_s	3.73	MΩ	[R6]

Parameter	Symbol	Value	Unit	Source
Unloaded Q	Q_0	32,000	—	[R6]
Geometry factor	$R/Q = R_s/Q_0$	116.6	Ω	Derived; [R7]
Coupling coefficient	$\beta = Q_0/Q_{\text{ext}}$	3.78	—	See note
Loaded Q	$Q_L = Q_0/(1+\beta)$	6,700	—	Derived
Loaded shunt impedance	$R_L = R_s/(1+\beta)$	0.78	MΩ	Derived
Operating gap voltage	V_{gap}	~712	kV	$V_{\text{RF}}/4$; [R5]

Note on β: [R6] gives $\beta_{\text{actual}} = 3.72$ and $\beta_{\text{optimum}} = 3.84$ for the PEP-II HER cavity ($Q_L = 6{,}780$); this document uses $\beta = 3.78$, and the 1% difference is immaterial to every result below. The IOC uses $\beta = 3.8$ for all four cavities (`SRF1:CAVn:BETA:CTRL`), and the reflected power measured on 7 October 2026 gives 3.56–4.27, median 3.74 (§8.8). [R1] quotes 3.8 MΩ in the same convention (800 kV at 85 kW gives 3.76 MΩ). Cavity geometry and measured impedance spectra are in [R7] and [R15].

2.2.2 Cavity Transfer Function

The cavity responds to the RF envelope (the baseband I/Q signal) of the current that drives it. For a cavity detuned by $\Delta\omega = \omega_0 - \omega_{\text{RF}}$, the envelope transfer function from driving current to gap voltage is

$$H_{\text{cav}}(s) = \frac{R_L\,\omega_{1/2}}{s + \omega_{1/2} - j\Delta\omega}, \qquad R_L = \frac{R_s}{1+\beta}, \tag{2.1}$$

where s is the Laplace variable of the envelope ($s = j2\pi f$ at an offset f from the carrier) and $\omega_{1/2} = \omega_0/(2Q_L)$ is the loaded half-bandwidth; the response peaks at the cavity's own resonance, $f = \Delta\omega/2\pi$. The **impedance seen by the beam** at an absolute frequency ω near ω_0 is the same resonance written in the RF domain:

$$Z_{\text{cav}}(\omega) = \frac{R_L}{1 + j\,2Q_L\,(\omega - \omega_0)/\omega_0} \tag{2.2}$$

Both carry the *loaded* shunt impedance $R_L = 0.78\,\text{M}\Omega$: the coupler and the matched circulator load appear in parallel with the cavity wall losses.

Cavity half-bandwidth:

$$\Delta f_{1/2} = \frac{f_0}{2Q_L} = \frac{476.3\ \text{MHz}}{2 \times 6700} = 35.5\ \text{kHz} \tag{2.3}$$

This determines the cavity natural response time $\tau_{\text{cav}} = 1/(2\pi\Delta f_{1/2}) \approx 4.5\ \mu\text{s}$.

2.2.3 Beam Loading — Steady-State Phasor Analysis

A bunched beam of DC current I_b presents a fundamental RF current $I_{b1} = 2I_b$ to the cavity, so at 500 mA the cavity sees 1.0 A at the RF frequency. Its steady-state induced voltage in the *loaded* cavity, on resonance, is

$$V_{b,\text{res}} = I_{b1}R_L = \frac{I_{b1}R_s}{1+\beta} = \frac{1.0\ \text{A} \times 3.73\ \text{M}\Omega}{4.78} = 0.78\ \text{MV} \tag{2.4}$$

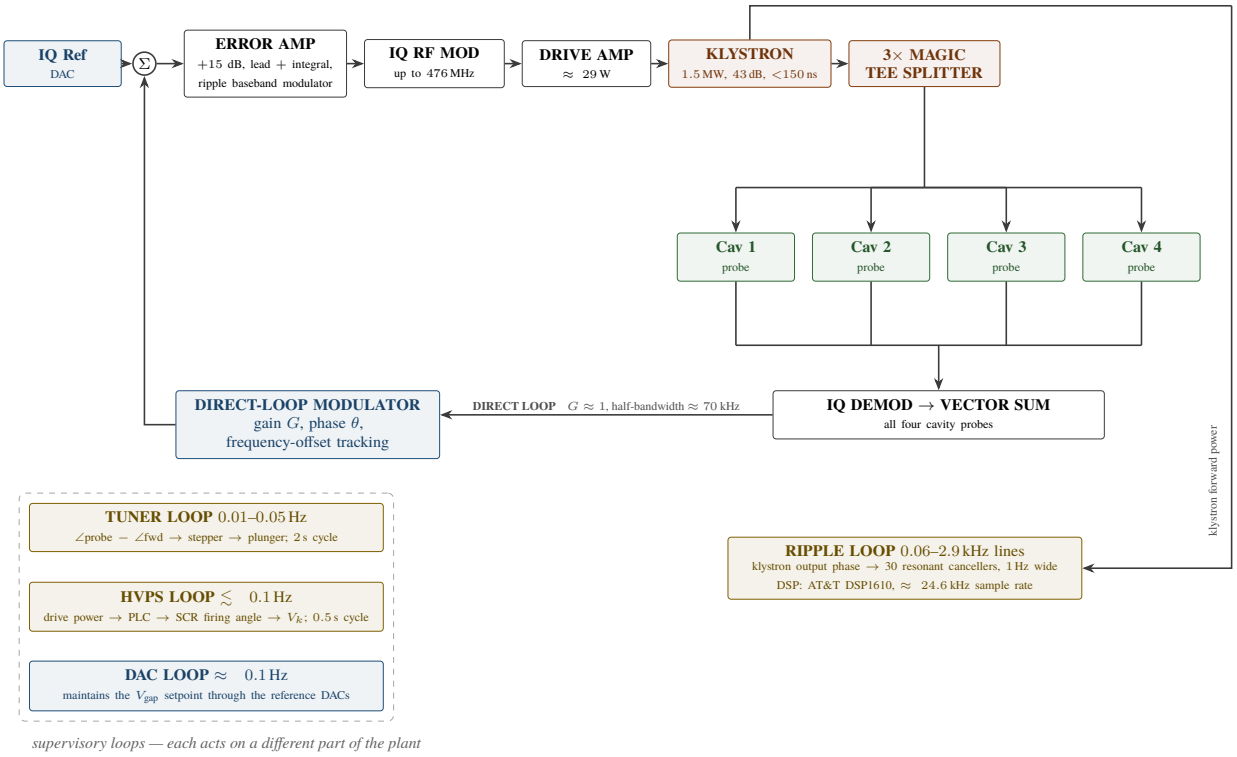


Figure 1: The SPEAR3 RF plant as a control system: the klystron and cavity chain is the plant, the LLRF processor is the controller, and the beam is the principal disturbance. About EPICS Loops: the tuner loop is triggered every 2 s (0.5 Hz), subject to measurement and motor-readiness checks, while the HVPS loop is event-driven, nominally every 0.5 s (2 Hz) with a 10 s fallback timeout; the DAC loop is reported to use similar timing.

and the beam draws twice as much power as the cavity walls dissipate (Eqs. (2.12) to (2.13)). **Beam loading, not wall loss, is what sets the design of this station.**

Synchronous phase — In the convention where ϕ_s is measured from the voltage crest (SLAC convention, used throughout this document):

$$V_{\mathrm{RF}} \cos\phi_s = U_0 \tag{2.5}$$

$$\cos\phi_s = \frac{U_0}{V_{\mathrm{RF}}} = \frac{1.09\ \mathrm{MeV}}{2.85\ \mathrm{MV}} = 0.382 \implies \phi_s \approx 67.5^\circ \tag{2.6}$$

Note: $\sin\phi_s = \sin(67.5^\circ) = 0.924$, which appears in the beam loading compensation formulas below.

Synchrotron frequency:

$$\nu_s = \sqrt{\frac{h\,\alpha_c\,V_{\mathrm{RF}}\,\sin\phi_s}{2\pi\,E_0}} \tag{2.7}$$

With $h = 372$, $V_{\mathrm{RF}} = 2.85$ MV, $\sin\phi_s = 0.924$ and $E_0 = 3000$ MeV: $\nu_s \approx 0.0078$, giving $f_s = \nu_s\, f_{\mathrm{rev}} \approx 10.0$ kHz. Published values: $Q_s = 0.008$ [R1], $\nu_s = 0.007$ [R3] — the range reflects differences in operational V_{RF} and α_c across lattice configurations. For control design, the key fact is that f_s lies well inside the cavity half-bandwidth (35.5 kHz), so the cavity impedance at the two synchrotron sidebands, together with the direct loop, sets the damping of the dipole synchrotron mode (§8).

Optimum detuning — The cavity is detuned so that its susceptance cancels the reactive part of the beam-induced current, leaving the generator driving a real load. Writing $G_c = 1/R_s$ for the cavity conductance (so that $P_{\mathrm{wall}} = \frac{1}{2}V_{\mathrm{gap}}^2 G_c$, consistent with Eq. (2.12)), the reactive balance $V_{\mathrm{gap}}\, G_c \tan\psi_0 + I_{b1}\sin\phi_s = 0$ gives

$$\tan\psi_{0,\mathrm{opt}} = -\frac{I_{b1}\, R_s\, \sin\phi_s}{V_{\mathrm{gap}}} = -\frac{1.0 \times 3.73 \times 10^6 \times 0.924}{712 \times 10^3} = -4.84 \tag{2.8}$$

Here ψ_0 is referred to the *unloaded* bandwidth. The angle the tuner loop actually measures — the phase between the cavity probe and the forward wave — is referred to the *loaded* bandwidth, and the two are related by $\tan\psi_L = \tan\psi_0/(1+\beta)$:

$$\tan\psi_{L,\mathrm{opt}} = \frac{-4.84}{1+3.78} = -1.01 \quad\Longrightarrow\quad \psi_{L,\mathrm{opt}} \approx -45.4^\circ \tag{2.9}$$

The optimum detuning is proportional to the beam current, so the tuner setpoint, the coherent synchrotron frequency and the best direct-loop setting all change with current (§7.2 and §8).

Derivation note: Equation (2.8) is the condition that the reactive component of the generator current vanish, which minimizes the power reflected at the input coupler. It contains the fundamental current $I_{b1} = 2I_b$, not I_b. Written with the geometry factor $R/Q = R_s/Q_0 = 116.6\,\Omega$, as in Eq. (2.10), it no longer depends on Q.

Optimum frequency detuning — converting the unloaded angle to a frequency offset with $\tan\psi_0 = 2Q_0\Delta f/f_0$:

$$\begin{aligned}\Delta f_{\mathrm{opt}} &= \frac{f_0}{2Q_0}\tan\psi_{0,\mathrm{opt}} = -\frac{(R/Q)\, I_{b1}\, \sin\phi_s\, f_0}{2\, V_{\mathrm{gap}}} \\ &= -\frac{116.6\ \Omega \times 1.0\ \mathrm{A} \times 0.924 \times 476.3\ \mathrm{MHz}}{2 \times 712\ \mathrm{kV}} \approx -36.0\ \mathrm{kHz}\end{aligned} \tag{2.10}$$

The cavity must therefore be tuned about 36 kHz **below** f_{RF} at 500 mA — roughly one cavity half-bandwidth (2.3), which is why the detuning angle comes out near -45°. Figure 2 shows the resulting steady-state phasor balance; §7 follows what happens when the beam current or f_{RF} moves away from this operating point.

Required generator power per cavity — At optimum coupling (β_{opt}) and optimum detuning (ψ_{opt}), the minimum generator power per cavity is the sum of cavity wall losses and beam power:

$$P_{\mathrm{gen/cav}} = P_{\mathrm{wall}} + \frac{P_{\mathrm{beam}}}{n_{\mathrm{cav}}} = \frac{V_{\mathrm{gap}}^2}{2R_s} + \frac{I_b\, U_0}{n_{\mathrm{cav}}} \tag{2.11}$$

Evaluation:

$$P_{\mathrm{wall}} = \frac{V_{\mathrm{gap}}^2}{2R_s} = \frac{(712\ \mathrm{kV})^2}{2 \times 3.73\ \mathrm{M\Omega}} = 68\ \mathrm{kW/cavity} \tag{2.12}$$

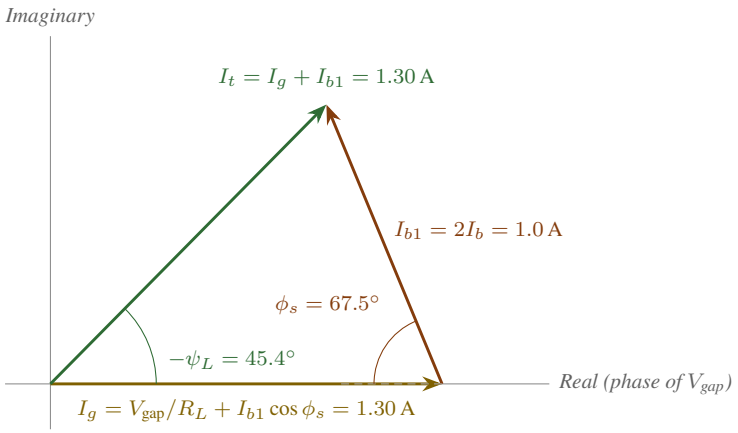


Figure 2: Beam-loading phasors per cavity at 500 mA, drawn to scale and referred to the gap, with the gap voltage on the real axis. At the optimum detuning the generator current I_g is in phase with the voltage, so the generator sees a real load. The quadrature part of the beam current is carried by the cavity itself: its impedance angle $\psi_L = -45.4°$ turns the total current I_t back onto the voltage, $V_{\mathrm{gap}} = R_L \cos\psi_L \, e^{j\psi_L} I_t$.

$$\frac{P_{\mathrm{beam}}}{n_{\mathrm{cav}}} = \frac{I_b\, U_0}{n_{\mathrm{cav}}} = \frac{0.5\ \mathrm{A} \times 1.09\ \mathrm{MeV}}{4} = 136.3\ \mathrm{kW/cavity} \tag{2.13}$$

$$P_{\mathrm{gen/cav}} = 68 + 136.3 \approx 204\ \mathrm{kW/cavity} \tag{2.14}$$

Total RF power: $4 \times 204.3 \approx 817$ kW — within the 1.5 MW klystron capacity, leaving $\sim 46\%$ headroom. The measured forward power on 7 October 2026 (499 mA) was 193–214 kW per cavity, 829 kW in total, consistent with this calculation.

Note: Eq. (2.11) is the minimum-power condition at optimum coupling. In the general case with arbitrary coupling β and detuning ψ, the generator power includes reflected power terms. The important identity is that beam power per cavity $= I_b V_{\mathrm{gap}} \cos\phi_s = I_b\, U_0/n_{\mathrm{cav}}$.

2.3 Klystron as the Actuator

For control analysis, the klystron is modeled as:

$$G_{\mathrm{kly}}(s) = K_{\mathrm{kly}} \cdot e^{-s\tau_{\mathrm{kly}}} \tag{2.15}$$

where K_{kly} is the small-signal gain and $\tau_{\mathrm{kly}} < 150$ ns.

Parameter	Value	Unit	Note	Source
Type	Marconi/CPI K3512S	—	—	[R1]
Maximum output power	1.5	MW	CW	[R1]
Gain	43 (min)	dB	—	[R1]
Bandwidth	5	MHz	-3 dB	[R1]
Group delay	< 150	ns	—	[R1]

Parameter	Value	Unit	Note	Source
Drive power at 500 mA	~ 29	W	—	[R5]; 29.0 W on 7 Oct 2026
Perveance $K = I_k/\lvert V_k\rvert^{3/2}$	$\sim 1.0 \times 10^{-6}$	A/V$^{3/2}$	—	HVPS readbacks (below)

The perveance follows from the HVPS operating point (§2.4.2): 75.5 kV at 22.4 A on 7 October 2026 gives $K = 22.4/(7.55 \times 10^4)^{3/2} = 1.08 \times 10^{-6}$ A/V$^{3/2}$.

Saturation model:

At fixed operating conditions, the static klystron power-transfer characteristic is approximated by

$$P_{\text{out}} = P_\infty \frac{P_{\text{in}}/P_{1/2}}{1 + P_{\text{in}}/P_{1/2}}, \tag{2.16}$$

where P_∞ is the asymptotic output-power parameter and $P_{1/2}$ is the input power at which $P_{\text{out}} = P_\infty/2$, not the input power at saturation. The small-signal power gain is $G_0 = P_\infty/P_{1/2}$. This phenomenological model requires validation against measured power-transfer data and does not describe AM/PM conversion.

AM-PM conversion — klystron output phase sensitivity to cathode voltage [R13]:

$$\Delta\phi_{\text{kly}} \propto \frac{\Delta V_k}{V_k} \tag{2.17}$$

2.4 HVPS Plant Model and Dynamics

2.4.1 Power Supply Architecture

The SPEAR3 klystron HVPS is a PEP-II design: a 12-pulse, primary-side SCR phase-controlled rectifier fed from a 12.47 kV substation feeder [R17], [R18]. Two complete units (SPEAR1 and SPEAR2) are installed; one is energized at a time.

2.4.2 HVPS Electrical Parameters

Parameter	Value	Unit
Input voltage	12,470 V RMS L-L	3-phase 60 Hz
Output voltage range	0 to -90 kV	DC
Maximum current	27 A	DC
Typical operating	-72 to -76 kV, 19–23 A	DC
Complete-supply rating	2,750 kVA (12.5 kV at 127 A RMS)	nameplate
Voltage regulation	$< \pm 0.5\%$	at > 65 kV
Voltage ripple	$< 1\%$ P-P, $< 0.2\%$ RMS	at > 60 kV
Filter inductors (L_1, L_2)	0.3 H each	85 A DC
Filter capacitors (C)	8 μF per stage $\times$ 4 series stages	at 30 kV
SCR crowbar arc energy	< 5 J (with), < 20 J (without)	—

2.4.3 SCR Phase Control Dynamics

$$V_{\text{dc}} = V_{\text{dc,max}} \cos \alpha \tag{2.18}$$

The installed firing electronics is an Enerpro FCOG6100 six-pulse board with an FCOAUX60 auxiliary gate-drive board [R20]; the FCOG1200 is the *replacement* part and is not in service. The board's phase-locked loop free-runs at $384 \times f_{\text{line}} = 23.04$ kHz and locks to the AC mains zero crossings, so the delay angle α is referenced to the line rather than to an internal ramp.

For small-signal control analysis the firing chain is modeled as a single-pole lag:

$$H_{\text{Enerpro}}(s) = \frac{1}{1 + s/\omega_{\text{Enerpro}}} \tag{2.19}$$

with $\omega_{\text{Enerpro}} \approx 415$ rad/s, i.e. -3 dB at 66 Hz. That figure comes from the published loop-filter analysis of this firing-circuit family [R19], which also gives unity DC gain, a $+0.81$ dB peak at 60 rad/s and $-45°$ phase margin at 339 rad/s. The single pole is therefore a simplification: the response is mildly underdamped rather than exactly first order.

Small-signal bandwidth and step settling are different numbers. The 415 rad/s pole corresponds to $\tau = 1/\omega_{\text{Enerpro}} = 2.4$ ms, while a *step* in the SIG HI command settles in **2–3 AC line cycles**, 33–50 ms [R19]. The firing interval does not explain the difference. The 12-pulse rectifier fires 12 times per line cycle, once every $1/(12 f_{\text{line}}) = 1.39$ ms, and a new delay angle takes effect at the next firing, so sampling delays a change by at most 1.39 ms; 2–3 cycles are 24–36 firings. Both numbers belong to the phase-locked loop that times the firings: SIG HI is summed into the PLL's loop filter, so α follows a command through the closed PLL — a small change with the 415 rad/s pole, a large step settling completely only after 2–3 cycles.

2.4.4 PLC Voltage Regulation Loop

The SLC-500 PLC implements a first-order Infinite Impulse Response (IIR) filter with scan period $T = 10$ ms:

$$y[m] = (1 - \lambda)\, y[m-1] + \lambda\, x[m] \tag{2.20}$$

with $\lambda = 0.4$, giving time constant $\tau = -T/\ln(1-\lambda) \approx 20$ ms [R21].

2.4.5 Klystron Arc Protection

Four mechanisms limit the energy an arc can deposit in the klystron: the SCR crowbar, the star-point bypass, the $500\ \Omega$ filter isolation resistors and the cable termination inductors. The published PEP-II result is < 5 J to the klystron with the crowbar operating and $I^2t < 40$ A^{2}s [R17].

Timing and component caveats: the crowbar conduction delay is $\approx 10\ \mu$s and primary current interruption takes 4–8 ms; the termination inductors are 350 μH as built, against the 200 μH design figure of [R17]. The energy and I^2t figures above are published PEP-II design and test results.

2.4.6 HVPS Dynamic Response Summary

The elements below are listed fastest first. Each row states the physical time it describes; they are *not* interchangeable, and the equivalent frequency is quoted only where a single-pole or resonant interpretation is meaningful.

Element	Characteristic time	Equivalent frequency	What the number is
SCR commutation overlap	~ 100 μs	~ 1.6 kHz	Current transfer between thyristors; not a control-loop limit
Output LC filter	$\sqrt{LC} = 775$ μs	$f_0 \approx 205$ Hz	0.3 H per leg into the 2 μF net series bank (§2.4.2); lightly damped resonance
12-pulse firing interval	1.39 ms between firings; $\approx$ 0.69 ms mean transport lag	~ 230 Hz	Sampling limit of phase control: α can change only at a firing instant
Enerpro small-signal pole	$\tau = 2.4$ ms	66 Hz (-3 dB)	Equation (2.19), $\omega_{\text{Enerpro}} = 415$ rad/s [R19]
Enerpro step settling	2–3 line cycles, 33–50 ms	—	Large-signal settling after a SIG HI step [R19]
PLC digital filter	$\tau = 20$ ms	~ 8 Hz	Equation (2.20), $\lambda = 0.4$ at a 10 ms scan
Open-loop voltage step	$\approx$ 50–70 ms	~ 3 Hz	Enerpro settling and PLC filter in series
EPICS HVPS loop	≈ 0.5 s cycle	$\lesssim 0.1$ Hz	Closed-loop bandwidth (§6.5)

Nothing in the plant limits the closed HVPS loop. The loop is closed in EPICS software with a cycle of about 0.5 s (§6.5), far slower than every plant element above. The HVPS is therefore a quasi-static actuator for the RF control system, and any HVPS ripple (D5, §4.5) must be rejected downstream, by the ripple loop and the direct loop, and cannot be regulated away by the HVPS controller itself.

2.5 Tuner Mechanics

Mechanism and motor data are from the tuner documentation set [R23]–[R27]; the 2023 mechanical inspection findings are in [R22].

Parameter	Value
Motor	Superior Electric Slo-Syn M093-FC11 (NEMA 34D)
Drive mechanism	Worm gear (self-locking)
Tuning sensitivity	$p_1 \approx$ 20–30 kHz/mm (Eq. (6.7))
Tuning range	$\sim \pm 200$ kHz
Plunger travel per motor step	3.175 μm (legacy), 0.05 μm (Galil, not yet in service)
Frequency per motor step	$\approx$ 60–100 Hz (legacy)

The cavity resonant frequency is a smooth, monotonic function of plunger insertion depth over the working range. The control software represents it by the cubic-plus-V^2 model of Eq. (6.7), whose coefficients come from the frequency calibration of §9.4.4.

2.6 Loop Delay Budget — The Fundamental Bandwidth Ceiling

The total delay around the direct loop limits the usable proportional gain. For an on-resonance, single-pole cavity, the open-loop response is

$$L(f) = \frac{G}{1 + jf/\Delta f_{1/2}}\, e^{-j2\pi f \tau_d}, \tag{2.21}$$

where G is the low-frequency loop gain. The unity-gain condition gives $f_c = \Delta f_{1/2}\sqrt{G^2-1} \approx G\,\Delta f_{1/2}$ for $G \gg 1$. At frequencies well above the cavity half-bandwidth, the cavity phase approaches $-90°$, while the delay adds a phase lag of $2\pi f \tau_d$ radians. A phase margin of at least $45°$ therefore requires, approximately,

$$2\pi f_c \tau_d \le \frac{\pi}{4}, \qquad G \lesssim \frac{1}{8\tau_d\,\Delta f_{1/2}}. \tag{2.22}$$

$$\boxed{f_{c,\max} \approx \frac{1}{8\tau_d}, \qquad G_{\max} \approx \frac{\pi}{4\,\omega_{1/2}\,\tau_d}} \tag{2.23}$$

The second form is Gamp's maximum-gain rule [R9]. It bounds the impedance reduction that any direct loop can provide. The often-quoted $1/(4\tau_d)$ is the frequency at which the delay alone has consumed $90°$; a crossover there needs a lead network to supply the phase margin. Table 3 checks Eq. (2.23) against the full loop model.

Component	Legacy (analog VXI)	LLRF9 (digital)
Klystron group delay	< 150 ns [R1]	same tube
Drive amplifier	≤ 50 ns (vendor, Doc L)	same amplifier
Controller	analog RFP: a few ns per stage (baseband modulators < 5 ns [R13]); ≈ 30 ns in total (estimate)	≈ 270 ns internal (vendor) [R4]
Probe and drive cables to the tunnel	the balance, ≈ 0.4 μs	same cables
Total τ_d	$\approx$ 625 **ns, measured** (by Jim Sebek on Oct. 2, 2026; 624 ± 4 ns from the closed-loop fits of that day, Table 7). The IOC's tracking parameter `PHASE.B`, 699.5 ns, is the delay that the RFguiInvasive Configure Direct routine fitted to its stored open-loop record [R41], [R42], most likely with the klystron replaced in September 2020 (the closed-loop fits of May 2020 give 700–780 ns [R42]); it has not been updated for the present tube	≈ 870 **ns** (estimate: 270 ns plus the same ≈ 0.6 μs external path)
$G_{\max}$ at $45°$ margin	≈ 15 dB	≈ 12 dB
$f_{c,\max}$	≈ 0.2 MHz	≈ 0.15 MHz

The klystron, the drive amplifier and the cables are common to both controllers, and the LLRF9's 270 ns of internal processing replaces a few tens of nanoseconds of analog RFP. The LLRF9 loop will therefore be slower than the legacy loop, and its delay-limited gain about 2.5 dB lower (§8.5); its total delay must be measured at commissioning.

2.7 Complete Open-Loop Plant Transfer Function

$$G_{\text{plant}}(s) = G_0 \cdot \frac{H_{\text{cav}}(s)}{R_L} \cdot e^{-s\tau_d} = \frac{G_0\,\omega_{1/2}}{s + \omega_{1/2} - j\Delta\omega}\; e^{-s\tau_d} \tag{2.24}$$

where G_0 is the gain of the drive chain (modulator, drive amplifier and klystron) at low frequency; the bandwidth of the chain is added in Eq. (6.1). The cavity half-bandwidth and the loop delay together set the limits of what feedback can achieve.

3 I/Q Signal Processing and the LLRF Signal Chain

3.1 Baseband I/Q Representation

All RF feedback loops act on the complex envelope of the RF signals, written in in-phase and quadrature (I/Q) components:

$$V_{\mathrm{RF}}(t) = A(t)\cos[\omega_{\mathrm{RF}}t + \phi(t)] = I(t)\cos(\omega_{\mathrm{RF}}t) - Q(t)\sin(\omega_{\mathrm{RF}}t) \tag{3.1}$$

where $I(t) = A(t)\cos\phi(t)$ and $Q(t) = A(t)\sin\phi(t)$, with inverse relations:

$$A(t) = \sqrt{I^2 + Q^2}\,, \qquad \phi(t) = \mathrm{atan2}(Q, I) \tag{3.2}$$

A gain change scales the I/Q vector and a phase shift rotates it, so every operation in the loop — weighting a cavity probe, setting the loop gain and phase, comparing with the reference — is a real 2×2 matrix acting on (I, Q). Every loop transfer function in this document is written for the complex envelope.

3.2 The LLRF Signal Chain

Figure 3 follows the signals through the legacy RF processor (RFP), an analog VXI module of PEP-II design [R13], [R36]; the LLRF9 performs the same operations digitally. In order:

1. the four cavity probe signals are demodulated to I/Q (§3.3);
2. a combining network weights each probe and forms the vector sum (§3.4);
3. the direct-loop modulator sets the loop gain and phase on the vector sum, and can rotate the phase with the mean cavity detuning (frequency-offset tracking, §8.8);
4. difference amplifiers subtract the station reference — a pair of RFP DACs set by the DAC loop [R36] — and amplify the error by 15 dB [R13];
5. lead and integral compensation, common to I and Q, and a diode limiter shape the error;
6. an I/Q RF modulator converts it to the klystron drive [R13].

The ripple DSP works alongside this chain: it compares the klystron output with the reference and applies its phase corrections, and the gain-tracking coefficient, through a further baseband modulator in the drive path (§6.3 and §6.4).

3.3 I/Q Demodulation

Each probe signal is converted to baseband by a high-level ($+13$ dBm) I/Q demodulator; the mixer outputs are low-pass filtered ($F_c = 225$ MHz) and amplified by 17 dB to ± 1 V full scale, the input range of the multipliers that follow [R13].

3.4 Cavity Probe Vector Sum

A programmable combining network — four baseband modulators and two summing amplifiers — weights each demodulated probe with a complex gain set by DACs, so that the sum represents the total accelerating voltage of the station [R13].

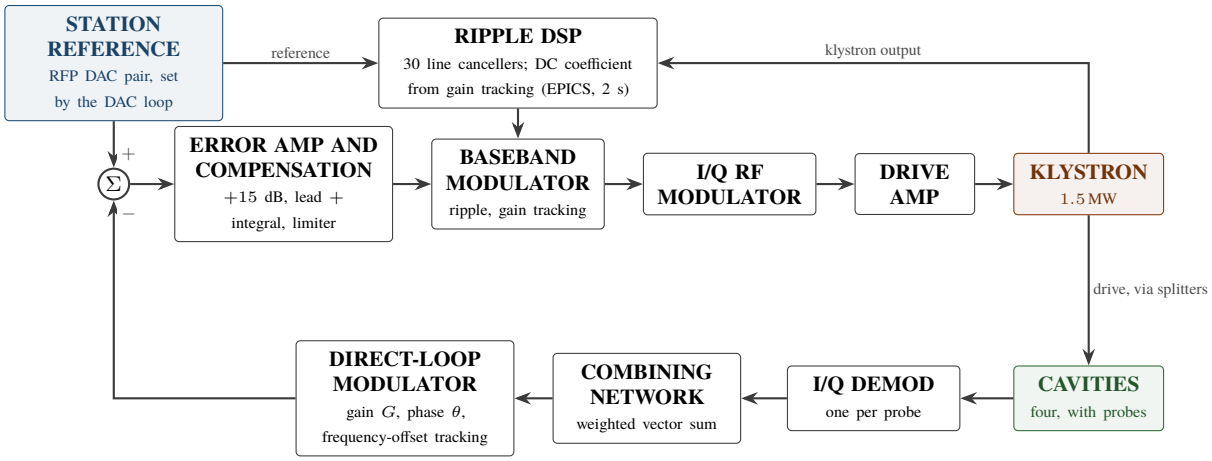


Figure 3: LLRF signal chain of the legacy RFP: forward path along the top, measurement path along the bottom. Every block acts on baseband I/Q; the LLRF9 performs the same operations digitally, with an independent phase for the integral path.

3.5 Error Signal

The loop acts on the difference between the reference and the measured vector sum:

$$\vec{E} = \vec{V}_{\text{ref}} - \vec{V}_{\text{sum}}, \qquad \vec{V}_{\text{drive}}(s) = C(s)\,\vec{E}(s), \tag{3.3}$$

where $C(s)$ is the controller of Eq. (5.1): loop gain and phase, lead and integral compensation.

3.6 Baseband I/Q Modulator

Every gain and phase operation in the chain is a baseband modulator, a scaled rotation

$$\begin{pmatrix} I_{\text{out}} \\ Q_{\text{out}} \end{pmatrix} = G_m \begin{pmatrix} \cos\theta_m & -\sin\theta_m \\ \sin\theta_m & \cos\theta_m \end{pmatrix} \begin{pmatrix} I_{\text{in}} \\ Q_{\text{in}} \end{pmatrix}. \tag{3.4}$$

In the RFP each is built from four AD834 four-quadrant multipliers [R16] and two EL2073 summing amplifiers, with < 5 ns group delay, > 40 MHz full-power bandwidth and > 50 dB dynamic range; the four matrix weights come from 12-bit DACs [R13]. Unlike an RF phase shifter, such a modulator rotates the phase continuously, without steps. The PEP-II RFP uses seven of them: four in the combining network and one each for the direct-loop gain and phase, the comb loop and the ripple loop [R13].

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4 Disturbance Analysis and Control Problem Statement

The ten disturbance classes below are referred to as D1–D10 throughout the rest of this document. Most of them are the reason a particular loop exists; D3, D4 and D7 need no loop of their own at SPEAR3, and D10, the noise of the RF system itself, is shaped by the direct loop rather than removed by it.

4.1 Disturbance Taxonomy

#	Disturbance	Frequency	Magnitude	Impact
D1	Beam loading (steady-state)	DC	$V_{b,\mathrm{res}} = 0.78$ MV/cavity	Dominates voltage budget
D2	Beam loading (transient)	DC–100 kHz (Broadband baseband transients; cavity-field response shaped by the cavity and direct-loop dynamics)	$\approx 0.8\%$ of V_{gap} per 1% current step	Field transient
D3	Robinson (dipole) mode	$f_s \sim 10$ kHz (2.7)	Damped; rate set by the direct loop	Loss of damping if the loop is mis-set
D4	Coupled-bunch modes (fundamental)	$f_{\mathrm{RF}} + n f_{\mathrm{rev}} \pm f_s$	$\lesssim 5\ \mathrm{s}^{-1}$ growth	Negligible at SPEAR3
D5	HVPS ripple	360, 720, 1080… Hz	$< 1\%$ P-P voltage	Phase modulation
D6	Klystron gain drift	0.01–1 Hz	up to 7 dB	Loop gain variation
D7	Microphonics	1–300 Hz	$\Delta f \sim$ 1–10 Hz	Negligible against $\Delta f_{1/2}$
D8	Thermal detuning (cavity)	< 0.01 Hz	$\Delta f \sim$ 1–100 Hz	Slow frequency drift
D9	Ring circumference change	Tidal (12–24 h) to seasonal	f_{RF}: up to $\approx$ 100 Hz within a day, $\approx$ 4 kHz/year	Tuner setpoint migration
D10	RF noise (klystron, drive chain, probes, reference)	Broadband; the beam responds below and near $f_{\mathrm{coh}} \approx 8.8$ kHz	-106 to -103 dBc/Hz beside the synchrotron line (cavity probe, 2026)	Drives the dipole mode: beam timing and energy jitter

4.2 D1/D2: Beam Loading — The Dominant Disturbance

D1 — Steady-state beam loading: The beam draws twice the wall-loss power. Detuning the cavity (§2.2.3) cancels the reactive part of this loading at the operating current; any change of current re-creates it (§7).

D2 — Transient beam loading: Injection, top-off, beam loss, and nonuniform fills produce changes in beam loading faster than the tuner can follow. The disturbance spectrum can extend to f_{rev} and higher revolution harmonics, but the resulting cavity-field perturbation is shaped by the effective cavity impedance, $Z_{\mathrm{eff}} = Z_{\mathrm{cav}}/(1+L)$, with components far from the cavity resonance generally attenuated.

How feedback acts on D1/D2: The direct loop divides the cavity impedance seen by the beam by the return difference $1 + G_{\mathrm{OL}}$ (Eq. (5.4)). The PEP-II design applied $\approx$ 15 dB [R13]; the SPEAR3 legacy loop runs at $G \approx 1$ (0 dB, §8.8), for which $|1 + G_{\mathrm{OL}}|$ is only ≈ 1.6 at the carrier and ≈ 2 at the cavity resonance. A 1% (5 mA) current step changes the beam-induced voltage by $\Delta I_{b1} R_L \cos\psi_L \approx 5.5\,\mathrm{kV}$ per cavity, 0.8% of V_{gap}; the loop divides this residual by $|1 + G_{\mathrm{OL}}(0)|$, and an integral path removes it in steady state. The same impedance, sampled at the synchrotron sidebands, sets the damping of the dipole (D3) and coupled-bunch (D4) modes.

4.3 D3: Robinson Mode

Every bunch oscillates about the synchronous phase at f_s, and the in-phase (dipole, $n = 0$) motion of all bunches drives the cavities at the two sidebands $f_{\mathrm{RF}} \pm f_s$. Above transition the motion is damped when

$$\mathrm{Re}\{Z_{\mathrm{eff}}(\omega_{\mathrm{RF}}+\omega_s)\} < \mathrm{Re}\{Z_{\mathrm{eff}}(\omega_{\mathrm{RF}}-\omega_s)\} \tag{4.1}$$

and the rate at which the cavities add to radiation damping is, to first order [R10], [R14],

$$\frac{1}{\tau_{\mathrm{Rob}}} = -\frac{\alpha_c\,\omega_{\mathrm{RF}}\,I_b}{2\,\omega_s\,(E_0/e)\,T_{\mathrm{rev}}}\left[\mathrm{Re}\{Z_{\mathrm{eff}}(\omega_{\mathrm{RF}}+\omega_s)\} - \mathrm{Re}\{Z_{\mathrm{eff}}(\omega_{\mathrm{RF}}-\omega_s)\}\right] \tag{4.2}$$

(positive means damping), with Z_{eff} the total impedance of the four cavities. A cavity tuned *below* f_{RF} — exactly what beam-loading compensation requires (2.10) — has more resistance at the lower sideband, so at SPEAR3 the Robinson mode is *damped*, and strongly so without feedback: 5.8×10^{3} s^{-1} at 500 mA, 14 times radiation damping. The hazard is therefore not an instability of the beam–cavity system itself but a *loss* of this damping caused by the direct loop. Feedback lowers and widens the resonance, so the damping falls roughly as $(1+G)^{-3}$, and a wrongly phased loop or integral path can reverse its sign [R9], [R11]. At 500 mA the coherent frequency without feedback is pulled from 10.0 kHz down to 6.4 kHz, too large a shift for the first-order Eq. (4.2); §8 solves the full problem and turns it into tuning guidance. The mode is also a resonator that every disturbance near f_s drives — the HVPS lines (D5), beam transients (D2) and the noise of the RF system (D10) — and the motion they cause grows as its damping falls (§8.7).

4.4 D4: Coupled-Bunch Modes

With $h = 372$ RF buckets the beam supports 372 coupled-bunch modes $n = 0,\ldots,h-1$ (mode $n=-1$ is mode $h-1$). Mode n samples the impedance at the synchrotron sidebands of the revolution harmonics $(ph+n)\,f_{\mathrm{rev}}$, and grows at the rate [R10]

$$\frac{1}{\tau_n} = \frac{\alpha_c\,\omega_{\mathrm{RF}}\,I_b}{2\,\omega_s\,(E_0/e)\,T_{\mathrm{rev}}}\sum_p\left[\mathrm{Re}\{Z((ph+n)\omega_{\mathrm{rev}}+\omega_s)\} - \mathrm{Re}\{Z((ph+n)\omega_{\mathrm{rev}}-\omega_s)\}\right] \tag{4.3}$$

(positive means growth); $n=0$ is the Robinson mode of Eq. (4.2). At SPEAR3 the revolution frequency, 1.28 MHz, is 36 cavity half-bandwidths: only the RF harmonic itself falls inside the fundamental resonance, and every mode $n\neq0$ samples it at least one revolution harmonic away. The model of §8 gives a growth rate of at most ≈ 5 s^{-1} (mode $n=-1$, from no feedback to 15 dB of loop gain), against 424 s^{-1} of radiation damping. At PEP-II, with $f_{\mathrm{rev}} = 136$ kHz, more than 1 A of beam and a detuning that moved the cavity resonance past the first revolution harmonic, the same rates were faster than one revolution without feedback [R13]; that is what the PEP-II comb and woofer loops were built for, and why SPEAR3 needs neither (§6.8).

4.5 D5: HVPS Ripple

The 12-pulse SCR rectifier produces two families of harmonics:

12-pulse fundamental harmonics (ideal 12-pulse cancellation):

$$f_{12\mathrm{p}} = 12k\times f_{\mathrm{line}} = 720,\ 1440,\ 2160,\ \ldots\ \mathrm{Hz} \tag{4.4}$$

Residual 6-pulse harmonics (from imperfect phase balance between the two 6-pulse bridges):

$$f_{6\mathrm{p}} = 6k\times f_{\mathrm{line}} = 360,\ 720,\ 1080,\ \ldots\ \mathrm{Hz} \tag{4.5}$$

The 360 Hz component (the first 6-pulse harmonic, absent in ideal 12-pulse operation) is the **dominant residual** and the primary target of the ripple loop. Additional low-order harmonics at 60, 120, 180 and 240 Hz may appear at reduced levels ($\sim$ 20–40 dB below 360 Hz) depending on SCR firing symmetry

[R17], [R18]. The ripple reaches the RF field mainly as phase modulation of the klystron output (2.17). The harmonics continue far above the 2.88 kHz reach of the ripple loop (§6.3): on 8 October 2026 the cavity-probe spectrum showed them up to 10 kHz, and those at 8640 and 9000 Hz, which flank the coherent synchrotron frequency at 500 mA, at -57 and -55 dBc, 16 dB above the noise beside them. They drive most of the beam's dipole motion (§8.8).

4.6 D6: Klystron Gain Variation

As the beam current changes from 0 to 500 mA, the HVPS loop raises the cathode voltage V_k to keep the klystron drive power, and with it the margin below saturation, constant (§6.5). As V_k changes, the small-signal gain of the klystron (the slope of P_{out} against P_{in}) varies by up to ~ 7 dB, and with it the gain of the direct loop. Gain tracking (§6.4) compensates for it.

4.7 D7/D8: Microphonics and Thermal Detuning

For normal-conducting copper cavities: microphonic excursions < 10 Hz (negligible vs. $\Delta f_{1/2} = 35.5$ kHz).

Thermal detuning estimation: The resonant frequency of a copper cavity scales inversely with its linear dimensions: $\Delta f/f = -\alpha_{\text{Cu}} \cdot \Delta T$, where $\alpha_{\text{Cu}} \approx 16.5 \times 10^{-6}$ /°C is the linear thermal expansion coefficient of copper. For a simple cavity at 476.3 MHz, this gives:

$$\frac{\partial f_0}{\partial T} \approx -f_0 \cdot \alpha_{\text{Cu}} = -476.3\ \text{MHz} \times 16.5 \times 10^{-6} \approx -7.9\ \text{kHz/°C} \tag{4.6}$$

4.8 D9: Ring Circumference Change — RF Frequency Migration

The RF frequency must remain an integer multiple of the revolution frequency:

$$f_{\text{RF}} = h \cdot f_{\text{rev}} = \frac{h \cdot c}{C} \tag{4.7}$$

The circumference C follows the tunnel and its floor, and a feedback system updates the RF master-oscillator frequency to track it. Two motions add. Within a day the ground moves with the tides, and f_{RF} varies — back and forth, not as a drift — by up to about 100 Hz in the operating history. Over a year the tunnel temperature changes, and f_{RF} migrates by about 4 kHz. With $\Delta C/C = -\Delta f_{\text{RF}}/f_{\text{RF}}$,

$$\frac{\Delta f_{\text{RF}}}{f_{\text{RF}}} = \frac{100\,\text{Hz}}{476.3\ \text{MHz}} = 2.1 \times 10^{-7}\ \text{(daily, tidal)}, \qquad \frac{4\,\text{kHz}}{476.3\ \text{MHz}} = 8.4 \times 10^{-6}\ \text{(yearly, thermal)}, \tag{4.8}$$

that is up to about 50 μm of circumference within a day and about 2.0 mm over a year, the latter consistent with a seasonal tunnel temperature change of ~ 0.8°C ($\alpha_{\text{struct}} \approx 10$–$12 \times 10^{-6}$/°C). A longer ring needs a lower f_{RF}.

Contrast with D8 (cavity body thermal detuning):

	D8: Cavity thermal detuning	D9: Ring circumference change
Physical origin	Cavity copper temperature $\rightarrow f_0$ shifts	Tides (daily) and tunnel temperature (seasonal) $\rightarrow C$ changes $\rightarrow$ required f_{RF} shifts
Rate	-7.9 kHz/°C	Up to ≈ 100 Hz within a day; ≈ 4 kHz/year (operational data)

	D8: Cavity thermal detuning	D9: Ring circumference change
Timescale	Minutes to hours	Tidal (12–24 h) to seasonal
Correction	Tuner re-tracks f_0	RF frequency feedback moves f_{RF}; the tuners follow

Impact on the control system: The operating detuning $\Delta f = f_0 - f_{\mathrm{RF}}$ (§B.9) defines the tuner equilibrium. D8 moves f_0 and D9 moves f_{RF}. Because the tuner loop (§6.6) is closed around the detuning angle $\psi = \arctan(2Q_L\Delta f/f_0)$, it re-converges to ψ_{opt} after either, provided the tuner range is large enough for the accumulated seasonal shift.

4.9 D10: RF Noise

Every element of the chain adds broadband noise to the field: the klystron and its HVPS (between and above the ripple-loop lines), the drive amplifier, the cavity probes and demodulators that form the vector sum, and the reference. The direct loop divides noise that enters ahead of the cavities by the return difference and copies noise in its measurement and reference onto the cavities through T (Eq. (5.3)). For the field, D10 is the floor of the regulation. For the beam it is the drive of the dipole mode, which follows it below the coherent synchrotron frequency and amplifies it near that frequency by up to $Q\,(f_s/f_{\mathrm{coh}})^2$ (§8.7). No loop removes D10: the gain and phase of the direct loop decide how much of it reaches the beam, and with the cavities detuned they do so differently for amplitude and phase noise. On 8 October 2026 the cavity-probe spectrum showed -106 and -103 dBc/Hz beside the two sidebands of the synchrotron line; this broadband noise moved the beam's phase by 0.14–0.19° rms at f_{coh}, less than the HVPS harmonics beside the line (D5) did (§8.8).

4.10 Frequency-Domain Summary: The Disturbance Spectrum

Figure 4 summarises the disturbances on one frequency axis. Each band that threatens the field has one loop that owns it (§6); D3 is damped by the detuned cavities, and the direct loop must be set so as not to undo that (§8). D10 has no band of its own: it is broadband, and the beam responds to the part of it below and near the synchrotron frequency (§8.7).

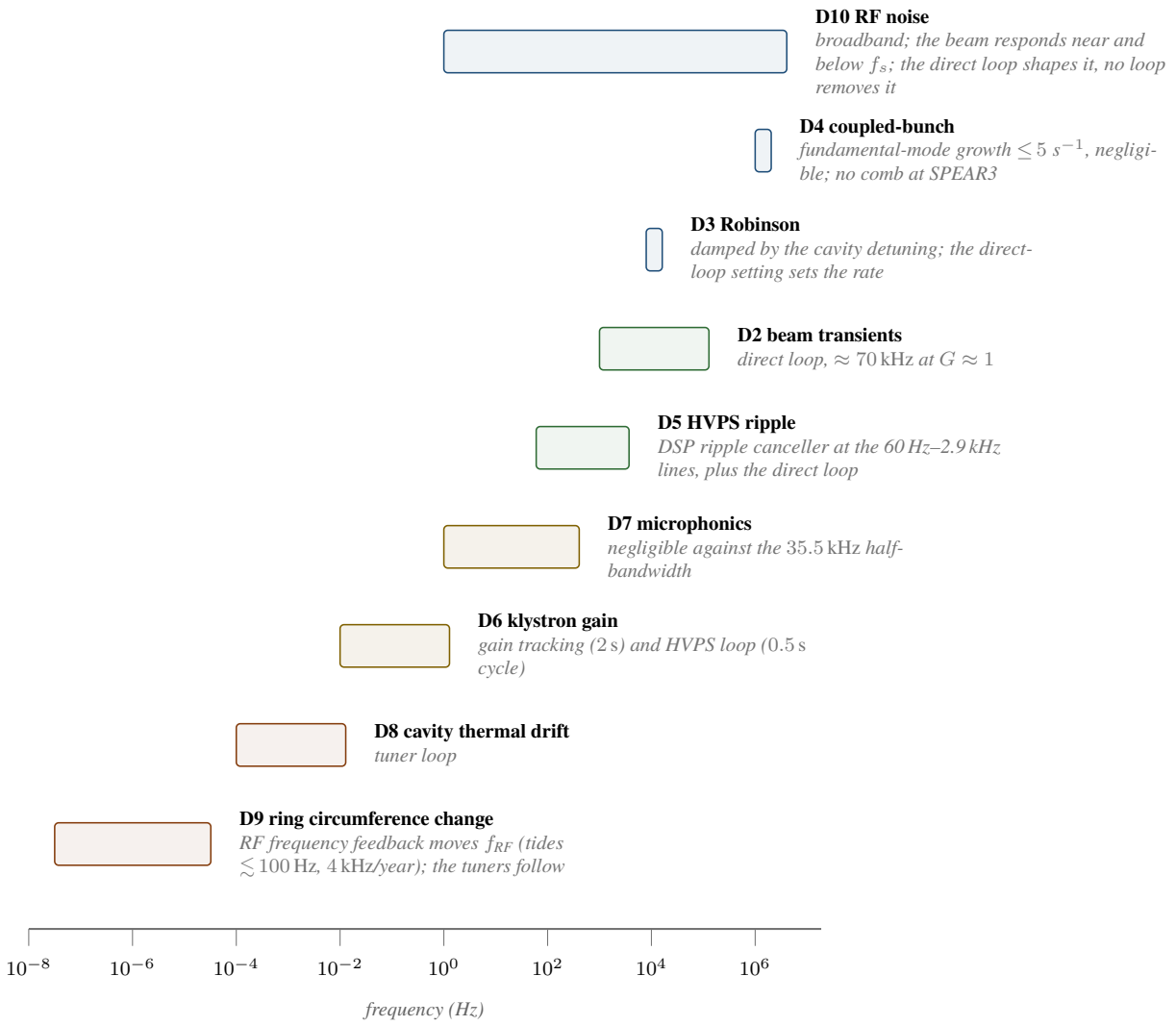


Figure 4: Disturbance spectrum and the loop that answers each band.

5 Control Architecture: From Disturbances to Loops

5.1 The Design Logic: Bandwidth Matching

The design principle is to **match each loop's bandwidth to the frequency content of the disturbance it answers**. For SPEAR3:

$$\text{Tuner } (0.01\text{–}0.05 \text{ Hz}) \lesssim \text{HVPS, DAC, gain tracking } (\lesssim 0.1 \text{ Hz}) \ll \text{Ripple (1 Hz-wide lines at 60 Hz–2.9 kHz)} \ll \text{Direct } (\approx 70 \text{ kHz})$$

Figure 5 shows what each loop measures and what it drives. Every loop has an actuator of its own, and only the direct and ripple loops act on the RF field within a synchrotron period.

5.2 Feedback Fundamentals: Loop Gain, Closed Loop and Effective Impedance

The three relations below apply to the RF field loops; §6 gives the block-by-block form and the parameters of each loop.

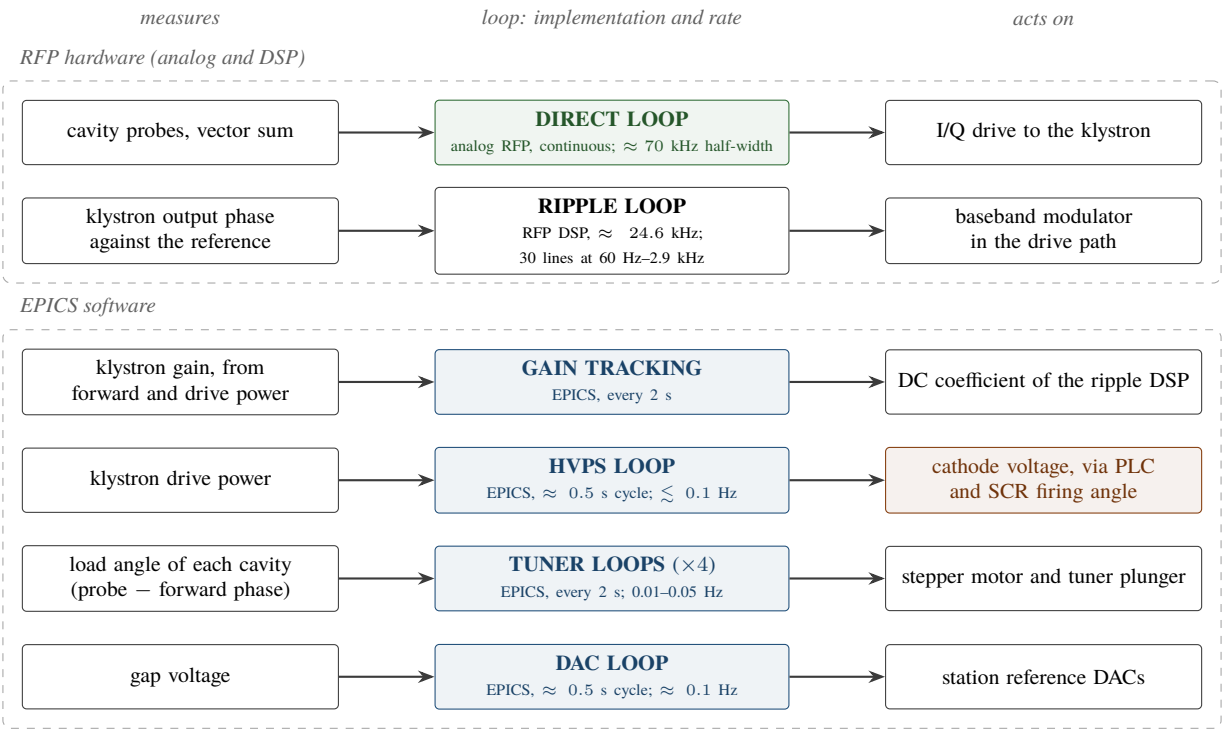


Figure 5: The active SPEAR3 control loops, in the order of §6: what each loop measures (left), how and how often it acts (middle), and what it drives (right). Green: analog RFP; gray: RFP DSP; blue: EPICS software; brown: high-power actuator.

Open-loop transfer function:

$$G_{\mathrm{OL}}(s) = C(s)\, G_{\mathrm{plant}}(s), \qquad C(s) = G\, e^{j\theta}\, G_{\mathrm{lead}}(s)\, G_{\mathrm{int}}(s), \tag{5.1}$$

with the plant of Eq. (2.24). The controller $C(s)$ has four parts: the loop gain G, the loop phase error θ, the lead network $G_{\mathrm{lead}}(s)$ and the integral path $G_{\mathrm{int}}(s)$. Their explicit forms, the stability margins and what is known of their SPEAR3 values are in §6.2.

Closed-loop transfer function:

$$T(s) = \frac{G_{\mathrm{OL}}(s)}{1 + G_{\mathrm{OL}}(s)} \tag{5.2}$$

At frequencies well below f_c, $|G_{\mathrm{OL}}| \gg 1$ and $T \approx 1$; above f_c, $T \to 0$.

Disturbance rejection and noise transfer:

$$\vec{V}_{\mathrm{err}} = \frac{\vec{V}_{\mathrm{d}}}{1 + G_{\mathrm{OL}}} - T\left(\delta\vec{V}_{\mathrm{m}} - \delta\vec{V}_{\mathrm{ref}}\right), \tag{5.3}$$

where $\vec{V}_{\mathrm{d}}$ is the field error that a disturbance entering ahead of the cavities — beam loading, klystron, HVPS — would cause with the loop open, $\delta\vec{V}_{\mathrm{m}}$ the noise in the measured vector sum and $\delta\vec{V}_{\mathrm{ref}}$ that of the reference. The loop suppresses the first by the return difference and copies the second onto the cavities: more gain cannot remove measurement or reference noise (D10). With the cavities detuned, G_{OL} and T are not symmetric about the carrier (Figure 6). Acting differently on the upper and the lower sideband of an error, they turn part of an amplitude error into a phase error and back. For the size of the field error that

does not matter; for the beam, which senses one combination of the two (§8.1), it does, and §8.7 treats the two quadratures separately.

Effective cavity impedance seen by the beam:

$$Z_{\text{eff}}(\omega) = \frac{Z_{\text{cav}}(\omega)}{1 + G_{\text{OL}}(\omega)} \tag{5.4}$$

This is the central result. The direct loop **lowers and widens the cavity resonance**: for a proportional loop of gain G at the correct phase and negligible delay, $R_L \to R_L/(1+G)$ and $\omega_{1/2} \to (1+G)\,\omega_{1/2}$ [R13], [R12], [R9]. The reduction is set by the gain:

- At the cavity resonance: $20\log_{10}(1+G)$ — about 16 dB for the 15 dB PEP-II design, 6 dB for the present SPEAR3 legacy setting $G \approx 1$
- Near the carrier: an integral path adds gain without limit as $f \to 0$
- Above the crossover f_c: no reduction

Less impedance is not automatically more stability. The Robinson damping of §4.3 comes from the *difference* of the resistive impedance at the two synchrotron sidebands. The loop that reduces the impedance also reduces that difference, and a loop phase error rotates reactive impedance into resistive impedance of either sign. §8 quantifies both effects.

5.3 Multi-Loop Stability: The Bandwidth Separation Principle

Several loops act on one RF station, and two loops interact only where both have gain. If their crossover frequencies differ by at least a factor of ten,

$$\frac{f_{c,\text{fast}}}{f_{c,\text{slow}}} \geq 10 \quad \text{for adjacent loops acting on the same quantity,} \tag{5.5}$$

the faster loop sees the slower one as frozen, the slower loop sees the faster one as a closed loop of unit gain, and each can be designed and checked on its own [R33]. At SPEAR3 the EPICS loops are about a million times slower than the direct loop. The ripple resonators lie inside the direct loop's band but are compatible with it for another reason: each is only ≈ 1 Hz wide (§6.3), so it adds gain only on its own line. The EPICS loops are not separated in frequency among themselves, but each drives a different actuator, and the direct loop, which holds the field, decouples them.

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6 Loop-by-Loop Transfer Functions and Design

Each subsection follows the same order: purpose, signal flow, control law and parameters.

6.1 Loop Overview

The timescale of each disturbance dictates the technology of the loop: analog hardware for the microsecond-scale beam loading, a DSP for the narrow HVPS ripple lines, and EPICS software for everything slower (1s or more).

Loop	Implementation	Controller	Rate and bandwidth	Addresses	SPEAR3 status
Direct (§6.2)	Analog RFP (VXI)	Proportional + lead + integral	Continuous; $\approx$ 70 kHz closed-loop half-width at $G \approx 1$	D1, D2, D5, D6; sets D3 damping	Active
Ripple (§6.3)	DSP1610 on the RFP	Resonant phase cancellers at 30 line harmonics	$\approx$ 24.6 kHz sampling; lines at 60 Hz–2.9 kHz, $\approx$ 1 Hz wide	D5; slow klystron phase	Active
Gain tracking (§6.4)	EPICS subroutine	Ratio maintenance	Every 2 s	D6	Active
HVPS (§6.5)	EPICS SNL, PLC, SCR firing	Proportional with deadband	≈ 0.5 s cycle; $\lesssim 0.1$ Hz	D6 (drive margin)	Active
Tuner (§6.6)	EPICS SNL, four stepper motors	Bang-bang with deadband	Every 2 s; 0.01–0.05 Hz	D1 detuning, D8, D9	Active
DAC (§6.7)	EPICS SNL	Proportional with deadband	≈ 0.5 s cycle; ≈ 0.1 Hz	Slow amplitude drift	Active
Comb, woofer, gap feedforward (§6.8)	PEP-II digital modules	—	—	D2, D4	Not installed

6.2 Direct Loop — Wideband RF Field Regulation

Purpose: Regulates the vector sum of the four cavity fields against beam loading, klystron and HVPS disturbances (D1, D2, D5, D6) and divides the cavity impedance seen by the beam by the return difference $1 + G_{\mathrm{OL}}$. At SPEAR3 it is *not* needed to cure an instability: the Robinson mode is damped by the detuned cavity and the fundamental-mode coupled-bunch modes are negligible (§4.3 and §4.4). Its gain and phase instead decide how much of the Robinson damping survives and how strongly the noise of the RF system (D10) moves the beam, the subject of §8.

Signal flow: §3.2 and Figure 3. Because the gain and phase are applied once, before the compensation networks, the integral path is rotated together with the proportional path.

Open-loop transfer function (Eq. (5.1) written out, normalized so that G is the loop gain at the resonance of an on-tune cavity):

$$G_{\mathrm{OL}}(s) = G\,e^{j\theta} \cdot \frac{1 + s/\omega_z}{1 + s/\omega_p} \cdot \left(1 + \frac{\omega_i}{s}\right) \cdot \frac{\omega_a}{s + \omega_a} \cdot \frac{\omega_{1/2}}{s + \omega_{1/2} - j\Delta\omega} \cdot e^{-s\tau_d} \tag{6.1}$$

where the terms are (left to right): loop gain and phase error, lead compensator, PI integrator, drive amplifier and klystron bandwidth, cavity response (2.1), and transport delay. At SPEAR3 both networks are on (Table 6). The RFgui loop models [R41] give the RFP lead network as $\omega_z = 2\pi \times 944$ kHz and $\omega_p = 2\pi \times 2.39$ MHz, and the integral network as a lag with a zero at 3.19 kHz and a pole at 0.26 kHz, 21.8 dB more gain at DC than at high frequency: in the form of Eq. (8.2), $\omega_i = 2\pi \times 2.9$ kHz with a leak $\omega_\ell = 2\pi \times 0.26$ kHz. The measured closed-loop responses and the loop on/off measurements of June 2020 agree with this lag network (§8.8); the measured Robinson damping, however, matches the model only with an effective $\omega_i \approx 2\pi \times 1$ kHz (§8.6 and §8.8). Table 1 lists the values that the analysis of §8 uses.

Stability margins: The design targets are a phase margin of at least 45° and a gain margin of at least 6 dB. With the cavity contributing -90° above its half-bandwidth and the delay $-2\pi f \tau_d$, a proportional loop meets the phase-margin target up to ≈ 15 dB at the legacy delay of 625 ns (Eq. (2.23), Table 3). PEP-II

applied 15 dB with less than 500 ns of delay and used its lead-lag network for extra phase margin with the cavities detuned [R13]. At the present SPEAR3 setting $G \approx 1$ the proportional path alone has no unity-gain crossover ($|G_{\mathrm{OL}}| \leq 1$); the lag network adds one at $\approx \pm 6$ kHz on tune and $\approx \pm 3$ kHz at 500 mA, with at least 89° of phase margin. The binding margin is then the ≈ 20 dB of gain left before the delay-limited instability. The legacy configuration routine (RFguiInvasive Configure Direct) sets the gain for a 66° phase margin [R41].

Closed-loop response: Figure 6 shows $T(s)$ of Eq. (5.2) for the model of Table 1, with the RFP lead network added, under two conditions, together with the response measured on 2 October 2026 with the RFgui closed-loop routine [R38]. With the cavities on tune and no beam the response is symmetric, and the proportional path alone would peak at $G/(1+G)$ on resonance. A delay-free proportional loop would widen the cavity's half-bandwidth by $1+G$, to ± 67 kHz at $G = 0.875$; the 625 ns inside the loop, partly offset by the lead network, widen it further, to half-power points at ± 77 kHz. The response measured at 22 mA follows the model with the integral path within 1 dB and 11° out to ± 100 kHz but falls off faster beyond. Fitted with the full loop model, lag network included, and with gain, phase, delay, Q_L and detuning free, the six runs at 20–22 mA give $G = 0.88 \pm 0.03$, 624 ± 4 ns and $Q_L = 6800 \pm 250$, close to the 6700 of Table 1 (Table 7). The routine's own resonator-times-delay fit gives $Q = 3447$, with half-power points at ± 69 kHz, and 621 ns.

At 500 mA the cavities sit 36 kHz below f_{RF} and the response is asymmetric: $|T|$ peaks at -44 kHz, and at the carrier it is 0.45 with a phase of -27°. The response measured at 497 mA has a similar phase but lies 3–7 dB below the model. The difference needs no change of gain, delay or Q_L: with them held at their low-current values, a klystron whose small-signal response to the drive amplitude differs from that to the drive phase reproduces the measurement (dotted; §8.8). The integral path changes mainly the region within about ± 10 kHz of the carrier. The lag network raises $|T|$ at the carrier to 0.92 and the rejection $|1 + G_{\mathrm{OL}}|$ at 60 Hz to 22 dB on tune and 19 dB at 500 mA. At the synchrotron sidebands, ± 8.8 kHz (panels c and d), it changes $|T|$ by at most 0.06 and turns the phase by about 11°, in opposite directions at the two sidebands. In the model this is enough to make the Robinson mode unstable at 500 mA (§8.6), because the damping is a small difference between the two sidebands; the beam is in fact well damped, and §8.8 discusses the discrepancy. The routine leaves out ± 18 kHz around the carrier, but the recorded signals cover it: fitted within ± 25 kHz with the proportional part held, the 22 mA records give $f_i = 3.0$–4.2 kHz and a leak below 0.15 kHz.

Parameter	Value	Notes
Addresses	D1, D2, D5, D6; sets D3 damping	Field regulation
Measurement	Cavity probe vector sum (I/Q)	After combining network
Actuator	I/Q modulator on klystron drive	Direct analog path
Total loop delay τ_d	≈ 625 ns (measured, legacy)	§2.6
Loop gain G	PEP-II design 15 dB [R13]; SPEAR3 legacy now ≈ 0 dB	`COUNTS.A` $= 1.3$ dB; §8.8
Bandwidth	$\approx$ 70 kHz closed-loop half-width at $G \approx 1$ (77 kHz at $G = 0.875$ with the loop delay and lead network, Figure 6); ≈ 0.2 MHz crossover at 15 dB	Equation (2.23)

Parameter	Value	Notes
Integral path	PEP-II: "30 kHz bandwidth integrator" [R13]; legacy RFP: lag network, zero 3.19 kHz, pole 0.26 kHz (RFgui model [R41], consistent with the closed-loop measurements); the Robinson damping matches an effective $f_i \approx 1$ kHz	Robinson-safe only if weak at f_s (§8.6 and §8.8)
Lead compensation	On; zero 944 kHz, pole 2.39 MHz (RFgui model [R41])	Adds phase margin at high gain
Impedance reduction	$20\log_{10}(1+G)$: 6 dB now, 16 dB at the design gain	Unlimited near DC with an ideal integral path; about 22 dB with the RFP lag network
SPEAR3 status	**Active**	—

6.3 Ripple Loop — HVPS Harmonic Rejection

Purpose: Cancels the RF phase modulation caused by HVPS switching ripple (D5) and holds the slow phase across the klystron and drive amplifier as the cathode voltage moves (D6). The ripple appears as harmonics of 60 Hz on the klystron cathode voltage and reaches the RF through the voltage sensitivity of the klystron phase (2.17).

Signal flow: klystron output and station reference I/Q (digitized on the RFP itself since the Rev 4 module of 2006) → DSP phase error $e_\phi[m] = \phi_{\text{ref}} - \phi_{\text{kly}}$ → a DC term plus a bank of resonant filters at line harmonics → phase correction applied through a baseband modulator in the drive path [R13], [R30].

Key design choice: The loop measures the **klystron output** rather than the cavity probes, because the ripple frequencies (< 3 kHz) lie well inside the cavity bandwidth and the correction must be applied upstream.

Algorithm (SPEAR3 firmware `sp3ripple.s` and coefficient table `Sp3PhsCoefs.tbl`, W. Ross, 2006 [R30]): each harmonic k is a second-order resonator acting on the phase error,

$$H_k(z) = \frac{c_{0k}\, z^{-1} + c_{1k}\, z^{-2}}{1 - 2r_k \cos(2\pi f_k T_s)\, z^{-1} + r_k^2\, z^{-2}}, \qquad u_\phi = \Big(H_{\text{DC}} + \sum_k H_k\Big)\, e_\phi, \tag{6.2}$$

where $e_\phi = \phi_{\text{ref}} - \phi_{\text{kly}}$ is the phase error of the klystron output against the station reference, u_ϕ is the phase correction that the DSP applies through its baseband modulator in the drive path, H_{DC} is the DC term that holds the slow phase, and $T_s \approx 1/24.6$ kHz is the sampling period (one sample per 52 turns). The pole radius $r_k^2 = 0.999838$ makes every line about 1 Hz wide, so the loop has very high gain exactly at the line harmonics and essentially none in between: it is an internal-model (repetitive) controller, not a broadband loop. The numerator coefficients c_{0k}, c_{1k} set the gain and phase of each line to suit the drive-path phase at that frequency. The 2006 revision removed the amplitude estimator to make room for double-precision phase arithmetic, so **the loop corrects phase ripple only**.

Harmonics: four "slow" lines at 60, 120, 180 and 240 Hz, updated in turn at a quarter of the sample rate, and 26 "fast" lines between 300 Hz and 2.88 kHz (every 60 Hz up to 960 Hz, sparser above). The table also lists eight lines between 5.76 and 7.2 kHz that the SPEAR3 build does not process.

Parameter	Value	Notes
Addresses	D5 (HVPS ripple), slow klystron phase (D6)	Phase only
Measurement	Klystron output against station reference (I/Q)	Upstream of the cavities

Parameter	Value	Notes
Line width	≈ 1 Hz per harmonic	$r_k^2 = 0.999838$ (fast), 0.999350 (slow)
Harmonics	4 slow (60–240 Hz) + 26 fast (0.3–2.88 kHz)	`Sp3PhsCoefs.tbl`
Sample rate	≈ 24.6 kHz	Ripple clock on the RFP module
DSP	AT&T DSP1610 (16-bit fixed-point)	On RFP VXI module
SPEAR3 status	**Active** (`STN:RFP:RIPPLELOOP` on, 7 October 2026)	DC-coefficient tracking also on

What the ripple loop does at SPEAR3. The legacy documents disagree: [R12] (1999) describes it only as a slow phase tracker, [R36] (2004) states that harmonic removal is not implemented, and [R1] (2004) describes adaptive cancellation between 60 Hz and 10 kHz. The firmware settles the question: since its 2006 revision the DSP cancels klystron phase ripple at 30 line harmonics up to 2.88 kHz, in addition to tracking the slow phase. Records of 2 June 2020 show both loops at work on the 360 and 720 Hz phase lines of the cavity field: the ripple loop alone reduced them by 26 and 18 dB, the direct loop alone, at the gain of the time and with its lag network, by 35 and 29 dB, and the two together by 39 and 33 dB [R42]. The legacy system rejected these sidebands by ≈ 25 dB in 2021 [R4]; since the direct loop's proportional path alone provides only ≈ 4 dB there (Table 4), that rejection comes from the ripple loop and the lag network. The LLRF9 has no separate ripple loop: line-harmonic rejection is part of its digital field loop and is set by that loop's proportional and integral gains (16–21 dB measured in 2021). At the gains SPEAR3 can use, most of it must come from the integral path, which is constrained by Robinson stability (§8.6).

6.4 Gain Tracking

Purpose: Keeps the gain of the direct loop's forward path constant while the klystron gain changes with cathode voltage by up to ~ 7 dB (D6).

Implementation: Every 2 s (`STNRIPPLE:LOOP:READY`) the EPICS subroutine `subSysDCcoeff` [R37] reads the klystron gain K_{kly} in dB, from the forward and drive power monitors, and moves the DC coefficient of the ripple DSP towards

$$c_{\mathrm{DC}} = c_{\max}\, 10^{(K_{\mathrm{kly,set}} - K_{\mathrm{kly}})/20}, \tag{6.3}$$

so that c_{DC} times the klystron gain stays constant. Each step is limited by a deadband in dB and by a maximum change of c_{DC}, and tracking pauses during a beam abort or when the klystron gain is low (park). The DSP applies this coefficient, together with its harmonic phase corrections, through its baseband modulator in the drive path [R13]. The RFguiInvasive Configure Direct routine (`ConfDirect` in [R12]) sets the loop gain, the loop phase and the starting point of the gain tracking [R41].

6.5 HVPS Loop — Klystron Operating Point

Purpose: Adjusts the cathode voltage V_k to keep the klystron drive power constant, so that the klystron stays $\sim 10\%$ below saturation as the beam current changes (D6) and the direct loop keeps headroom for fast corrections.

Signal flow: klystron drive power (or the station gap voltage) $\rightarrow$ EPICS SNL `rf_hvps_loop.st,v` $\rightarrow$ PLC voltage setpoint $\rightarrow$ Enerpro SCR firing angle $\rightarrow V_k$ [R12].

Control law: In processing mode (`STATION_PROC`) the program steps V_k up or down while it watches the cavity vacuum, to condition the cavities. In operation (`STATION_ON_CW`) it is a proportional controller with deadband and rate limit,

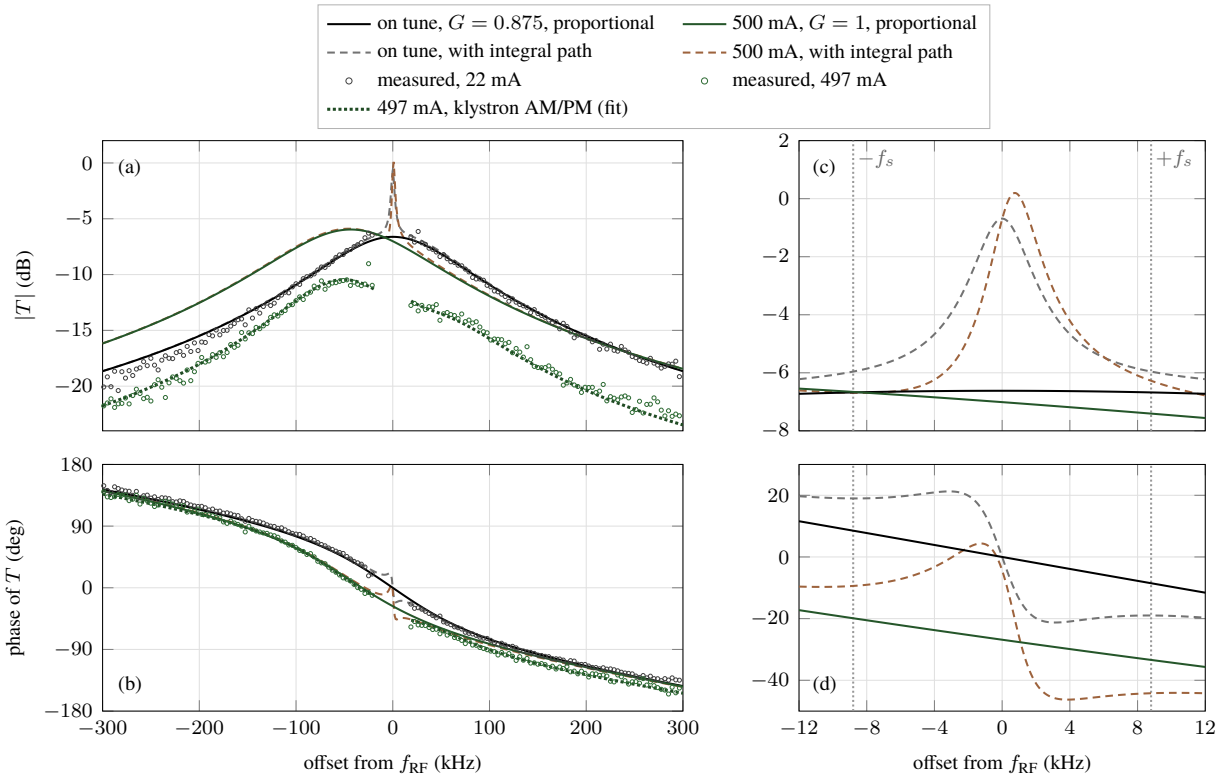


Figure 6: Closed-loop response T of the direct loop (Eq. (5.2)), model of Table 1 with the RFP lead network, and measurements: (a) magnitude and (b) phase within ± 300 kHz of f_{RF}; (c), (d) the model within ± 12 kHz. Black and gray: cavities on tune, no beam, $G = 0.875$ (the gain fitted at 20–22 mA, Table 7). Green and brown: 500 mA, cavities at the optimum detuning (-36 kHz), $G = 1$. Solid: proportional loop; dashed: with the integral path, the RFP lag network ($f_i = 2.9$ kHz, leak 0.26 kHz, sharing the loop phase). Circles: measured on 2 October 2026 [R38] at 22 mA (black, 09:47) and 497 mA (green, 13:51), both at `PHASE.C` $= -20°$; every third point is shown, and ± 18 kHz around the carrier is left out. Dotted, (a) and (b): the 497 mA run fitted with the klystron's amplitude gain and AM/PM conversion free and G, τ_d and Q_L held at the 20–22 mA values (§8.8). Dotted, (c) and (d): the synchrotron sidebands at 500 mA. Phase in the convention of Eq. (5.2), zero on resonance for $\theta = 0$; the RFgui tools quote the same ideal phase as $180°$, and the measured phases are converted accordingly.

$$\Delta V_k[m] = \begin{cases} K_{\text{HVPS}}\,(P_{\text{drive}} - P_{\text{drive,set}}) & \text{if } |P_{\text{drive}} - P_{\text{drive,set}}| > P_{\text{deadband}} \\ 0 & \text{otherwise,} \end{cases} \tag{6.4}$$

clamped to a maximum step per cycle. A cycle runs each time the station-voltage readback updates — "usually every 0.5 second" in the program's own description — and at least every 10 s.

Parameter	Value	Notes
Addresses	D6	Klystron operating point
Measurement	Klystron drive power or station gap voltage	Mode-selectable
Actuator	HVPS cathode voltage V_k	Via PLC and Enerpro firing angle
Cycle	≈ 0.5 s, event-driven; at most 10 s	`HVPS:LOOP:READY`
Bandwidth	$\lesssim 0.1$ Hz	Set by the cycle and the deadband, not by the plant (§2.4.6)
Implementation	EPICS SNL `rf_hvps_loop.st,v` (343 lines, 4 states)	init, off, proc, on
SPEAR3 status	**Active**	

6.6 Tuner Loop — Cavity Resonant Frequency Tracking

Purpose: Moves each cavity tuner to hold the optimum detuning angle ψ_{opt} (2.9) as the beam current changes (the dominant term), and follows cavity thermal drift (D8) and the migration of f_{RF} (D9).

Signal flow: cavity probe phase and forward phase → difference (the load angle) → comparison with ψ_{target} → deadband → stepper motor → tuner plunger (§2.5).

Control law (`rf_tuner_loop.st,v` [R28], executed every 2 s):

$$\varepsilon = \left[\angle(V_{\text{probe}}) - \angle(V_{\text{fwd}})\right] - \psi_{\text{target}} \tag{6.5}$$

$$\text{Step command} = \begin{cases} +N_{\text{step}} & \text{if } \varepsilon > \varepsilon_{\text{deadband}} \\ -N_{\text{step}} & \text{if } \varepsilon < -\varepsilon_{\text{deadband}} \\ 0 & \text{otherwise} \end{cases} \tag{6.6}$$

Frequency offset estimate (`subSysFreqOff` [R37]):

$$\Delta f_{\text{est}}(x, V) = p_0 + p_1(x - x_{\text{home}}) + p_2(x - x_{\text{home}})^2 + p_3(x - x_{\text{home}})^3 + t_1 V^2 \tag{6.7}$$

where x and x_{home} are the tuner position and its home position in mm, $p_0 \ldots p_3$ are calibration coefficients ($p_1 \approx$ 20–30 kHz/mm) and $t_1 V^2$ is a term in the cavity voltage (named a temperature coefficient in the code). Both are written by the RFguiInvasive Make Polynomials routine, which measures each cavity's resonance over the tuner range with the direct loop open, fits the cubic, and derives t_1 from the shift once the cavities are at voltage [R41]. The estimate is smoothed exponentially. The tuner loop itself servos on the load angle; the estimate is used to park the cavities and by the direct loop's frequency-offset tracking (§8.8).

Parameter	Value	Notes
Addresses	D1 detuning with current, D8, D9	One loop per cavity

Parameter	Value	Notes
Measurement	$\angle(\text{probe}) - \angle(\text{fwd})$	Load angle
Actuator	Stepper motor → tuner plunger	§2.5
Cycle	2 s	`CAVTUNR:LOOP:SEQ`
Bandwidth	≈ 0.01–0.05 Hz	Set by the cycle and the deadband
Control law	Bang-bang with deadband	N_{step} steps per cycle
Implementation	EPICS SNL `rf_tuner_loop.st,v` (555 lines)	States: init, unknown, reset, off, on
Motor controller	**AB 1746-HSTP1** (in service). Galil DMC-4143 Rev 1.3h is **in development, not installed** — test moves were made Aug 2025 [R29]; it goes online when the LLRF upgrade completes	—
SPEAR3 status	**Active**	

6.7 DAC Loop — Outermost Amplitude Regulation

Purpose: Holds the gap voltage (or the klystron drive power) at its software setpoint by adjusting the station reference DACs. It is the outermost amplitude loop and corrects slow drifts.

Signal flow: Gap voltage readback (cavity probe amplitude) or drive power readback → error calculation → proportional controller with deadband → DAC count adjustment → RFP reference DACs.

Control law (from `rf_dac_loop.st,v` and `subIQcounts` in `subIQ.c`):

$$\Delta C_{\text{DAC}}[m] = K_{\text{DAC}} \cdot (V_{\text{set}} - V_{\text{meas}}) \cdot D_{\text{conv}} \cdot (1 + G_{\text{loop}}) \tag{6.8}$$

where ΔC_{DAC} is the DAC count change, K_{DAC} is proportional gain (0–1), D_{conv} is the calibration conversion factor (counts/kV, Eq. (9.1)), and G_{loop} is the linear direct-loop gain (`COUNTS.L`). The output is subject to deadband, rate limiting ($|\Delta C| \leq \Delta C_{\max}$) and range limiting ($\pm 2047$ counts, 12-bit DAC). The loop runs on the same trigger as the HVPS loop, at least every 10 s.

Operating modes (station-state dependent):

Station State	DAC Loop Mode	Controlled Variable
`STATION_OFF` / `STATION_PARK`	Inactive	—
`STATION_TUNE`	Drive power regulation	Tune-mode DAC
`STATION_ON_CW`, direct loop OFF	Drive power regulation	RFP DAC
`STATION_ON_CW`, direct loop ON	Gap voltage regulation	RFP DAC

Parameter	Value	Notes
Bandwidth	~ 0.1 Hz	Limited by cycle and smoothing
Cycle	≈ 0.5 s, event-driven; at most 10 s	`DAC_LOOP_MAX_INTERVAL = 10.0`
DAC range	± 2047 counts (12-bit)	Minimum delta: 0.5 counts
Implementation	EPICS SNL: `rf_dac_loop.st,v` (290 lines, 4 states)	init, off, tune, on

Parameter	Value	Notes
SPEAR3 status	**Active**	

6.8 PEP-II Loops Not Installed at SPEAR3

The PEP-II RFP serves three more loops [R13]. Their VXI modules — two comb modules and the gap-voltage feed-forward (GVF) module — are absent from the SPEAR3 crate [R36], and SPEAR3 does not need them.

- **Comb loop.** Second-order IIR filters, one each for I and Q, with peaks at the synchrotron sidebands of the low-order revolution harmonics and a zero at the harmonics themselves, followed by a 32-tap FIR group-delay equalizer (phase linear to $< 10^\circ$ over 4 MHz) and a one-turn delay with a 25 ns vernier [R13]. Because it has gain only near each sideband it is not limited by the loop delay, and at PEP-II it added ≈ 20 dB of impedance reduction to the ≈ 16 dB of the direct loop. At SPEAR3 the fundamental-mode coupled-bunch growth is $\lesssim 5\ \mathrm{s}^{-1}$ (§4.4).
- **Longitudinal "woofer".** The bunch-by-bunch longitudinal feedback sent a band-limited (4 MHz) kick signal over a 10-bit, 10 MHz fiber link; equalized and delayed like the comb, it modulated the station reference and damped the low-order modes ($|n| < 10$) with 30 dB of loop gain [R13], [R34]. SPEAR3 has no longitudinal feedback system.
- **Gap feed-forward.** A DSP in the GVF module generated a station reference that varied within one turn to follow the transient of the PEP-II ion-clearing gap, adapting over 100 ms [R13]. SPEAR3 has no ion-clearing gap; its reference comes from a pair of RFP DACs set by the DAC loop [R36].

7 Transient Operation: Matching Under Changing Beam Current and RF Frequency

Everything in §2.2.3 is a *steady-state* result: it describes the station at one beam current, with the cavity already sitting at the detuning that matches it. Operationally the station is almost never in that state. Three things move the operating point, on three very different timescales:

- **Beam current** changes over a fill, a top-off shot or a beam loss (D1, D2). It fixes the fundamental beam current $I_{b1} = 2I_b$ and hence the detuning the cavity needs.
- **The RF frequency** follows the ring circumference (D9, §4.8): up to about 100 Hz within a day as the ground moves with the tides, and about 4 kHz over a year as the tunnel temperature changes. The cavity has to follow f_{RF}, not the other way round.
- **The cavity resonance** itself drifts with copper temperature and moves with microphonics (D7, D8).

All three arrive at the coupler as the same single quantity — a detuning error — and all three are corrected by the same single actuator, the mechanical tuner. What distinguishes them is only how fast they move and how far.

7.1 Reflected Power as a Function of Detuning and Beam Current

With the direct loop holding the gap voltage at V_{gap}, the load presented to the input coupler is the cavity in parallel with the beam. Normalizing the total gap admittance to the external (coupler) conductance βG_c, and introducing the two dimensionless beam-loading parameters together with the detuning variable

$$a \equiv \frac{I_{b1} R_s \cos\phi_s}{V_{\mathrm{gap}}} = \frac{P_{\mathrm{beam}}}{P_{\mathrm{wall}}}, \qquad b \equiv \frac{I_{b1} R_s \sin\phi_s}{V_{\mathrm{gap}}}, \qquad t \equiv \tan\psi_0 = \frac{2Q_0 \Delta f}{f_0}, \tag{7.1}$$

the normalized load admittance is $y_L = [(1+a) + j(t+b)]\,/\beta$, and the fraction of forward power reflected at the coupler is

$$|\Gamma|^2 = \left|\frac{1-y_L}{1+y_L}\right|^2 = \frac{(\beta-1-a)^2 + (t+b)^2}{(\beta+1+a)^2 + (t+b)^2} \tag{7.2}$$

At 500 mA, $a = 2.00$ and $b = 4.84$. Two consequences follow immediately.

The tuner owns the imaginary part. Equation (7.2) is minimized over tuning when $t = -b$, which is precisely the optimum detuning of Eq. (2.8). Driving $t + b$ to zero is the entire job of the tuner loop, and it is the *only* loop that can do it.

The coupler sets the floor. At $t = -b$ the residual reflection is

$$|\Gamma|^2_{\mathrm{min}} = \left(\frac{\beta-1-a}{\beta+1+a}\right)^2 \tag{7.3}$$

which vanishes only when $\beta = 1 + a$. Perfect matching at 500 mA would need $\beta_{\mathrm{opt}} = 3.00$; the installed PEP-II coupler gives $\beta = 3.78$. The station is therefore **over-coupled at its design current** and carries a residual $|\Gamma|^2_{\mathrm{min}} = 1.3\,\%$.

Forward power with the coupler floor. With the field held at V_{gap}, the cavity and beam absorb $P_{\mathrm{wall}} + P_{\mathrm{beam}}/n_{\mathrm{cav}} = 204.3\,\mathrm{kW}$ per cavity (2.14), so the klystron must deliver $P_{\mathrm{fwd}} = 204.3/(1 - |\Gamma|^2_{\mathrm{min}}) = 204.3/0.987 = 207\,\mathrm{kW}$ per cavity. On 7 October 2026 the four cavities took 193–214 kW at 499 mA, 207 kW on average, with 0.9–3.1% reflected; the coupling factors inferred from those reflections, $\beta = 3.6$–4.3 with a median of 3.74, agree with the PEP-II value (§8.8).

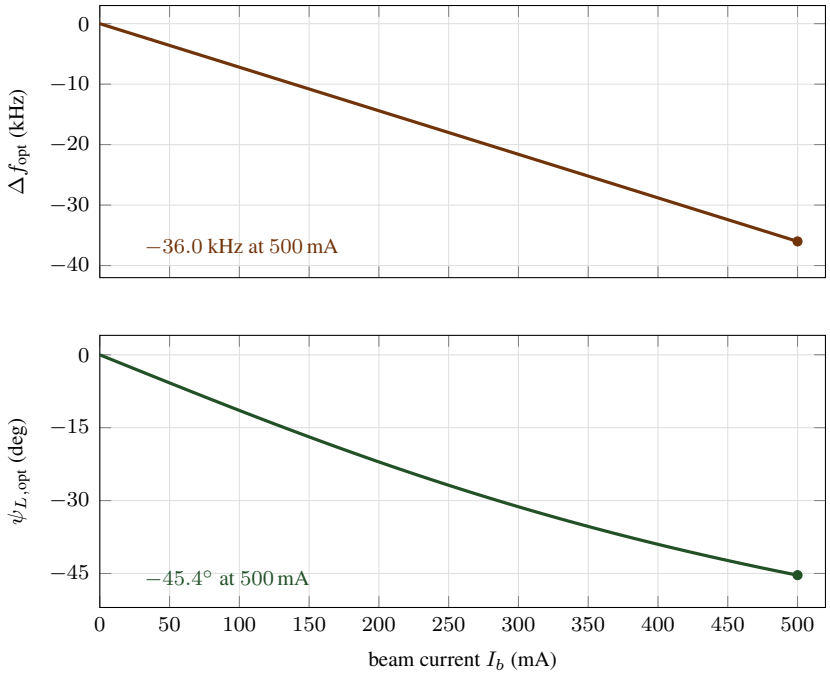


Figure 7: Cavity detuning required to cancel the reactive beam loading, as a function of beam current. The frequency offset (top) is linear in current; the loaded detuning angle measured by the tuner loop (bottom) is its arctangent and so compresses at high current.

7.2 Required Detuning Versus Beam Current

Because ϕ_s depends only on U_0 and V_{RF}, and not on current, the detuning the cavity needs is **linear in beam current**: $\Delta f_{\text{opt}} = -72.0\,\text{Hz/mA} \times I_b$. Figure 7 shows both the frequency offset and the loaded angle the tuner servos on. Filling from zero to 500 mA therefore requires the tuner to walk the cavity through the full $36.0\,\text{kHz}$, about one cavity half-bandwidth; the loaded angle swings from $0°$ to $-45.4°$.

The practical consequence is that the tuner setpoint is never static. It tracks the stored current continuously, and superimposed on that are the slow drifts of D7–D9.

7.3 What Happens When the Tuner Lags

Figure 8 evaluates Eq. (7.2) against cavity tuning for the two extreme beam conditions. The two curves have minima in different places, and that displacement is the whole story of transient matching.

Event	Detuning error	$\lvert\Gamma\rvert^2$	P_{fwd}/cavity	Consequence
Steady 500 mA, tuner converged	0	1.3%	207 kW	The normal operating point
Top-off shot, ±5 mA	0.4 kHz	1.3%	208 kW	Below the tuner deadband; invisible
Circumference drift, one year	4 kHz	1.9%	208 kW	Absorbed as a slow migration of the tuner setpoint (D9)
Fill 0 → 500 mA, tuner still on resonance	36.0 kHz	35%	312 kW	$4 \times 312 = 1.25\,\text{MW}$, or 83% of the klystron rating

Event	Detuning error	$\lvert\Gamma\rvert^2$	P_{fwd}/cavity	Consequence
Beam loss 500 → 0 mA, cavity still detuned	36.0 kHz	67%	208 kW	Only wall loss is delivered, but the circulator load takes 140 kW/cavity
No beam, cavity on resonance	0	34%	103 kW	The over-coupled floor; cannot be improved without beam

The two rows that matter operationally are the fill and the beam loss. Both leave the cavity a full half-bandwidth from its matching point, and in both cases a third or more of the forward power goes into the circulator load until the tuner catches up. This is why a fill is done at reduced gap voltage or slowly enough for the tuner to track, and why the circulator load, not the klystron, is the component sized by the transient.

7.4 Division of Labor: the Direct Loop Holds the Field, the Tuner Restores the Match

Two loops respond to these transients, and it is worth stating plainly that neither can do the other's job.

The direct loop regulates the field, and it is fast. With a closed-loop half-bandwidth of about 70 kHz at the present legacy gain (up to ≈ 0.2 MHz at the delay limit, Eq. (2.23)) it corrects a beam-current step within a few microseconds — faster than the $4.5\,\mu$s cavity fill time (2.3), and far faster than any mechanical motion. But it holds V_{gap} by *demanding more generator current*. It does not change y_L: the cavity is exactly as mismatched as before, and the extra forward power is delivered, reflected, and dumped in the circulator load. **A field-regulation loop cannot reduce reflected power.**

The tuner restores the match, and it is slow. Only the tuner changes t, and it does so through a stepper motor and a mechanical plunger, with a 2 s cycle and a bandwidth of 0.01–0.05 Hz (§2.5 and §6.6) — six to seven decades slower than the direct loop.

The resulting sequence for any current transient is therefore always the same:

1. The beam current changes, which changes a and b essentially instantaneously.
2. The direct loop holds V_{gap} through the change. Field amplitude and phase never leave specification.
3. The reflected power jumps to the value given by Eq. (7.2) at the *old* tuning, and the klystron carries the extra forward power.
4. The tuner walks t back to $-b$ over many 2 s cycles, and the reflected power decays to the coupler floor.

The separation of timescales is what makes this safe. Because the direct loop is more than 10^6 times faster than the tuner, the two never interact dynamically — the tuner sees a cavity whose field is already regulated, and the direct loop sees a detuning that is constant on its own timescale. That is the same bandwidth-separation argument as Eq. (5.5), applied to the two loops that act on the same cavity. The cost of the separation is that mismatch is corrected slowly, so transient reflected power must be budgeted as a hardware rating rather than regulated away.

Not covered here: this analysis is quasi-static in the tuner variable — it evaluates Eq. (7.2) at each tuning, but does not model the tuner's own closed-loop trajectory, its deadband limit cycle, or the coupled dynamics of four independently tuned cavities driven from one klystron. A measured tuner step response and a fill record with synchronized current, forward and reflected power would let those be added; neither is currently in the repository.

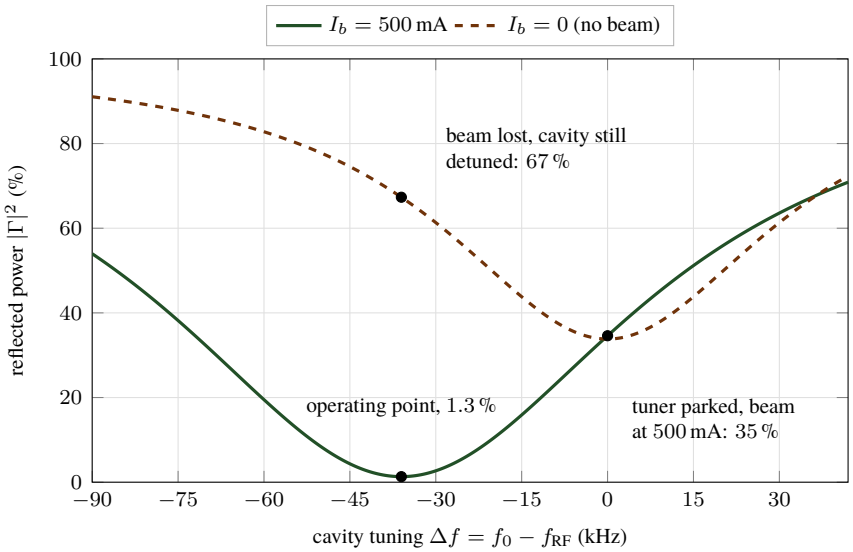


Figure 8: Reflected power at the input coupler against cavity tuning, with and without beam. The minima sit at different detunings, so a cavity tuned for one beam condition is badly mismatched in the other. With no beam the over-coupled cavity cannot be matched at all: the best it can do on resonance is 34 % reflected.

8 Synchrotron Damping and Direct-Loop Optimization

The dipole synchrotron mode of the stored beam — all bunches oscillating in phase, coupled-bunch mode $n = 0$ — is coupled to the fundamental mode of the four cavities and, through them, to the direct loop. Its coherent frequency and damping rate are therefore the most direct beam-based measure of how the direct loop is set. A correctly phased loop of moderate gain leaves the mode strongly damped; a mis-phased or over-strong loop, or an integral path with the wrong phase, removes the damping that the detuned cavity would otherwise provide. The mode is also a driven oscillator: the noise of the RF system moves the beam by an amount that grows as the damping falls, and the same loop decides how much of that noise reaches it. This section states that picture (§8.1), gives the complete small-signal model (§8.2), its predictions as functions of beam current, loop gain, loop phase, delay and integral path (§§8.3–8.6) and the beam's response to noise (§8.7), compares them with what is known about the legacy station today (§8.8), and turns the result into a procedure for setting the loop (§8.9). Every number is computed from the model of §8.2 with the parameters of §1.2 and §2.2.1, unless it is marked as measured.

8.1 Beam Dipole Motion as a Driven, Damped Oscillator

Linearized about the synchronous phase, the dipole motion of the beam — the phase $\Delta\phi$ of the bunch centroids, positive when late — obeys the equation of a driven, damped oscillator (Appendix C):

$$\frac{d^2\Delta\phi}{dt^2} + 2\sigma\,\frac{d\Delta\phi}{dt} + \left(\omega_{\mathrm{coh}}^2 + \sigma^2\right)\Delta\phi \simeq \omega_s^2\, n_{\mathrm{eff}}(t), \qquad n_{\mathrm{eff}} = \varepsilon_A \cot\phi_s - \varepsilon_\phi. \tag{8.1}$$

The left side is the beam together with the cavities and the loop. Its damping rate σ and coherent frequency ω_{coh} are the root of the characteristic equation of §8.2, which contains the cavity impedance, the loop and the beam current; exactly, $D_0(s)\,\Delta\phi = \omega_s^2\, n_{\mathrm{eff}}$ with D_0 of Eq. (8.4), and Eq. (8.1) keeps its dominant root. The right side is the drive: the relative amplitude error ε_A and the phase error ε_ϕ (positive = advance, Eq. (3.1)) of the total gap voltage that the RF system produces with the bunches held at the synchronous phase. The

voltage that the oscillation induces itself is already in D_0, through Z_b, and must not be counted again as drive; a gap error measured with beam contains both. At $\phi_s = 67.5°$, $\cot\phi_s = 0.41$: a 0.1% amplitude error moves the beam as much as a $0.024°$ phase error, and once the errors are at the gap the beam cannot tell them apart.

A bunch executing phase oscillations at ω_s modulates the beam current at $\omega_{\text{RF}} \pm \omega_s$, and the cavity answers at both sidebands. The resistive part of that answer exchanges energy with the oscillation: above transition the motion is damped when the cavity presents less resistance at the upper sideband than at the lower one (4.1), which is the case for a cavity tuned below f_{RF}. The reactive part shifts the coherent frequency, and at SPEAR3 lowers it. Because beam-loading compensation already tunes each cavity about one half-bandwidth below f_{RF} (2.10), the SPEAR3 beam is strongly Robinson-damped without any feedback. The second Robinson limit — the restoring force cancelled at zero frequency by the reactive beam loading — would be reached only at 1.19 A with the cavities at the optimum detuning.

The direct loop acts on both sides of Eq. (8.1). Through the impedance it presents to the beam, $Z_{\text{eff}} = Z_{\text{cav}}/(1 + G_{\text{OL}})$ (5.4), it sets σ and ω_{coh}; through the way it passes the noise of the RF system to the gap (5.3), it sets n_{eff}. Four consequences set every rule in §8.9:

1. The loop **lowers and widens** the resonance, so the impedance difference between the two sidebands — and with it the damping — falls steeply with gain, approaching pure radiation damping at high gain.
2. A loop **phase error**, and the loop **delay**, rotate Z_{eff} and convert reactive impedance into resistive impedance of one sign or the other; whether that adds or removes damping depends on the sign of the error.
3. An **integral path** adds large gain close to the carrier with a phase fixed by the loop. Around a detuned cavity that phase is not the one that damps the beam, so a strong integral path can make the Robinson mode unstable while it improves field regulation.
4. The beam's response to a drive near ω_{coh} grows as $1/\sigma$, so the loop's **noise transfer** and the damping it leaves cannot be chosen separately. With the cavities detuned the loop treats amplitude and phase noise differently, and more gain can leave the beam moving more although the field is quieter (§8.7).

8.2 Small-Signal Model

Cavities and loop. Each cavity k is the baseband resonator of Eq. (2.1), $Z_k(s) = R_L\omega_{1/2}/(s+\omega_{1/2}-j\Delta\omega_k)$, with s the Laplace variable of the RF envelope. One klystron drives all four cavities, and the loop measures their vector sum with weights w_k ($\sum_k w_k = 1$). Its open-loop transfer function is

$$G_{\text{OL}}(s) = C(s)\,\frac{\omega_a}{s+\omega_a}\,e^{-s\tau_d}\,\frac{1}{R_L}\sum_k w_k Z_k(s), \qquad C(s) = Ge^{j\theta}\left[1 + e^{j\Delta\theta_i}\,\frac{\omega_i}{s+\omega_\ell}\right]G_{\text{lead}}(s), \quad (8.2)$$

where G is the loop gain at the resonance of an on-tune cavity, θ the loop phase error there (positive = lead), ω_a the bandwidth of drive amplifier and klystron (2.5 MHz assumed), τ_d the total delay and G_{lead} the lead network. The integral path equals the proportional path in magnitude at $f_i = \omega_i/2\pi$; $\Delta\theta_i$ is its phase *relative* to the proportional path — zero in the legacy RFP, where one modulator rotates both paths (§6.2), but an independent setting in the LLRF9 — and ω_ℓ allows for a leaky integrator. Equation (8.2) is Eq. (6.1) generalized to four cavities that need not share one detuning.

Impedance seen by the beam. With the field loop closed the beam drives the cavities through

$$Z_b(s) = \frac{\sum_k Z_k(s)}{1 + G_{\text{OL}}(s)}, \quad (8.3)$$

which is exact for one generator current shared by all cavities: the loop cannot tell beam-induced from generator-induced voltage, so it divides the total by the return difference.

Beam dynamics. For rigid bunches in a uniform fill, coupled-bunch mode n evolves as e^{st}, where s is a root of

$$D_n(s) \equiv s^2 + 2\alpha_{\mathrm{rad}}\, s + \omega_s^2 - j\kappa \left[Z_b(s + jn\omega_{\mathrm{rev}}) - \overline{Z_b(\bar{s} - jn\omega_{\mathrm{rev}})} \right] = 0 \tag{8.4}$$

with

$$\kappa = \frac{\alpha_c\, \omega_{\mathrm{RF}}\, I_b}{(E_0/e)\, T_{\mathrm{rev}}}, \qquad \omega_s^2 = \frac{\alpha_c\, \omega_{\mathrm{RF}}\, V_{\mathrm{RF}} \sin\phi_s}{(E_0/e)\, T_{\mathrm{rev}}}, \qquad \alpha_{\mathrm{rad}} = \frac{J_s U_0}{2E_0 T_{\mathrm{rev}}}. \tag{8.5}$$

Equation (8.4) follows from the linearized equation of synchrotron motion [R8], with the beam-induced voltage fed back through Z_b; Appendix C gives the derivation.

At 500 mA, $\kappa = 754\ \Omega^{-1}\mathrm{s}^{-2}$, $\omega_s/2\pi = 10.03\,\mathrm{kHz}$ and $\alpha_{\mathrm{rad}} = 424\ \mathrm{s}^{-1}$. The root connected continuously to the incoherent oscillator, $s = -\sigma + j\omega_{\mathrm{coh}}$, gives the coherent frequency and the amplitude damping rate σ ($\sigma > 0$: damped). The line seen on a probe spectrum has $Q = \omega_{\mathrm{coh}}/2\sigma$, so a Lorentzian fit of f and Q gives $\sigma = \pi f/Q$.

Three limits make Eq. (8.4) familiar:

- Without feedback it is exactly Pedersen's quartic for a beam-loaded cavity [R35], and at low current it reproduces the Robinson rate of [R10].
- When the coherent shift is small it reduces to the first-order sideband formula, the baseband form of

Table 1: Loop, cavity and beam parameters used in the calculations of this section.

Parameter	Value used	Basis
Loop gain G	0 dB (linear 1) for the legacy station; scanned in Figure 10 and Table 2	Measured (§8.8); 15 dB is the PEP-II design [R13]
Loop phase error θ	0 unless stated, including any frequency-offset tracking; $\pm 15^\circ$, $\pm 30^\circ$ in Figure 10; -90° to $+90^\circ$ in Figure 11	Loop assumed correctly phased
Total delay τ_d	625 ns; 270–870 ns in Table 3	Measured (§2.6)
Drive amplifier and klystron ω_a	One pole at $2\pi \times 2.5$ MHz	Assumed: the 5 MHz klystron bandwidth [R1] as a baseband pole
Lead network ω_z, ω_p	Not modeled ($G_{\mathrm{lead}} = 1$), except in Figure 6: $2\pi \times 944$ kHz, $2\pi \times 2.39$ MHz	RFgui loop models [R41]; adds 2% to the damping at 500 mA
Integral path $\omega_i = 2\pi f_i$, $\Delta\theta_i$, ω_ℓ	Off, except in Table 4: $f_i = 0.3$–10 kHz, $\Delta\theta_i = 0$ (legacy), $+45^\circ$, $+90^\circ$; in Table 8: $f_i = 1$ kHz, $\Delta\theta_i = 0$, both with $\omega_\ell = 0$ (ideal integrator); in Figure 6: the RFP lag network, $f_i = 2.93$ kHz, $\omega_\ell = 2\pi \times 0.26$ kHz, $\Delta\theta_i = 0$	Varied; the measured loop responses indicate the lag network, the measured damping an effective 1 kHz (§8.6 and §8.8)
Vector-sum weights w_k	$1/4$ each	Equal weights
Detuning $\Delta\omega_k$	Optimum for the beam current (2.10), the same in every cavity	Tuners converged
Cavity, beam and ring	$R_L = 0.78\,\mathrm{M}\Omega$, $Q_L = 6700$, $\beta = 3.78$; $I_b = 500$ mA (0–800 mA in Figure 9), $V_{\mathrm{RF}} = 2.85$ MV, $U_0 = 1.09$ MeV, $\alpha_c = 1.18 \times 10^{-3}$	§1.2 and §2.2.1; the closed-loop fits at 20–22 mA give $Q_L = 6800 \pm 250$ (Table 7)

Eq. (4.2):

$$\sigma \simeq \alpha_{\rm rad} - \frac{\kappa}{2\omega_s}\big[\mathrm{Re}\, Z_b(j\omega_s) - \mathrm{Re}\, Z_b(-j\omega_s)\big], \qquad \omega_{\rm coh} \simeq \omega_s + \frac{\kappa}{2\omega_s}\big[\mathrm{Im}\, Z_b(j\omega_s) + \mathrm{Im}\, Z_b(-j\omega_s)\big] \tag{8.6}$$

- At $s = 0$ it gives the second (static) Robinson condition

$$D_0(0) = \omega_s^2 + 2\kappa\, \mathrm{Im}\, Z_b(0) > 0, \tag{8.7}$$

which any integral path satisfies automatically, since it makes $Z_b(0) = 0$.

A time-domain simulation of the same system — rigid bunch, four cavities, delayed loop — agrees with the roots of Eq. (8.4) to 0.2% in frequency and 1% in damping. Equation (8.6) is adequate once the loop gain is 0 dB or more; without feedback at 500 mA it gets the damping right to within 3% but puts the coherent line at 7.2 kHz instead of 6.4 kHz, so the full equation is used throughout.

Why the damping falls with gain. For an ideal proportional loop (no delay, $\theta = 0$, identical cavities) the closed loop simply replaces $R_L \to R_L/(1+G)$ and $\omega_{1/2} \to (1+G)\,\omega_{1/2}$ at unchanged $\Delta\omega$. Equation (8.6) then gives, for $\omega_s \ll (1+G)\,\omega_{1/2}$,

$$\sigma - \alpha_{\rm rad} \simeq -\frac{2\,\kappa\, n_{\rm cav}\, R_L}{(1+G)^2\,\omega_{1/2}}\, \sin\psi' \cos^3\psi', \qquad \tan\psi' = \frac{\tan\psi_L}{1+G}, \tag{8.8}$$

which is positive for $\psi_L < 0$ and falls as $(1+G)^{-3}$ once $G \gg |\tan\psi_L|$. At 500 mA, with $\alpha_{\rm rad}$ added back, it gives $\sigma = 5.6 \times 10^3$ s^{-1} without feedback and $\sigma = 2.1 \times 10^3$ s^{-1} at $G = 1$; the delay of the real loop removes a further 10–15%.

Parameters used. The lead and integral networks of the legacy RFP are switched on. The measured loop responses agree with the component values of the RFgui loop models (§6.2 and §8.8), but with that integral network the model gives far less Robinson damping than is measured (§8.6). The calculations therefore leave out the lead network, which changes the damping by about 2%, and include an integral path only where it is varied deliberately; Table 1 lists every value used.

8.3 Dependence on Beam Current

Figure 9 follows the Robinson mode from zero to 800 mA with every cavity held at the optimum detuning for the current — the tuner loop's job — without feedback, at the present legacy gain (0 dB) and at the 15 dB PEP-II design gain. Table 2 lists the values at 500 mA.

Table 2: Robinson mode at 500 mA (proportional loop, $\theta = 0$, $\tau_d = 625$ ns). Radiation damping 424 s^{-1}; incoherent $f_s = 10.03$ kHz.

Loop setting	Coherent f (kHz)	σ (s^{-1})	Q	Static ratio $D_0(0)/\omega_s^2$	$n = -1$ growth from the cavities (s^{-1})
No feedback	6.41	5760	3.5	0.41	1.7
$G = 0$ dB (legacy now)	8.77	1850	15	0.76	2.5
$G = 6$ dB	9.41	866	34	0.88	3.3
$G = 10$ dB	9.69	560	54	0.94	4.0
$G = 15$ dB (PEP-II design)	9.89	434	72	0.97	5.3

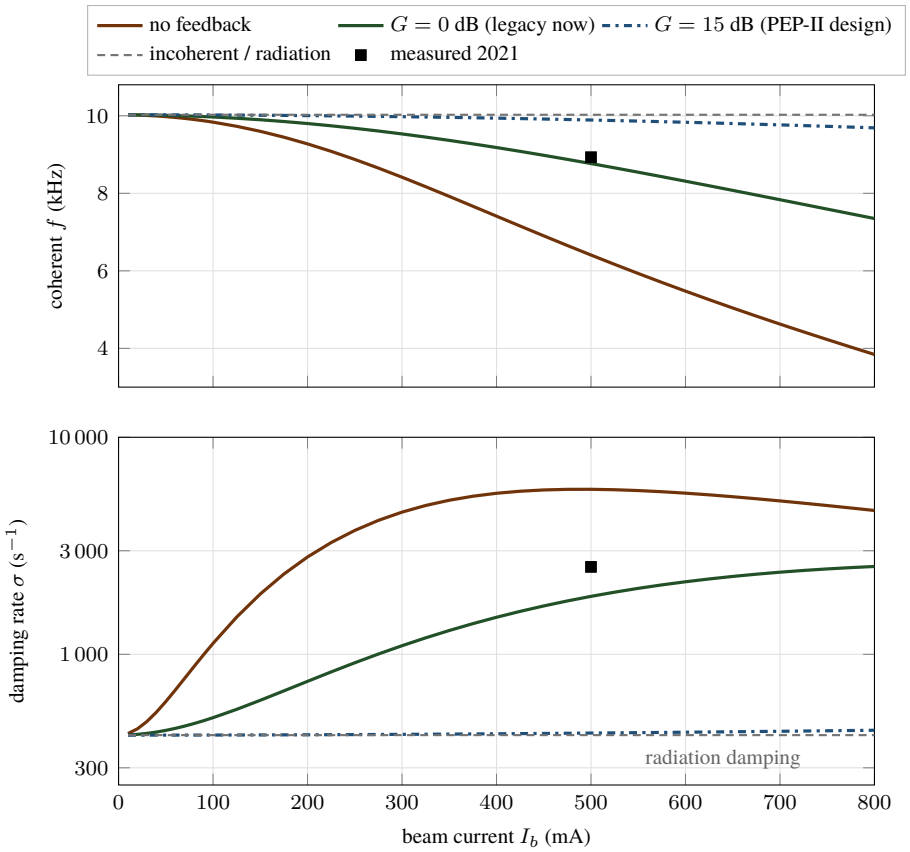


Figure 9: Coherent dipole (Robinson) mode versus beam current with the cavities at the optimum detuning: coherent frequency (top) and amplitude damping rate (bottom, logarithmic), without feedback and with a proportional direct loop of 0 dB (present legacy gain) and 15 dB (PEP-II design), both with the measured 625 ns delay and a 2.5 MHz amplifier pole. Dashed: incoherent synchrotron frequency (top) and radiation damping (bottom). Square: the 2021 measurement on the legacy system at 500 mA [R4].

Without feedback the damping at 500 mA is 14 times radiation damping, but the cavities pull the line down to 6.4 kHz, and the static ratio of 0.41 is far from zero: with the detuning following the current, the second Robinson limit is 1.19 A, a factor of 2.4 away. Each step in gain moves the coherent line back towards the incoherent 10.0 kHz and gives up damping; at 15 dB essentially only radiation damping is left ($Q \approx 70$). The coupled-bunch modes $n = \pm 1$ are negligible at every setting.

8.4 Dependence on Loop Gain and Phase

Figure 10 shows the damping at 500 mA as the gain is raised, for loop-phase errors up to $\pm 30^\circ$; each curve stops where the field loop itself becomes unstable. Figure 11 shows the damping at the present gain of 0 dB as a function of the phase error, from 100 to 500 mA. Five rules follow.

1. **Gain buys impedance reduction and costs damping.** At the correct phase the damping falls from 3.2×10^3 s^{-1} at -6 dB to 1.9×10^3 at 0 dB, 866 at 6 dB and 434 s^{-1} — radiation damping — at 15 dB.
2. **Above about -4 dB, a lead error adds damping and a lag error removes it, and the asymmetry grows with gain.** At 500 mA and 0 dB a 30° error changes the damping by -20% (lag) or $+9\%$ (lead); at 6 dB a 30° lag leaves 220 s^{-1}, less than radiation damping, so the cavities then *anti*-damp the beam.

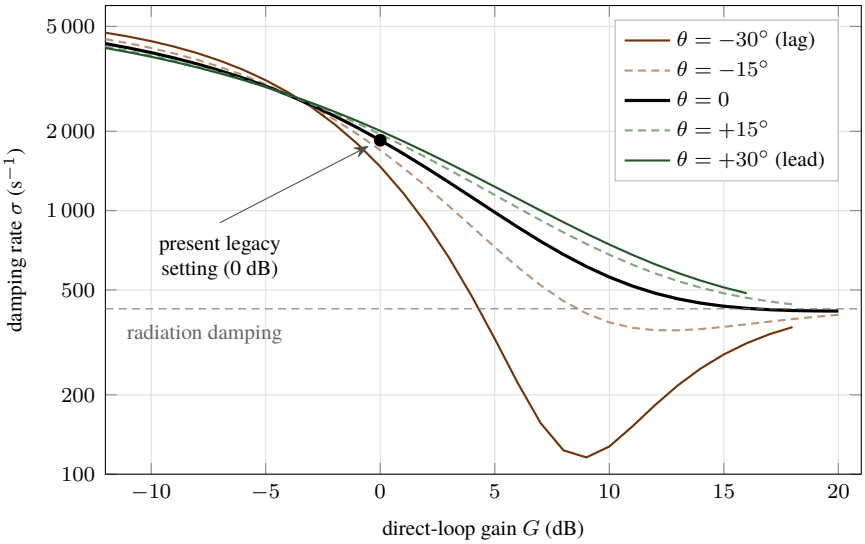


Figure 10: Robinson damping rate at 500 mA versus direct-loop gain for loop-phase errors θ from -30° (lag) to $+30^\circ$ (lead), with the measured 625 ns delay and a 2.5 MHz amplifier pole. Each curve ends where the field loop itself becomes unstable. Dashed: radiation damping. Dot: the present legacy setting (0 dB).

3. **A lag error is most dangerous at low current**, where the detuning, and with it the Robinson damping, is small. At 0 dB the cavities stop adding damping ($\sigma = \alpha_{\text{rad}}$) at $\theta \approx -9^\circ$ for 100 mA, -21° for 200 mA, -31° for 300 mA, -45° for 400 mA and -65° for 500 mA, and the beam becomes Robinson-unstable ($\sigma < 0$) beyond -40° to -55° below 500 mA (-75° at 500 mA). A lead of up to $+90^\circ$ increases the damping at every current.
4. **The delay acts as a frequency-dependent phase error** $-\omega\tau_d$: about -2° at the synchrotron sidebands, but $+8^\circ$ at the operating detuning of -36 kHz, so a loop phased on an on-tune cavity runs slightly ahead at the operating point. The legacy frequency-offset tracking is designed to remove such shifts (§8.8); at the present gain either sign changes the damping by less than 5%.
5. **The field loop goes unstable first, at 17–20 dB** with this delay (§8.5), well above any gain worth running at SPEAR3.

8.5 Delay and the Maximum Usable Gain

Table 3 finds, with the full model at 500 mA, the gain that leaves 45° of phase margin for several total delays, and compares it with Gamp's rule (2.23).

Table 3: Maximum proportional gain for a 45° phase margin at 500 mA ($\theta = 0$, 2.5 MHz amplifier pole).

Total delay τ_d	$G_{\max}$, full model	$\pi/(4\omega_{1/2}\tau_d)$	Crossover f_c	Gain margin	σ at $G_{\max}$ (s^{-1})
270 ns (LLRF9 internal only)	20.8 dB	22.3 dB	419 kHz	6.0 dB	427
500 ns	16.3 dB	16.9 dB	265 kHz	5.5 dB	433
625 ns (legacy, measured)	14.7 dB	15.0 dB	225 kHz	5.3 dB	437

Total delay τ_d	$G_{\max}$, full model	$\pi/(4\omega_{1/2}\tau_d)$	Crossover f_c	Gain margin	σ at $G_{\max}$ (s^{-1})
700 ns (legacy IOC parameter)	13.8 dB	14.0 dB	207 kHz	5.3 dB	440
870 ns (LLRF9, estimate)	12.2 dB	12.1 dB	176 kHz	5.1 dB	447

Gamp's rule is within 0.6 dB for delays of 500 ns and more. At 625 ns the legacy station could run up to ≈ 15 dB — the PEP-II design value [R13] — but would then keep only radiation damping. The LLRF9 adds its 270 ns of internal processing to the same ≈ 0.6 μs external path (§2.6); at ≈ 870 ns in total its delay-limited gain is ≈ 12 dB, so **the upgrade does not by itself allow a higher direct-loop gain**. At SPEAR3 that is no loss: no instability needs the impedance reduction that high gain would buy (§4.4), and the damping it would cost is real.

8.6 The Integral Path

An integral path makes the field error vanish in steady state and rejects low-frequency disturbances, the HVPS ripple lines (D5) in particular. Its effect on the beam depends on its phase, and the detuned cavity makes the obvious choice the wrong one. Table 4 adds an integral path to the present 0 dB proportional loop at 500 mA.

Table 4: Integral path added to the 0 dB proportional loop at 500 mA (625 ns). σ in s^{-1}; negative = unstable. Rejection $20\log_{10}|1+G_{\mathrm{OL}}|$ for $\Delta\theta_i = 0$ at the lower/upper line.

f_i	σ, $\Delta\theta_i = 0$ (legacy)	σ, $+45°$	σ, $+90°$	Rejection at ∓ 360 Hz (dB)	Rejection at ∓ 720 Hz (dB)
none	1850	1850	1850	4.0 / 3.9	4.0 / 3.9
0.3 kHz	1621	1813	2026	5.7 / 2.9	4.8 / 3.3
1 kHz	1074	1724	2452	9.6 / 5.4	6.9 / 3.0
2 kHz	278	1584	3099	13.7 / 10.8	9.6 / 5.3
3 kHz	-503	1419	3803	16.6 / 14.5	11.9 / 8.2
10 kHz	-4064	-808	14189	26.2 / 25.5	20.6 / 19.1

Three conclusions follow.

- **Shared phase (legacy architecture).** An integral path rotated together with the proportional path ($\Delta\theta_i = 0$) removes damping fast: 1.1×10^3 s^{-1} at $f_i = 1$ kHz, 280 s^{-1} at 2 kHz, and the beam is Robinson-unstable at 3 kHz. A scan of the common loop phase from $-75°$ to $+90°$ finds no setting that rescues a 3 kHz integral path at this current. In this model, a beam that is stable with `INTCOMP` on needs an integral network that is weak at the synchrotron frequency, equivalent to $f_i \lesssim 2$ kHz, and the measured damping corresponds to about 1 kHz (§8.8). The measured loop responses, however, indicate the lag network of the RFgui loop models [R41], equivalent to $f_i \approx 2.9$ kHz with a 0.26 kHz leak (§8.8). With it the model makes the beam Robinson-unstable ($\sigma \approx -390$ s^{-1} at $\theta = 0$, and no damping even at a $40°$ lead), whereas the measured damping is about 10^3 s^{-1}. The model therefore misses a damping contribution at high current, and 1 kHz is an effective value (§8.10). Read as $f_i = 30$ kHz, the PEP-II "30 kHz bandwidth integrator" [R13] would be unstable here.

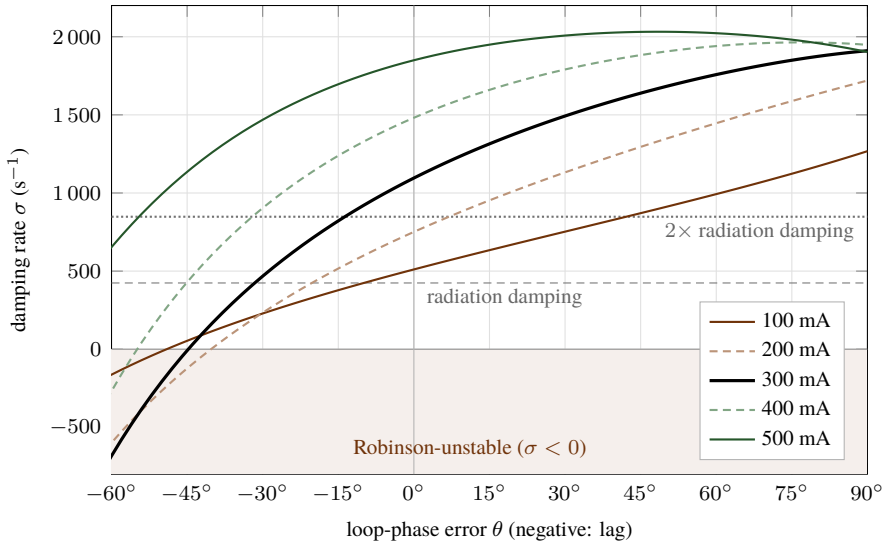


Figure 11: Robinson damping rate versus loop-phase error θ (negative: lag) at the present legacy gain of 0 dB, for beam currents from 100 to 500 mA with the cavities at the optimum detuning for each current; 625 ns delay, 2.5 MHz amplifier pole. Dashed: radiation damping. Dotted: twice radiation damping, the acceptance level of §8.9. Shaded: Robinson-unstable ($\sigma < 0$). Thin lines: $\theta = 0$ and $\sigma = 0$.

- **Rotated phase.** Rotating the integral path ahead of the proportional path by about the load angle ($\Delta\theta_i = +45°$) keeps a 3 kHz path damped, and $+90°$ turns even a 10 kHz path into strong additional damping. This freedom exists only where the integral path has its own phase control, as in the LLRF9.
- **Ripple rejection needs a strong integral path or a canceller.** The proportional loop alone rejects the 360 and 720 Hz lines by only ≈ 4 dB, and 20–26 dB needs $f_i \approx 10$ kHz. The legacy station combines its lag network with the narrowband DSP canceller (§6.3). The LLRF9 must get it from its integral path, and must therefore set that path's phase with the Robinson mode in mind.
- **The integral path also sets two parts of the beam's jitter** (§8.7). Its gain at the line harmonics lowers the timing jitter the beam inherits from them, which no damping reduces; a shared phase lowers σ and so raises the resonant motion near f_{coh}. Equation (8.13) weighs the two, and with an independent phase $\Delta\theta_i$ the LLRF9 can improve both.

8.7 Noise-Driven Motion and the Gain Trade-off

Equation (8.1) turns the damping into a noise problem. This subsection gives the beam's response to a drive, how the noise of the RF system reaches the drive through the loop, and what follows for the loop gain. Spectral densities are one-sided and per hertz: $S_x(f)$ for a signal x, so that $\langle x^2 \rangle = \int_0^\infty S_x \, df$.

Response. The beam's phase follows the drive through

$$\chi(s) = \frac{\Delta\phi(s)}{n_{\text{eff}}(s)} = \frac{\omega_s^2}{D_0(s)}, \qquad S_{\Delta\phi}(f) = |\chi(j2\pi f)|^2 \, S_n(f). \tag{8.9}$$

Near the coherent frequency $|\chi|^2$ is a Lorentzian of full width σ/π — the line of a probe spectrum (§8.2) — with a peak $|\chi|_{\max} \simeq Q\,(\omega_s/\omega_{\text{coh}})^2$. For a drive whose spectral density is flat across the line,

$$\boxed{\langle \Delta\phi^2 \rangle = \frac{\omega_s^4}{8\,\omega_{\text{coh}}^2} \, \frac{S_n(f_{\text{coh}})}{\sigma}} \tag{8.10}$$

that is, $|\chi|^2_{\max}\, S_n$ over a noise bandwidth of $\sigma/2$ (in hertz, with σ in s^{-1}). At 500 mA it agrees with the integral of the exact $|\chi|^2$ to 3% without feedback ($Q = 3.5$) and to 1% from 0 dB of gain upwards. For the line of 8 October 2026, $|\chi|_{\max} = 33$ and the noise bandwidth is 540 Hz (§8.8).

Below the line the beam follows the drive: $\chi(0) = \omega_s^2/D_0(0)$ is the inverse of the static ratio of Table 2 (2.46 without feedback, 1.31 at 0 dB) and exactly 1 with an integral path. The line harmonics of D5, all below 2.9 kHz, therefore move the beam's phase as much as they move the gap's, whatever the damping; only their rejection helps. A line of rms drive n_L at ω_L gives $\Delta\phi_L = |\chi(j\omega_L)|\, n_L \simeq \omega_s^2 n_L/\sqrt{(\omega_{\text{coh}}^2 - \omega_L^2)^2 + 4\sigma^2\omega_L^2}$, which depends on where the coherent frequency, moving with current and gain, lies relative to the line.

What the motion means. Each frequency component of $\Delta\phi$ carries an energy deviation $\Delta E/E_0 = (\omega/\alpha_c\omega_{\text{RF}})\,\Delta\phi$, from $d\Delta\phi/dt = \alpha_c\omega_{\text{RF}}\,\Delta E/E_0$ (Appendix C), so

$$S_{\Delta E/E_0}(f) = \Big(\frac{2\pi f}{\alpha_c\,\omega_{\text{RF}}}\Big)^2 S_{\Delta\phi}(f), \tag{8.11}$$

0.016 per radian at 8.8 kHz; where the dispersion η is not zero the orbit moves by $\eta\,\Delta E/E_0$. The timing jitter is $\Delta\phi/\omega_{\text{RF}}$, 0.58 ps per 0.1°. The two weight the spectrum differently: the timing contains everything the beam follows below the line, the energy deviation essentially only the resonant part, and for a flat drive $\langle(\Delta E/E_0)^2\rangle = \omega_s^4\, S_n(f_{\text{coh}})/[8\sigma(\alpha_c\omega_{\text{RF}})^2]$ does not depend on ω_{coh}. The photon-energy jitter, $2\,\Delta E/E_0$, is of order 10^{-4} and negligible; timing jitter and source motion are the effects that users see.

Where the drive comes from. The drive is the gap error that the RF system produces with the bunches held fixed. Each noise source x — amplitude or phase noise at the klystron output, noise ahead of the klystron, noise in the measured vector sum, noise of the reference — reaches it through a transfer function h_x of its own, and for independent sources

$$n_{\text{eff}}(s) = \sum_x h_x(s)\, u_x(s), \qquad S_n(f) = \sum_x |h_x(j2\pi f)|^2\, S_{u_x}(f). \tag{8.12}$$

Drives that are not RF noise — injection transients and changes of the fill — add to it, and a measurement must avoid them. For a loop symmetric about the carrier, h_x is the open-loop path times $1/(1 + G_{\text{OL}})$ for noise ahead of the cavities and T for measurement and reference noise (5.3). At SPEAR3 the loop is not symmetric. The cavities sit 36 kHz below f_{RF}, so the detuned cavity turns klystron amplitude noise into gap phase noise and back through its load angle ($\psi_L = -45^\circ$); the loop acts differently on the two sidebands of every error; and near saturation the klystron passes amplitude and phase corrections with different gains (§8.10). Appendix C gives the two-quadrature form of D_0 and h_x that includes all three; for a linear klystron its D_0 is Eq. (8.4). Table 5 compares it with the scalar estimate for noise at the klystron output.

Table 5: Beam drive at the coherent frequency at 500 mA (proportional loop, $\theta = 0$, 625 ns): spectral density of n_{eff} for noise at the klystron output with the loop closed, relative to the same noise without feedback, against the scalar estimate $1/|1 + G_{\text{OL}}|^2$ averaged over the two sidebands.

Gain	f_{coh} (kHz)	Scalar $1/\lvert1 + G_{\text{OL}}\rvert^2$	Klystron phase noise	Klystron amplitude noise
0 dB	8.77	0.41	1.03	0.27
6 dB	9.41	0.21	0.67	0.10
15 dB	9.89	0.047	0.19	0.014

Without feedback, klystron amplitude noise drives the beam 5.5 times more strongly, in power, than the same klystron phase noise: the load angle turns it into gap phase. Feedback reduces the amplitude drive by

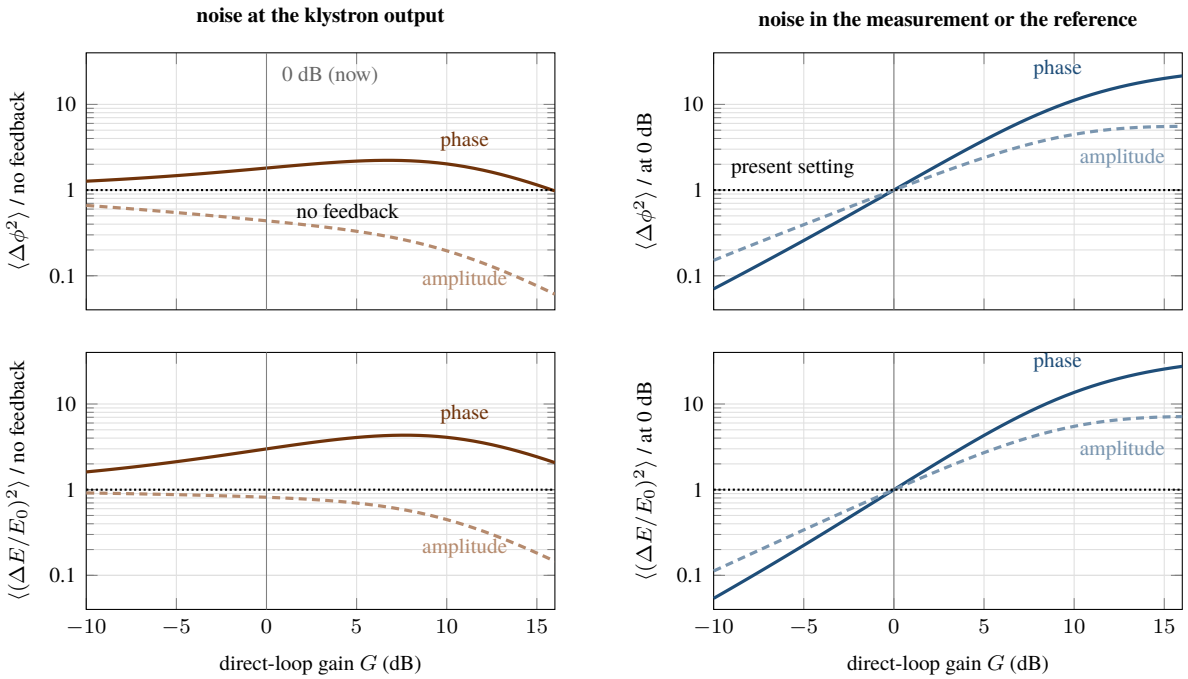


Figure 12: Beam phase (top) and energy (bottom) jitter at 500 mA caused by one white noise source at a time, versus the proportional gain ($\theta = 0$, 625 ns, 2.5 MHz amplifier pole, linear klystron). Left: phase (solid) and amplitude (dashed) noise at the klystron output, relative to the same noise without feedback. Right: phase and amplitude noise in the measured vector sum or in the reference, which reaches the beam only through the loop, relative to the present gain of 0 dB. Above the dotted line the gain makes the beam move more. Independent sources add in power.

more than the scalar estimate and the phase drive by much less — at 0 dB not at all. For measurement noise the scalar estimate is $|T|^2 = 0.20$ at 0 dB; the exact drive is 0.10 of what the same error at the gap would cause for a phase error and 0.78 for an amplitude error.

The trade-off. Figure 12 combines the drive with the response: the beam's phase and energy jitter at 500 mA for white noise from each source, as functions of the proportional gain. Three regimes follow.

1. **Measurement and reference noise always moves the beam more as the gain rises**: $T \to 1$ while σ falls. From 0 to 6 dB the timing variance it causes grows 4.9-fold for phase noise and 2.8-fold for amplitude noise, and by 15 dB 20-fold and 5.5-fold.
2. **Klystron amplitude noise is reduced at every gain**: to 0.44 of its timing variance without feedback at 0 dB and to 0.08 at 15 dB (energy: 0.82 and 0.18).
3. **Klystron phase noise moves the beam more with feedback than without**, at every gain up to the delay limit: 1.8 times the timing variance without feedback at 0 dB, 2.2 times at 6 dB and 1.15 times at 15 dB, and 3.0, 4.2 and 2.4 times the energy variance. The loop removes damping faster than it removes this drive.

With the klystron measured in normal operation ($g_A = 0.40$, $k_{\mathrm{AP}} = 0.10$, §8.10) the third regime changes sign for the timing (0.84 at 0 dB) but hardly for the energy (1.13), and the first is unchanged. The resonant estimate (8.10) is within 1% of the full integral at 0 dB and above, and a time-domain simulation with white noise injected at the klystron output reproduces the jitter without feedback, at 0 dB and at 6 dB to within 11%. As a scale, a white klystron phase noise of 10^{-11} rad^2/Hz (-113 dBc/Hz) moves the beam by 0.05° rms at 0 dB.

Setting the loop for low jitter. The quantity to minimize at each current is the jitter that matters to users, subject to the damping limit $\sigma \geq 2\alpha_{\text{rad}}$ of §8.9. For the timing,

$$\langle \Delta\phi^2 \rangle \simeq \frac{\omega_s^4}{8\,\omega_{\text{coh}}^2\,\sigma} \sum_x |h_x(j\omega_{\text{coh}})|^2\, S_{u_x}(f_{\text{coh}}) + \sum_L |\chi(j\omega_L)|^2\, n_L^2; \tag{8.13}$$

for the energy and the orbit, the first term with ω_{coh}^2 replaced by $(\alpha_c\omega_{\text{RF}})^2$, and without the second. Every factor depends on the gain, the loop phase and the integral path; the source spectra S_{u_x} are the unknowns. Because the three regimes pull the gain in opposite directions, the best setting depends on which sources dominate near f_{coh}, which has not been measured (§8.10). The integral path of the LLRF9 is the clearest case: its gain at the line harmonics lowers the second term, and its phase decides whether it also lowers σ and so raises the first (§8.6).

8.8 The Legacy Station Today

The station is still run by the legacy VXI system; the LLRF9 has so far operated only in the 2021 parallel test [R4]. Table 6 is a snapshot of the station taken on 7 October 2026, with the quantities derived from it.

Table 6: Legacy station, 7 October 2026, 16:36.

Quantity	Value	Source or derivation
Beam current	499 mA	`SRF1:STN:BEAM:CURR`
RF frequency	476.303964 MHz	`SPEAR:RFFreq`
Station voltage	2847 kV; cavities 672–732 kV	Probe readbacks
Klystron	795 kW out, 29.0 W drive; HVPS 75.5 kV, 22.4 A	99% of the IOC limit `KLYSOUTFRWD:POWER:MAX` (800 kW); drive 4.4 dB below the saturation setting
Cavity forward, reflected	193–214 kW; 0.9–3.1%	Sum of forward 829 kW, 4.2% above the klystron reading, within the calibration of the power monitors
Energy loss per turn	1.09 MeV	$\sum_k (P_{\text{fwd}} - P_{\text{refl}} - V^2/2R_s)_k / I_b$
Synchronous phase	63–71° per cavity	$\cos\phi = P_{\text{beam}}/(I_b V)$
Coupling	$\beta = 3.56$–4.27, median 3.74 ($Q_L = 6760$)	Measured reflection against Eq. (7.3)
Detuning, from the power balance	-34.9 to -36.8 kHz	Equation (2.10) per cavity
Detuning, IOC tuner model	-38 to -98 kHz	`CAVn:FREQ:OFFS`, Eq. (6.7)
Direct loop gain	`COUNTS.A` $= 1.3$ dB, `COUNTS.L` $= 1.161$	$G \approx 0.9$–1.0, see below
Direct loop phase	$-8.4°$ = `PHASE.C` $(-20°)$ + tracking ϕ_{tr} $(+11.6°)$	`PHASE.L`; `VAL` reads $-8.3°$
Integral and lead compensation	Both on	Network values: §6.2
Frequency-offset tracking	On	[R12]: diagnostic, normally off
DSP ripple loop	On, DC-coefficient tracking on	§6.3
Comb, gap feed-forward	Modules not installed	Doc L

What the loop gain really is. Three independent pieces of evidence put the present proportional gain at about 0 dB rather than the 15 dB of the design:

1. **The RFgui closed-loop measurement** (Measure Direct Closed, the MATLAB routine `MeasDirCls` of [R12]) injects noise through the RFP DACs and computes the transfer function to the summed feedback signal with the direct loop closed; it does not disturb stored beam [R12], [R41]. On 2 October 2026 at 22 mA, with `COUNTS.A` $= 0.9$ dB and `PHASE.C` $= -20°$, the routine's resonator-times-delay fit gave a peak of 0.489, $Q = 3447$, a delay of 621 ns and a phase within $2°$ of ideal [R38]. For a closed loop the fitted peak is $G/(1+G)$, giving $G = 0.957$; the full loop model, integral path included, fitted to the six runs at 20–22 mA gives $G = 0.88 \pm 0.03$ (Table 7). Scaled to today's 1.3 dB, $G = 0.92$. Reading the response as open-loop would require $Q_L = 3447$ and $\beta = 8.3$.
2. **The IOC's own model** divides the reference conversion by one plus `COUNTS.L` (record `STN:CONV:CONST`, routine `subIQampl2conv` [R37]); that is, it treats `COUNTS.L`, the linear gain computed from the dB setting (1.16 at 1.3 dB), as the dimensionless loop gain. The conversion count `COUNTS.E` $= 220.08$, set by the Configure Direct routine (`ConfDirect` in [R12]) [R41], therefore corresponds to unit gain.
3. **The 2021 synchrotron line** on the legacy system at 500 mA had $f = 8.93$ kHz and $Q = 11.1$ [R4], i.e. $\sigma = 2.5 \times 10^3$ s^{-1}. The proportional model at $G = 1$ gives 8.77 kHz and 1.9×10^3 s^{-1} ($Q = 15$); at the 15 dB design gain it would give 9.89 kHz and $Q = 72$, which the measurement excludes.

The frequency agrees to 2%. The model's damping is 27% low; the loop settings of 2021 (not recorded) and the Lorentzian fit could each contribute, while the lead network adds only 2%.

The closed-loop measurements of 2 October 2026. While the routine plays a noise sequence through the RFP DACs, it records the summed feedback signal in the RFP signal memory (I and Q, eight samples per turn, 10.24 MHz) and computes the transfer function from the noise reference to that record in 1.25 kHz bins. The same computation applied to the stored records and the station's noise references [R38] reproduces the routine's plots to 0.03% in magnitude and $0.5°$ in phase. Table 7 lists the ten runs, each fitted over 18.8–382 kHz with the closed-loop model of Figure 6 (lead and lag networks on, 2.5 MHz amplifier pole) with G, θ, τ_d, Q_L and the detuning free. The residuals are at the noise level of the data. Without the integral path the model fits about as well but returns 10% more gain and 11% more Q_L.

Table 7: RFgui closed-loop measurements of 2 October 2026 [R38] at `COUNTS.A` $= 0.9$ dB and 2.85 MV, fitted with the closed-loop model of Figure 6 (lead and lag networks on): loop gain G, loop phase θ (positive = lead; it includes the frequency-offset tracking then in use, which was not recorded), total delay τ_d, loaded Q and detuning Δf. Statistical errors are about 0.005 in G, $0.3°$, 4 ns, 50 in Q_L and 0.4 kHz; repeated runs differ by more.

Time	I_b (mA)	`PHASE.C`	G	θ	τ_d (ns)	Q_L	Δf (kHz)
09:33	22	$-14°$	0.91	$-6.1°$	619	6650	$+1.9$
09:41	22	$-17°$	0.89	$-2.8°$	626	6690	$+0.6$
09:47	22	$-20°$	0.88	$+1.4°$	620	6940	-1.1
09:49	22	$-23°$	0.87	$+5.4°$	625	7190	-2.1
10:09	22	$-23°$	0.85	$+5.6°$	629	7010	-1.1
13:44	363	$-23°$	0.53	$-6.4°$	684	4760	-24.0
13:49	494	$-23°$	0.39	$+0.8°$	690	4650	-34.7
13:51	497	$-20°$	0.42	$-5.3°$	673	4610	-35.9
13:58	20	$-20°$	0.84	$-1.7°$	625	6520	$+0.1$
17:20	395	$-20°$	0.58	$-18.5°$	664	4460	-33.7

- **Low current.** At 20–22 mA, $G = 0.88 \pm 0.03$ and $\tau_d = 624 \pm 4$ ns, in agreement with §2.6, and the cavities sit within 2 kHz of f_{RF}. The loaded Q is 6800 ± 250 (6500–7200), in agreement with the 6760 that the reflection data of Table 6 give with $Q_0 = 32{,}000$ and with the 6500 fitted to the open-loop record of the Configure Direct routine [R42]; the analysis keeps 6700 (Table 1).
- **Phase setting.** Stepping `PHASE.C` from $-14°$ to $-23°$ moved the fitted loop phase from $-6.1°$ to $+5.5°$: a more negative setting is a lead, by 1.2–1.3° per degree, and $\theta = 0$ at `PHASE.C` $\approx -19°$. The setting in use, $-20°$, is within 2° of ideal at low current. The phase shift applied to the direct-feedback loop is `PHASE.L` $=$ `PHASE.C` $+ \phi_{\mathrm{tr}}$, where ϕ_{tr} is the frequency-offset tracking correction.
- **High current.** At 497 mA the measured response lies 3–7 dB below the 500 mA model of Figure 6. At 363–497 mA the fits place the cavities within 6 kHz of the optimum detuning (2.10), but give $G = 0.39$–0.58, $Q_L = 4460$–4760 and delays of 664–690 ns; a run at 485 mA on 10 November 2025 gives $G = 0.60$, $Q_L = 4630$ and 595 ns [R42]. The beam's synchrotron motion (Appendix C) changes the response by at most 15% outside ± 18 kHz and 3% outside ± 40 kHz, and the same fit applied to a model that includes it returns that model's gain and Q_L within 1%. Since neither Q_L nor the delay depends on the beam current, the fitted values are effective parameters of a response that the linear loop model does not describe at high current. The klystron, near its power limit (99%, Table 6), explains it. Near saturation its small-signal response to a change of drive amplitude differs from that to a change of drive phase: the output amplitude follows with a gain g_A below one, and the output phase moves too (AM/PM conversion, k_{AP} radians per unit relative amplitude). Such a klystron makes the loop widely linear, and because the injected noise is one-dimensional — the station's noise references for I and Q are the same sequence (correlation 0.9998) — the routine records one mixture of the amplitude and phase responses. With G, τ_d and Q_L held at the 20–22 mA values and g_A, k_{AP}, the drive phase, θ and Δf free, this model fits the four high-current runs of 2 October to an rms of 0.012–0.015, at the noise level and better than the linear model with G, τ_d and Q_L free (0.014–0.018; 0.04–0.09 with them held), and Figure 6 shows the fit at 497 mA. The one-dimensional injection does not separate g_A from k_{AP}: the best fits pair $g_A = 0.6$–1.0 with $k_{\mathrm{AP}} = 1.1$–1.4, while $k_{\mathrm{AP}} = 0$ with g_A free leaves an rms of 0.018–0.027. Records of normal operation settle it (§8.10): $g_A = 0.40$ and $k_{\mathrm{AP}} \approx 0.1$. With these values, and with gain, delay and Q_L refitted at 20–22 mA with the same klystron, the model fits the high-current runs to 0.013–0.018 and places the cavities within 4 kHz of the power-balance detuning. As a cross-check, transfer curves measured in 2013 on a tube since replaced give $d \ln P_{\mathrm{out}} / d \ln P_{\mathrm{drive}} = 0.2$–$0.45$ near the top of the curves at 72–80 kV with 330–360 mA of beam [R42].
- **The Robinson mode.** Near $\pm f_s$ the high-current records, analyzed with 78 Hz bins, show the beam's coherent mode as a pole pair: 9.38 and 9.42 kHz with $\sigma \approx 1.0 \times 10^3$ s^{-1} at 363 and 395 mA, and 8.32–8.44 kHz with $\sigma = (0.97$–$1.46) \times 10^3$ s^{-1} at 494–497 mA; the run of 10 November 2025 gives 8.65 kHz and 1.6×10^3 s^{-1} at 485 mA. The damping is close to that of the synchrotron line of 8 October below; the frequency at 494–497 mA is 5% lower. The gain setting accounts for little of that: `COUNTS.A` was 0.9 dB on 2 October and 1.3 dB on 8 October, which in the model moves the frequency by only 50 Hz, a tenth of the difference. The klystron's small-signal response, which shifts the frequency by hundreds of hertz (§8.10), and the noise being injected on 2 October are the candidates.

The synchrotron line today. Two captures of the cavity-A probe spectrum on 8 October 2026 [R40], each fitted with one Lorentzian per sideband and paired with a snapshot taken within three minutes, give the line of Table 8. Its frequency has hardly changed since 2021, but its damping has more than halved: $\sigma = (1.09$–$1.16) \times 10^3$ s^{-1}, 2.5–2.7 times radiation damping. The proportional model is now 60–70% too high. An integral path that shares the loop phase lowers the damping (Table 4); with $f_i = 1$ kHz it reproduces the damping of both captures within 7% and the frequency within 2%. This value is an effective one. The corner trades against the loop-phase error: 0.8–1.0 kHz for θ between $-5°$, the value fitted at 497 mA (Table 7), and 0. It also depends on how the loop behaves at high current: the effective parameters fitted at 497 mA without the integral path ($G = 0.43$, $Q_L = 4800$, 680 ns) reproduce the frequency as

well (8.82–8.84 kHz) and need $f_i = 1.5$–1.8 kHz. The network actually installed, the lag network (§6.2), corresponds to 2.9 kHz with a 0.26 kHz leak; with it the model gives $\sigma \approx -390$ s^{-1} with the loop of Table 1 and 460 s^{-1}, barely above radiation damping, with the effective loop. The lead network, worth about 2%, is left out. Because an integral path can only lower the damping in this model, the 2021 line, 35% above the proportional model, suggests that the integral path was then off or weaker.

Table 8: Synchrotron line on 8 October 2026: measured on the cavity-A probe [R40] and modeled at the paired snapshots ($G = 1$, 625 ns). Each entry: coherent frequency (kHz), damping rate (s^{-1}), Q.

	13:31, 497 mA	15:35, 492 mA
Measured	8.79, 1160, 23.8	8.83, 1086, 25.5
Model, proportional, $\theta = 0$	8.75, 1876, 14.6	8.78, 1842, 15.0
Model with integral path, $f_i = 1$ kHz, $\theta = 0$	8.72, 1094, 25.0	8.75, 1080, 25.5
The same with $\theta = -5.3°$	8.80, 1019, 27.1	8.83, 1004, 27.6

What the line says about the beam's motion. The same captures give the size of the motion, not only its damping. Each sideband of the line carries the fraction

$$\frac{P_\pm}{P_\mathrm{c}} = \frac{I_{b1}^2\,|Z_{b,A}(\pm j\omega_\mathrm{coh})|^2}{2\,V_A^2}\,\langle\Delta\phi^2\rangle \tag{8.14}$$

of the carrier power, where $Z_{b,A} = Z_A/(1 + G_\mathrm{OL})$ is the closed-loop impedance of the probed cavity and V_A its voltage. The capture of 15:35, taken with the RMS detector, shows that broadband noise is not the main drive. A comb of 60 Hz harmonics runs through the line, strongest on the 6-pulse family of D5 (§4.5): the harmonics at 8640 and 9000 Hz, which flank f_coh, stand 16 dB above the noise beside them at -57 and -55 dBc. With the harmonics masked, the broadband line carries -58.5 and -59.8 dBc on the two sidebands. Two models reproduce the measured damping: the effective integral path of Table 8, with $|Z_{b,A}| = 343$ and 357 kΩ at the two sidebands, and the lag network with the klystron measured in normal operation (§8.10), for which the two-quadrature form of Appendix C gives the corresponding factor. From the broadband line they put the beam's phase motion at $0.19°$ and $0.14°$ rms, and by Eq. (8.10) the drive at $S_n(f_\mathrm{coh}) = 1.8$ and 1.1×10^{-11} rad^2/Hz. Each harmonic is the coherent sum of a field modulation and the beam's response to it; for the harmonics at 8640 and 9000 Hz the models put the beam's share of the power at 0.8–1.3, whether the HVPS modulates the klystron's amplitude or its phase. The harmonics that the beam dominates add $0.55°$ and $0.36°$, 93–99% of it from within 1 kHz of f_coh. In total the beam's phase moves by $0.58°$ and $0.39°$ rms (3.4 and 2.3 ps), four to six times the field-phase specification of §1.4, and its energy by $\Delta E/E_0 = 1.6$ and 1.0×10^{-4} rms (8.11). The probe-signal noise beside the line, -106 and -103 dBc/Hz on the upper and the lower side, corresponds to 4.1×10^{-11} rad^2/Hz if the two sidebands are uncorrelated, 2.3–3.9 times the broadband drive: the floor is not the drive, because below f_coh it contains the beam's own response and at 30 Hz resolution the skirts of the comb raise it. Two observations remain open: the upper sideband is 1.3 dB stronger than the lower one, where both models predict 0.35 dB weaker, and the passive spectrum shows neither which source drives the broadband line nor whether the harmonics are amplitude or phase modulation (§8.10).

The line harmonics between 60 and 1080 Hz, measured in the same capture with a 3 Hz resolution, carry -51.5 dBc in total, mostly at 60 and 720 Hz. The beam follows them (§8.7): they add $0.15°$ of timing jitter if they are phase modulation and $0.06°$ if they are amplitude modulation, but almost nothing to the energy. With the motion near f_coh they give 0.40–$0.60°$ (2.3–3.5 ps) of timing jitter. Since 2021 the

damping has fallen from 2.5 to 1.1×10^3 s^{-1}; for the same drive, that alone has raised the variance of the broadband motion 2.3-fold, and the variance driven by the harmonics at 9000 and 8640 Hz 2.2–2.9-fold and 3.8–4.4-fold.

Findings.

1. **The Robinson margin is adequate, but smaller than in 2021**: 2.5–2.7 times radiation damping today ($Q = 24$–26, within the acceptance limit $Q \leq 35$ of §8.9), against six times in 2021. In the model, the integral path at its effective 1 kHz costs about 40% of the damping that the proportional loop would leave. The present low gain is a sound choice for SPEAR3; raising it would buy impedance reduction that no instability requires and cost damping.
2. **The model does not reproduce the damping with the installed integral network.** Three measurements agree with the lag network of the RFgui loop models ($f_i = 2.9$ kHz, leak 0.26 kHz): with it in the model the low-current closed-loop fits return the Q_L of the reflection and open-loop data; fitted within ± 25 kHz of the carrier, the same records give $f_i = 3.0$–4.2 kHz; and in records of 2 June 2020 the ratio of drive modulation to cavity error at the HVPS lines between 0.36 and 2.9 kHz follows the network's shape, with $G \approx 2$ at the setting of the time [R42], and excludes a proportional controller. With that network the linear model leaves the beam unstable at 500 mA with the loop of Table 1 and barely damped with the effective loop, whereas the synchrotron line and the closed-loop records show $\sigma = (1.0$–$1.6) \times 10^3$ s^{-1} at 363–497 mA. The $f_i \approx 1$ kHz of Table 8 is therefore an effective value, and the model misses about 10^3 s^{-1} of damping; the klystron's amplitude compression, measured in normal operation, supplies most of it (§8.10). A synchrotron line with `INTCOMP` off at reduced current would measure the integral path's real cost: at 300 mA the model predicts $\sigma = 1.10 \times 10^3$ s^{-1} ($Q = 27$) without the integral path, 790 s^{-1} ($Q = 38$) with the effective 1 kHz path and 180 s^{-1} ($Q = 168$) with the lag network.
3. **Frequency-offset tracking is on**, although [R12] reserves it for diagnostics. It rotates the loop phase by

$$\phi_{\text{tr}} = -360^\circ \, F_{\text{tr}} \, \tau_d \, \overline{\Delta f}, \tag{8.15}$$

   with $F_{\text{tr}} = 0.8$ (`PHASE.F`), $\tau_d = 700$ ns (`PHASE.B`, §2.6) and $\overline{\Delta f}$ the mean of the IOC's tuner-model estimates `FREQ:OFFS`. Those estimates disagree with the detuning implied by the measured reflection and power balance — by more than 60 kHz for cavity 3 — so $\phi_{\text{tr}} = +11.6^\circ$, against $+7.2^\circ$ from the physical detuning. A positive ϕ_{tr} is a lag, because a more negative setting leads (Table 7). The excess, which drifted between $+3^\circ$ and $+6^\circ$ on 8 October, therefore lowers the damping by 5–11%, and the loop phase follows a stale calibration rather than the cavities. Either switch tracking off and re-trim `PHASE.C` by the ϕ_{tr} then in use, or recalibrate the tuner model (Make Polynomials, Eq. (6.7)). If tracking stays on, `PHASE.B` should also be set to the present delay, 625 ns: its 699.5 ns most likely dates from the klystron replaced in 2020 (§2.6) and makes ϕ_{tr} 12% too large.
4. **The klystron runs at 99% of the IOC's 800 kW forward-power limit**, with the HVPS loop holding the drive 4.4 dB below the saturation setting.
5. **The beam's phase moves by** 0.39**–**0.58° **rms near** f_{coh} (2.3–3.4 ps), four to six times the field-phase specification. Most of it is driven by the 6-pulse HVPS harmonics at 8640 and 9000 Hz, which flank f_{coh} above the reach of the ripple loop; broadband noise adds 0.14–0.19°. As the current rises, f_{coh} sweeps through this comb, and both models that reproduce the damping put a crossing of a 360 Hz harmonic at 220–230 mA, a candidate for the larger oscillation seen near 200 mA. Which source sets the broadband drive — the klystron, the measurement or the reference — and whether the harmonics modulate amplitude or phase decide which remedy works (§8.7); neither has been measured. The direct-loop study (`Studies/optLLRF`) plans the measurements.

8.9 Setting the Direct Loop: Procedure and Acceptance

The procedure applies to the legacy station now and to the LLRF9 at commissioning; steps that differ are marked. Steps 1–3 need no beam or only low current; steps 4 and 5 are machine-development work at operating current.

1. **Preconditions.** Tuners converged (load-angle errors below 1°), reflected power at the coupler floor of Eq. (7.3) (1–3%), klystron at least 10% below saturation, gain tracking active. Record a snapshot of the quantities in Table 6.
2. **Delay and phase** (no beam or below 50 mA, cavities near f_{RF}). Measure the loop response — on the legacy system with RFgui Measure Direct Closed (closed loop, beam undisturbed) or RFguiInvasive Configure Direct (open loop; it also sets the phase and gain and writes the fitted delay to `PHASE.B`) [R41], on the LLRF9 with its network analyzer — and fit gain, Q, delay and the phase at resonance. Set the phase so that the fitted phase at resonance is 180° within $\pm 5^\circ$ ($\theta = 0$). On the legacy system a more negative `PHASE.C` is a lead (Table 7); on the LLRF9, establish the sign once by stepping the phase setting by $+10^\circ$. A lag is the dangerous direction (Figure 11).
3. **Gain.** Choose G from Figure 10, which is for 500 mA; for a different maximum current, recompute it with the model of §8.2. At 500 mA, $G \leq 3$ dB keeps $\sigma \geq 2\alpha_{\mathrm{rad}}$ for phase errors within $\pm 15^\circ$ ($G \leq 2$ dB for $\pm 30^\circ$) and leaves more than 13 dB of margin to the delay-limited instability (Table 3). The present 0 dB lies inside this range. Within it the gain also sets how the noise of the RF system moves the beam (Figure 12): lower gain is better against measurement, reference and klystron phase noise, higher gain only against klystron amplitude noise. Do not raise the gain for field regulation alone; until the sources are measured (§8.10), stay near the present setting.
4. **Integral path.** Legacy: the network is fixed; measure its effect on the damping by comparing the synchrotron line with `INTCOMP` on and off at reduced current. LLRF9: start with $f_i \leq 1$ kHz, set $\Delta\theta_i$ by rotating it until the line is most strongly damped, and raise f_i only as far as the line-harmonic rejection requires (Table 4). Judge the setting by the beam's jitter (8.13) as well as by its damping.
5. **Verify with beam** (500 mA). Record a cavity-probe spectrum (20 kHz span, 30 Hz RBW, RMS detector, about 100 power-averaged sweeps, which take 15 s) and fit a Lorentzian to each synchrotron sideband [R40]. Accept if $Q \leq 35$ ($\sigma \geq 2\alpha_{\mathrm{rad}}$) and the frequency lies within 5% of the model run on a snapshot taken at the same time; record the 360 and 720 Hz lines, and the power in each sideband of the synchrotron line, which gives the beam's phase motion (8.14).
6. **Re-verify** after any change of loop gain or phase, klystron, HVPS, drive amplifier, cables or couplers, and when the maximum current changes by more than 20%.

Warning signs: at 500 mA, a synchrotron line with $Q \gtrsim 40$, or a coherent frequency close to the incoherent 10.0 kHz (too much gain, or a lag error). At lower current both are normal — the model gives $Q \approx 50$ and 9.8 kHz at 200 mA — so compare with the model at the measured current. Also: growing or splitting sidebands (Robinson anti-damping, often from the integral path); 360 or 720 Hz lines rising by more than 10 dB (ripple loop off or mis-set); a synchrotron line whose power rises by more than 3 dB at unchanged width (a stronger drive, D10); klystron drive approaching the saturation setting (no headroom left for the loop).

8.10 Model Limits and Open Questions

The model assumes rigid bunches, a uniform fill and equal vector-sum weights, and it treats the klystron as linear. Rigid bunches describe only the dipole motion: the quadrupole (bunch-length) mode, which amplitude noise near $2f_s$ drives, lies outside the model. For the dipole mode at the present operating point the first three assumptions do not matter. The last may: the closed-loop response at 363–497 mA does not follow the linear model (Table 7), and with the installed integral network the model lacks about 10^3 s^{-1} of damping at high current (§8.6). A klystron that responds differently to amplitude and phase perturbations

makes the loop act differently on the two quadratures, which a loop described by one complex gain cannot represent and the two-quadrature form of Appendix C can. The closed-loop records of the present tube at 363–497 mA require such a klystron (§8.8); as cross-checks, two data sets from tubes since replaced show the same behavior: the transfer curves of 2013 flatten near saturation, and in records of 2 June 2020, taken at operating power, switching on the direct loop reduced the 360 and 720 Hz phase lines of the cavity field by 35 and 29 dB but the amplitude lines by only 6 and 2 dB [R42].

The asymmetry also changes the Robinson mode. A time-domain version of the model, which agrees with the characteristic equation to within 0.3% for a linear klystron, includes the klystron's g_A and k_{AP}. At 497 mA, with the lag network and $G = 0.875$, $g_A = 0.75$ raises σ from -270 to $+140\ \mathrm{s}^{-1}$ and $k_{\mathrm{AP}} = 0.6$ raises it to $+340\ \mathrm{s}^{-1}$, while each lowers the frequency by 300–480 Hz. Klystron asymmetry can therefore supply the missing damping. The two-quadrature characteristic equation of Appendix C reproduces these values, and those below, to 1%.

Records of normal operation measure both parameters without disturbing the beam. At each trip the station freezes, together, the drive I/Q (RFP signal memory), the probe I/Q of one cavity and the klystron output amplitude (IQ&A history) [R42]. In the 4.9 ms of normal operation before the trip, ten such records give $g_A = 0.40 \pm 0.05$ from the output amplitude against the drive (coherence 0.93–0.98 at 50–200 kHz, above the HVPS lines), the same at 40% and at full output power, consistent with the HVPS loop holding the drive at a fixed point of the transfer curve. The probe's response to the drive amplitude, relative to its response to the drive phase, gives $k_{\mathrm{AP}} = 0.10 \pm 0.04$ whatever the cavity and beam; this estimate leans slightly towards a linear klystron (it returns $g_A = 0.50$). With $g_A = 0.40$ the model reaches the measured damping — $\sigma = 950\ \mathrm{s}^{-1}$ for a phase-path gain of 1.0, $450\ \mathrm{s}^{-1}$ for 1.3 — but at 7.9–8.2 kHz, 5–10% below the measured frequency. Amplitude compression in the klystron is therefore the main source of the missing damping. Open are the frequency offset and the phase-path gain: since $g_A \approx 0.4$ holds at low current too, the $G = 0.88$ of Table 7 is the gain that a one-dimensional injection sees, and the refits with $g_A = 0.40$ place the phase-path gain between 1.0 and 1.4 without fixing it. Seven measurements would remove the remaining uncertainties:

1. the synchrotron line at reduced current with `INTCOMP` and `LEADCOMP` switched off in turn, which measures the effect of each network on the damping directly;
2. the closed-loop measurement at low current and near 500 mA with independent I and Q noise, or with I-only and Q-only noise in turn, to separate the responses to amplitude and phase perturbations, together with the loop settings at the time (`PHASE.L`, `COUNTS.L`);
3. the amplitude and phase transfer curves (AM/AM and AM/PM) of the present klystron at the operating voltage and beam current, to confirm g_A and k_{AP} from the fault records and give their dependence on drive level and voltage, together with records that hold the klystron's output phase;
4. the synchrotron line at two or three currents (for example 100, 300 and 500 mA) with the loop settings recorded, to fit the effective loop phase directly;
5. the LLRF9 total loop delay, measured with its own network analyzer;
6. the noise spectra of the klystron output, of the cavity probes and of the reference near f_s, without beam or at low current and with the loop settings recorded, which give the spectral density of each source class of Eq. (8.12) near f_{coh};
7. the synchrotron line at operating current while a small noise band around f_{coh} is added to the drive, first in amplitude only and then in phase only, which measures the drive transfer of the klystron sources and, with the previous item, the gain and phase that minimize the beam's jitter (8.13); the same records should explain why the upper sideband of the line is 1.3 dB stronger than the lower one (§8.8).

9 Performance Requirements and Verification

9.1 Disturbance Rejection Summary

Disturbance	Frequency	Open-Loop Impact	Rejection	Residual
Beam loading (D1–D2)	DC–100 kHz	0.8% of V_{gap} per 1% current step	$\lvert 1 + G_{\text{OL}} \rvert$: 4 dB at $G \approx 1$; about 22 dB at DC with the legacy lag network	Regulated
Robinson (D3)	$\sim$ 10 kHz (2.7)	Damped, 5.8×10^3 s^{-1}	Loop *reduces* the damping	1.1×10^3 s^{-1} measured in 2026, 2.5×10^3 in 2021 (§8.8)
HVPS ripple (D5)	360–2000 Hz	-31 to -34 dBc lines [R4]	≈ 25 dB (legacy direct and ripple loops); 16–21 dB (LLRF9, 2021)	≈ -55 to -60 dBc
Thermal drift (D8)	< 0.01 Hz	~ 100 Hz/hr	Tuner tracks	Within the tuner deadband
Klystron gain (D6)	0.01–1 Hz	$\sim$ 7 dB var.	Gain tracking	< 0.5 dB
RF noise (D10)	Broadband; the beam responds below and near f_{coh}	Drives the dipole mode, amplified up to $Q\,(f_s/f_{\text{coh}})^2 \approx 33$	Shaped, not removed: klystron amplitude noise is reduced, klystron phase and measurement noise are not (§8.7)	-106 to -103 dBc/Hz beside the line; beam phase 0.14–0.19° rms at f_{coh}, against 0.36–0.55° from the D5 harmonics beside it (2026)

9.2 LLRF9 Commissioning Results

From [R4]: the LLRF9 controller [R39] achieves an improved noise floor relative to the legacy VXI system at 500 mA. Overall amplitude stability $< 0.05\%$ RMS and phase stability $< 0.05°$ RMS — both better than the requirements of §1.4.

9.3 Performance Margins

Parameter	Capacity	Typical	Margin
Klystron power	1.5 MW	$\sim$ 817 kW (4×204.3); 795 kW measured	$\sim 46\%$
Klystron forward-power limit (IOC)	800 kW	795 kW measured (Table 6)	$\approx 1\%$
HVPS voltage	90 kV	$\sim$ 72–76 kV (75.5 kV measured)	$\sim 16\%$
Gap voltage/cavity	1 MV	$\sim$ 712 kV	$\sim 40\%$
Direct-loop gain	$\approx$ 15 dB (45° margin, 625 ns)	$\approx$ 0 dB	$\approx$ 15 dB

Parameter	Capacity	Typical	Margin
Robinson damping	—	1.1×10^3 s^{-1} (measured, 492–497 mA)	2.5–2.7× radiation damping

9.4 Calibration Data

Calibration establishes the numerical relationships between raw hardware signals (ADC counts, DAC counts, detector voltages) and physical quantities (kV, kW, degrees, Hz). It is essential for closing all feedback loops accurately. This section is a high-level summary; the measured coefficient tables themselves live with the calibration data files [R31], [R32].

9.4.1 Calibration Sequence Overview

The master calibration is executed by `rf_calib.st,v` (3,345 lines, 28 states including setup and abort handling). It runs as an EPICS SNL program and requires the station to be OFF (no RF power). The sequence progresses through the following phases:

Phase	States	What It Does
Initialization	Startup, CombCheck, Setup	Registers with RFP module, verifies hardware, sets known baseline
Zero all multipliers	ZeroCavMults, ZeroDirMults, ZeroCombMults	Writes zero to all I/Q modulator DACs — establishes the ”no signal” baseline for offset measurement
Combiner calibration	Combiner	Calibrates the 4 cavity combining modulators: measures each probe independently, adjusts I/Q matrix for accurate vector sum
Direct loop calibration	Direct, SummingNodeI/Q, GainStageI/Q	Characterizes direct loop gain stages and summing node offsets. I and Q channels measured independently to resolve cross-coupling
Klystron chain	ZeroKlysMults, DiffNodeOffsets, KlysStage, CompStage, CombStage, KlysDemod	Calibrates klystron forward path: drive modulator offsets, klystron gain, compensator stages, demodulator phase alignment
Tuner & modulator	TuneStage, NullModulator	Calibrates tune-mode drive path; nulls RF modulator output (zero DAC $\rightarrow$ zero RF)
Completion	Finish/Done or Abort/Abend	Restores operational configuration, reports success/failure

Each state includes retry logic (`NUM_TRIES` iterations), severity checking on PV readbacks, and abort handling. The entire sequence takes ~2–5 minutes.

9.4.2 Drive Amplitude Calibration

Establishes the mapping from DAC counts to klystron input drive power. Data files: `driveAmpCalibration.xlsx`, `klystronCouplerDriveAmpCalibrations.xlsx`.

The key relationship is the **drive conversion constant** D_{conv} (counts/kV), computed by `subIQampl2conv`:

$$D_{\text{conv}} = \frac{A_{\text{ref}}}{V_{\text{gap}} \cdot (1 + G_{\text{loop}})} \tag{9.1}$$

where A_{ref} is the reference amplitude (counts), V_{gap} is the measured gap voltage (kV), and G_{loop} is the feedback loop gain. This constant is used by the DAC loop (6.8) and recomputed whenever the operating point changes.

9.4.3 Power Measurement Calibration

Maps RF detector voltages to physical power levels. Data files: `reflectedPowerCalibrations.xlsx`, `pulsarCouplerCalibration2049.xlsx`.

The **RF detector conversion loss** measured during calibration (`subIQampl2loss`):

$$L_{\text{det}} = 20 \cdot \log_{10}\left(\frac{E_{\text{conv}} \cdot \sqrt{P_{\text{cal}}}}{A_{\text{det}}}\right) \text{ dB} \tag{9.2}$$

where $E_{\text{conv}} = 0.31623$ (power conversion constant), P_{cal} is calibration power (mW), and A_{det} is measured detector amplitude (V). Accounts for cable attenuation, coupler directivity, and detector nonlinearity.

The **coupling factor** (VSWR-based, `subIQamplCplg`):

$$\beta_{\text{meas}} = \frac{1+r}{1-r}, \qquad r = \frac{A_{\text{refl}}}{A_{\text{fwd}}} \tag{9.3}$$

9.4.4 Frequency and Detuning Calibration

Maps tuner motor position to cavity resonant frequency offset. The polynomial model (6.7) is fitted from measured data: the tuner is stepped across its range and the resonant frequency measured at each position. Calibration coefficients $p_0, \ldots, p_3$ and the voltage coefficient t_1 are stored in EPICS PVs.

9.4.5 Tuner Mode DAC Calibration

Data file: `tuneModeDacCalibration.xlsx`. Establishes DAC counts-to-drive-power for the `STATION_TUNE` state (low-power cavity conditioning). Separate from operational drive calibration (§9.4.2) because the operating point differs.

9.4.6 Signal Routing and Patch Panel

Data file: `b132R11PatchPanel.xlsx`. Documents physical signal routing in Building 132 R11: cable lengths, attenuation values, connector types, and signal naming. Essential for verifying the delay budget (§2.6) and signal levels.

Upgrade note: Many calibrations will need adaptation for the LLRF9 system (different ADC/DAC ranges, signal levels, processing gains). The polynomial frequency model (6.7) and combiner matrix calibration are expected to transfer; drive and power detector calibrations require new measurements.

A Source Document Reference Index

A.1 Published Papers and Web Pages

Ref	Citation
[R1]	McIntosh, P. et al., “The SPEAR3 RF System,” SLAC-PUB-10983, EPAC 2004. `https://inspirehep.net/files/945e7ff73cc428af4c018fd1bdb6afa7`; OSTI record `https://www.osti.gov/biblio/839730`
[R2]	Hettel, R. et al., “Design of the SPEAR 3 Light Source,” PAC 1999
[R7]	Rimmer, R.A. et al., “RF Cavity Development for the PEP-II B Factory,” LBL-33360, 1992
[R9]	Gamp, A., “Beam-Cavity Interaction,” CAS 2011, arXiv:1112.3203. `https://export.arxiv.org/pdf/1112.3203v1.pdf`
[R10]	Wilson, P.B., “Fundamental-Mode RF Design,” SLAC-PUB-6062, 1993
[R11]	Boussard, D., “Control of Cavities with High Beam Loading,” PAC 1985. `https://proceedings.jacow.org/p85/PDF/PAC1985_1852.PDF`
[R13]	Corredoura, P., “Architecture and Performance of the PEP-II Low-Level RF System,” SLAC-PUB-8124, PAC 1999. Repository copy: `llrf/documentation/legacyArchitecture/architecture-and-performance-of-the-pep-ii-low-level-rf.pdf`
[R14]	Robinson, K.W., “Stability of Beam in RF Systems,” CEA-CEAL-1010, 1964
[R15]	Rimmer, R.A. et al., “Comparison of Calculated, Measured, and Beam Sampled Impedances,” PRSTAB 3, 102001, 2000
[R17]	Cassel, R. and Nguyen, M.N., “A Unique Power Supply for the PEP II Klystron,” SLAC-PUB-7591, PAC 1997. Repository copy: `hvps/architecture/originalDocuments/slac-pub-7591.pdf`
[R19]	Bourbeau, E.J., “Application of PLL Techniques to SCR Drives,” IEEE 1983
[R34]	Fox, J.D. et al., “Multi-Bunch Longitudinal Feedback — Woofer and Kicker Architecture,” PEP-II LFB system design note
[R35]	Pedersen, F., “Beam Loading Effects in the CERN PS Booster,” IEEE Trans. Nucl. Sci. NS-22 (1975) 1906–1909
[R39]	Dimtel, Inc., LLRF9 low-level RF controller (the upgrade controller), product page. `https://www.dimtel.com/products/llrf9`

A.2 Textbooks

Ref	Citation
[R3]	SSRL SPEAR Storage Ring Parameters, `https://www-ssrl.slac.stanford.edu/accphy/spear_parameters.html` (last updated 26 Apr 2002, H.-D. Nuhn)
[R8]	Wiedemann, H., *Particle Accelerator Physics*, 4th ed., Springer, 2015
[R16]	Analog Devices AD834 datasheet
[R33]	Åström, K.J. and Murray, R.M., *Feedback Systems: An Introduction for Scientists and Engineers*, 2nd ed., Princeton, 2021 (Ch. 12, loop interaction)

A.3 Repository Documents

Ref	Document	Path
[R4]	LLRF9 Commissioning Tests	`llrf/tests/llrf9Tests.pdf`
[R5]	System Design Report	`Designs/tex/doc0-system-design/0_system_design_report.pdf`

Ref	Document	Path
[R6]	RF System Description (Schwarz)	`llrf/documentation/legacyArchitecture/ps3403305100.pdf`
[R12]	Feedback Loop Description (Schwarz, PS-340-330-52-R0)	`llrf/documentation/legacyArchitecture/feedbackLoopDescriptionps3403305200.pdf`
[R18]	HVPS Technical Spec (PS-341-360-01-R2)	`hvps/architecture/originalDocuments/ps3413600102.pdf`
[R20]	Enerpro firing-board schematics and manuals (FCOG6100 in service, FCOG1200 for the upgrade)	`hvps/controls/enerpro/enerproDocuments/`
[R21]	Cassel PLC Code	`hvps/documentation/plc/CasselPLCCode.pdf`
[R22]	Cavity Tuner Inspection Report, 13 June 2023	`llrf/tuners/cavityTunerInspections20230613.docx`
[R23]	Tuner mechanical design paper, PAC 1997	`llrf/tuners/tunerDesignPac97HDS8P030.PDF`
[R24]	Superior Electric SLO-SYN stepper motor data	`llrf/tuners/SLO-SYN.pdf`
[R25]	SLO-SYN MD808 Stepper Drive Manual	`llrf/tuners/SLO-SYN_MD808_Stepper_Drive_Manual.pdf`
[R26]	SLO-SYN SS2000MD4M Step Drive / Translator Manual	`llrf/tuners/SLO-SYN_SS2000MD4M_Step_Drive_Translator_Manual.pdf`
[R27]	Superior Electric stepper catalog (legacy)	`llrf/tuners/Old Stepper Catalog_Superior Electric_0.pdf`
[R28]	Tuner Loop Source Code	`spear-rf-code-legacy/rfApp/src/seq/rf_tuner_loop.st,v`
[R29]	Galil Commissioning Notes	`llrf/tuners/galil/GalilCommissioning.docx`
[R30]	SPEAR3 RFP ripple-loop firmware (W. Ross, 2006) and its phase-harmonic coefficient table; `ripple.s` is the PEP-II original	`spear-rf-code-legacy/rfApp/src/dsp/rfpDsp/sp3ripple.s,v`; `spear-rf-code-legacy/iocBoot/tbl/Sp3PhsCoefs.tbl,v`
[R31]	RF Calibration Data	`llrf/calibrations/*.xlsx`
[R32]	RF Document Index	`llrf/documentation/RfSystemDocumentIndexR3.xlsx`
[R36]	The SPEAR3 Low Level RF System Description (J. Dusatko, v1.2, 2004, updated 2016)	`llrf/documentation/legacyArchitecture/SP3_writeup_V1d2.pdf`
[R37]	Legacy IOC direct-loop records and subroutines: gain and phase fields, reference conversion, frequency-offset tracking, tuner model	`spear-rf-code-legacy/rfApp/Db/rf_fbck.db,v`; `rf_stn_cav.db,v`; `rfApp/src/db/subIQ.c,v`; `subSys.c,v`
[R38]	RFgui closed-loop measurements (Measure Direct Closed), SRF1, 2 October 2026: ten runs at 20–497 mA with `COUNTS.A` $= 0.9$ dB and `PHASE.C` from -14° to -23°; MATLAB console output of the 22 mA run at 09:47: peak 0.489 V/V, $Q = 3447$, resonance 476.3025 MHz, delay 621 ns, phase -179°	`matlab_operations_prod/Measurement/ClosedLoopTF/2026-10-02/`: RFP signal-memory records of the summed feedback signal `DIRECT_I_CLOSED`, `DIRECT_Q_CLOSED` and figures `measureDirect*.fig`; noise references `NOISE_I_BL.out`, `NOISE_Q_BL.out` in `Measurement/OldData/testdata/`

Ref	Document	Path
[R40]	Cavity-A probe spectra, R&S FSW signal analyzer, 8 October 2026, 13:31 and 15:35: synchrotron line (20 kHz span, 30 Hz RBW, 60 and 100 averaged sweeps), line harmonics (3 Hz RBW) and a 3 MHz span; paired EPICS snapshots	Files `FSW_cavA_20261008T133059.json`, `FSW_cavA_matrix_20261008T153523.json`; snapshots `SRF1_20261008T133240`, `SRF1_20261008T153755`
[R41]	SPEAR3 RF MATLAB tools: RFgui (`measure_direct_closed.m`, `plot_TFC2.m`, loop models `compute_loop.m`, `rftfresponse.m`), RFguiInvasive (`configure_direct.m`, `cavityfit.m`, `measure_group_delay.m`, `make_polynomials.m`) and RFtoolbox; data directory `RF_DATA/SRF1`	`matlab_operations_prod/`
[R42]	Historical SRF1 measurements: closed-loop response of 4 May 2020 (no beam, `PHASE.C` $= -33^\circ$); cavity-probe and klystron-drive records of 2 June 2020 with the direct loop, ripple loop, integral and lead networks switched off in turn; klystron transfer curves of 11 March 2013 (a tube since replaced; the klystron failed on 3 September 2020 and a different one was installed on 7 September); saved IOC settings of December 2003; the station's noise references, the open-loop record of the Configure Direct routine and a closed-loop run at 485 mA of 10 November 2025; trip records of the present tube from 2021 (drive I/Q, cavity-probe I/Q and klystron output amplitude frozen together at trips; copied 13 November 2025)	`matlab_operations_prod/Measurement/OldData/`: `diags20200504/`, `feedback/`, `klystron/gainCurves/`, `SRF1/`, `testdata/`; trip records in `matlab_operations_prod/Measurement/FaultFiles/2021/`; the closed-loop response of 4 May 2020 and the run of 10 November 2025 in `matlab_operations_prod/Measurement/ClosedLoopTF/`

B Symbol and Notation Conventions

B.1 Cavity and Beam Parameters

Symbol	Definition	Unit	First Ref
f_0	Cavity resonant frequency	MHz	Eq. (2.2)

Symbol	Definition	Unit	First Ref
f_{RF}	RF operating frequency ($= h \cdot f_{\text{rev}}$)	MHz	§1.2
f_{rev}	Revolution frequency	MHz	§1.2
ω_0	Angular resonant frequency ($= 2\pi f_0$)	rad/s	Eq. (2.2)
$\omega_{1/2}$	Cavity half-bandwidth ($= \omega_0/2Q_L = 2\pi \times 35.5$ kHz)	rad/s	Eq. (2.1)
$\Delta f_{1/2}$	Cavity half-bandwidth (linear frequency)	kHz	Eq. (2.3)
Q_0	Unloaded quality factor	—	§2.2.1
Q_L	Loaded quality factor	—	§2.2.1
Q_{ext}	External quality factor (coupling port)	—	§2.2.1
β	Coupling coefficient ($= Q_0/Q_{\text{ext}}$)	—	§2.2.1
R_s	Shunt impedance (circuit convention: $V^2/2P$)	MΩ	§2.2.1
R_a	Shunt impedance (accelerator convention: $V^2/P = 2R_s$)	MΩ	§B.2
R_L	Loaded shunt impedance ($= R_s/(1+\beta) = 0.78$ MΩ), the impedance the beam and the loop see	MΩ	Eq. (2.1)
G_c	Cavity conductance ($= 1/R_s$)	S	§2.2.3
V_{gap}	Gap voltage per cavity	kV	§2.2.1
V_{RF}	Total RF voltage ($= n_{\text{cav}} \times V_{\text{gap}}$)	MV	§2.2.3
I_b	DC beam current	A	§2.2.3
I_{b1}	Fundamental RF component of the bunched beam current ($= 2I_b$)	A	Eq. (2.4)
$V_{b,\text{res}}$	Beam-induced voltage at resonance ($= I_{b1}R_L$)	MV	Eq. (2.4)
P_{wall}	Wall loss per cavity	kW	Eq. (2.12)
P_{beam}	Beam power ($= I_b U_0$)	kW	Eq. (2.13)
ϕ_s	Synchronous phase angle (from voltage crest)	degrees	Eq. (2.5)
ψ	Detuning angle	degrees	Eq. (2.8)
$\psi_0,\ \psi_L$	Detuning angle referred to the unloaded / loaded bandwidth ($\tan\psi_L = \tan\psi_0/(1+\beta)$)	degrees	Eq. (2.9)
ψ_{opt}	Optimum detuning angle	degrees	Eq. (2.9)
a	Beam loading parameter, real part ($= P_{\text{beam}}/P_{\text{wall}}$)	—	Eq. (7.1)
b	Beam loading parameter, reactive part	—	Eq. (7.1)
t	Normalized detuning ($= \tan\psi_0 = 2Q_0\Delta f/f_0$)	—	Eq. (7.1)
Γ	Voltage reflection coefficient at the input coupler	—	Eq. (7.2)
β_{opt}	Coupling that matches the loaded cavity ($= 1 + a$)	—	Eq. (7.3)
Δf	Frequency detuning ($= f_0 - f_{\text{RF}}$)	kHz	Eq. (2.10)
$\Delta\omega$	Angular frequency offset from RF	rad/s	Eq. (2.2)
U_0	Energy loss per turn (synchrotron radiation)	MeV	§1.2
h	Harmonic number ($= f_{\text{RF}}/f_{\text{rev}}$)	—	§1.2
α_c	Momentum compaction factor	—	Eq. (2.7)
ν_s	Synchrotron tune ($= f_s/f_{\text{rev}}$)	—	Eq. (2.7)

Symbol	Definition	Unit	First Ref
f_s	Synchrotron frequency	kHz	Eq. (2.7)
E_0	Beam energy (3.0 GeV; expressed in MeV when used with V_{RF} in MV)	GeV	Eq. (2.7)
n_{cav}	Number of cavities	—	Eq. (2.11)

B.2 Klystron and HVPS Parameters

Symbol	Definition	Unit	First Ref
K_{kly}	Klystron small-signal gain	— or dB	Eq. (2.15)
τ_{kly}	Klystron group delay	ns	Eq. (2.15)
g_A, k_{AP}	Klystron near saturation: small-signal gain of the output amplitude for a change of drive amplitude (relative units); AM/PM conversion	—, rad	§8.8
$G_{\text{kly}}(s)$	Klystron transfer function ($K_{\text{kly}}e^{-s\tau_{\text{kly}}}$)	—	Eq. (2.15)
K	Klystron perveance ($I_k/\lvert V_k\rvert^{3/2}$)	A/V$^{3/2}$	§2.3
P_{sat}	Klystron saturation power	MW	Eq. (2.16)
$P_{\text{in,sat}}$	Klystron input power at saturation	W	Eq. (2.16)
V_k, I_k	Klystron cathode voltage and current	kV, A	Eq. (2.17)
α	SCR firing delay angle	deg	Eq. (2.18)
ω_{Enerpro}	Small-signal pole of the firing chain (415 rad/s)	rad/s	Eq. (2.19)
λ, T	PLC filter coefficient (0.4) and scan period (10 ms)	—, ms	Eq. (2.20)
P_{fwd}, P_{drive}	Klystron forward and drive power	kW, W	§6.5

B.3 Loop Transfer Functions and Compensator Parameters

Symbol	Definition	Unit	First Ref
G_0	Low-frequency gain of the drive chain	—	Eq. (2.24)
$G_{\text{plant}}(s)$	Open-loop plant: drive chain, cavity and delay	—	Eq. (2.24)
$H_{\text{cav}}(s)$	Cavity transfer function	—	Eq. (2.1)
τ_d	Total loop delay	ns	Eq. (2.23)
f_c	Crossover frequency (where $\lvert G_{\text{OL}}\rvert = 1$)	kHz	Eq. (2.23)
$C(s)$	Direct-loop controller ($Ge^{j\theta}G_{\text{lead}}G_{\text{int}}$)	—	Eqs. (3.3) and (5.1)
$G_{\text{OL}}(s)$	Direct loop open-loop transfer function	—	Eq. (5.1)
G	Proportional loop gain, referred to the resonance of an on-tune cavity	— or dB	Eqs. (5.1) and (6.1)
θ	Loop phase error at that resonance (positive = lead)	deg	Eq. (6.1)
$G_{\text{lead}}(s)$	Lead compensator: $(1 + s/\omega_z)/(1 + s/\omega_p)$	—	Eq. (5.1)
$G_{\text{int}}(s)$	PI integrator: $1 + \omega_i/s$	—	Eq. (5.1)

Symbol	Definition	Unit	First Ref
ω_z, ω_p	Lead compensator zero and pole frequencies	rad/s	Eq. (6.1)
ω_i, f_i	Frequency at which the integral path equals the proportional path	rad/s, kHz	Eq. (6.1)
$\Delta\theta_i$	Phase of the integral path relative to the proportional path	deg	Eq. (8.2)
ω_ℓ	Leak frequency of a non-ideal integrator	rad/s	Eq. (8.2)
ω_a	Drive amplifier and klystron bandwidth (2.5 MHz assumed)	rad/s	Eq. (6.1)
$T(s)$	Closed-loop transfer function	—	Eq. (5.2)
$\vec{V}_\mathrm{d}$, $\delta\vec{V}_\mathrm{m}$, $\delta\vec{V}_\mathrm{ref}$	Open-loop field error of a disturbance ahead of the cavities; noise of the measured vector sum and of the reference	V	Eq. (5.3)
Z_cav	Cavity impedance without feedback	Ω	Eq. (2.2)
Z_eff	Effective impedance with feedback	Ω	Eq. (5.4)

B.4 Ripple Loop and Gain Tracking

Symbol	Definition	Unit	First Ref
$H_k(z)$	Resonator of harmonic k; H_DC the DC term	—	Eq. (6.2)
c_{0k}, c_{1k}	Numerator coefficients: gain and phase of line k	—	Eq. (6.2)
r_k	Pole radius ($r_k^2 = 0.999838$ fast, 0.999350 slow; line width ≈ 1 Hz)	—	Eq. (6.2)
f_k	Line harmonic: 60–240 Hz (slow), 300–2880 Hz (fast)	Hz	Eq. (6.2)
T_s	DSP sampling period ($\approx 1/24.6$ kHz)	s	Eq. (6.2)
e_ϕ, u_ϕ	Phase error $\phi_\mathrm{ref} - \phi_\mathrm{kly}$ and phase correction	rad	Eq. (6.2)
c_DC, c_max	DC coefficient of the ripple DSP; its calculated maximum	V	Eq. (6.3)
$K_\mathrm{kly,set}$	Klystron gain setpoint of the gain tracking	dB	Eq. (6.3)

B.5 Slow Loop Parameters

Symbol	Definition	Unit	First Ref
K_HVPS	HVPS loop proportional gain	V/W	Eq. (6.4)
$P_\mathrm{drive,set}$, P_deadband	HVPS loop drive-power setpoint and deadband	W	Eq. (6.4)
ε	Tuner load-angle error	deg	Eq. (6.5)
ψ_target	Target load angle	deg	Eq. (6.5)
N_step	Tuner steps per correction cycle	steps	Eq. (6.6)
$p_0 \ldots p_3$	Frequency offset polynomial coefficients	kHz, kHz/mm, …	Eq. (6.7)
t_1	Voltage-dependent detuning coefficient	kHz/kV^2	Eq. (6.7)
x, x_home	Tuner position, home position	mm	Eq. (6.7)
K_DAC	DAC loop proportional gain (0–1)	—	Eq. (6.8)

Symbol	Definition	Unit	First Ref
D_{conv}	Drive conversion constant	counts/kV	Eqs. (6.8) and (9.1)
G_{loop}	Linear direct-loop gain (`COUNTS.L`)	—	Eq. (6.8)

B.6 Calibration Parameters

Symbol	Definition	Unit	First Ref
A_{ref}	Reference DAC amplitude	counts	Eq. (9.1)
L_{det}	RF detector conversion loss	dB	Eq. (9.2)
E_{conv}	Power conversion constant (0.31623)	—	Eq. (9.2)
P_{cal}	Calibration power	mW	Eq. (9.2)
β_{meas}	Measured coupling factor (VSWR-based)	—	Eq. (9.3)
r	Ratio of reflected to forward amplitude ($= \lvert\Gamma\rvert$)	—	Eq. (9.3)

B.7 Synchrotron Dynamics and Direct-Loop Setting

Symbol	Definition	Unit	First Ref
n	Coupled-bunch mode number ($n = 0$: Robinson mode)	—	Eq. (4.3)
τ_n	Growth time of coupled-bunch mode n	s	Eq. (4.3)
$Z_k(s)$, w_k	Baseband impedance of cavity k; its vector-sum weight	Ω, —	Eq. (8.2)
$Z_b(s)$	Closed-loop impedance seen by the beam	Ω	Eq. (8.3)
$D_n(s)$	Characteristic function of mode n	s^{-2}	Eq. (8.4)
κ	Beam–cavity coupling $\alpha_c\omega_{\text{RF}}I_b/((E_0/e)T_{\text{rev}})$	$\Omega^{-1}\text{s}^{-2}$	Eq. (8.5)
ω_s	Incoherent synchrotron angular frequency	rad/s	Eq. (8.5)
α_{rad}, J_s	Radiation damping rate of the amplitude; longitudinal partition number	s^{-1}, —	Eq. (8.5)
σ, ω_{coh}	Damping rate and frequency of the coherent mode ($s = -\sigma + j\omega_{\text{coh}}$)	s^{-1}, rad/s	Eq. (8.4)
Q	Quality factor of the synchrotron line, $\omega_{\text{coh}}/2\sigma$	—	§8.2
n_{eff}	Drive of the dipole mode: the gap-voltage error with the bunches held fixed, as an equivalent phase	rad	Eq. (8.1)
ε_A, ε_ϕ	Relative amplitude error and phase error (positive = advance) of the total gap voltage	—, rad	Eq. (8.1)
$\chi(s)$	Response of the beam phase to the drive, $\omega_s^2/D_0(s)$	—	Eq. (8.9)
$S_x(f)$	One-sided spectral density of the signal x	$[x]^2$/Hz	Eq. (8.9)
$h_x(s)$, u_x	Transfer function from noise source x to the drive; that source	—	Eq. (8.12)
η	Dispersion at the source point	m	Eq. (8.11)

Symbol	Definition	Unit	First Ref
$P_{\pm}/P_{\mathrm{c}}$, $Z_{b,A}$, V_A	Power in a sideband of the synchrotron line relative to the carrier; closed-loop impedance and voltage of the probed cavity	—, Ω, V	Eq. (8.14)
ψ'	Detuning angle referred to the closed-loop bandwidth	deg	Eq. (8.8)
ϕ_{tr}, F_{tr}	Legacy frequency-offset tracking phase and its scale factor (`PHASE.K`, `PHASE.F`)	deg, —	Eq. (8.15)
$\Delta\phi$, ΔE	Phase deviation of a bunch from the synchronous phase (positive = late); its energy deviation	rad, eV	Eq. (C.1)
M, Φ	Number of equally spaced bunches; complex amplitude of a coupled-bunch mode	—, rad	Eq. (C.3)
$\delta\vec{I}_b$, $\delta\vec{V}$	Changes of the beam-current and gap-voltage envelopes caused by the bunch motion	A, V	Eq. (C.2)
$\mathbf{F}(s)$, $\mathbf{K}$	Real 2×2 form of a complex-coefficient transfer function; small-signal matrix of the klystron	—	Eq. (C.5)
$\hat{p}$, $\hat{b}$	Directions in the (I, Q) plane along which the bunch senses the voltage and changes the current	—	Eq. (C.6)

B.8 I/Q Signal Processing

Symbol	Definition	Unit	First Ref
$I(t), Q(t)$	In-phase, quadrature baseband components	V	Eq. (3.1)
$A(t)$	Amplitude $[= \sqrt{I^2 + Q^2}]$	V	Eq. (3.2)
$\phi(t)$	Phase $[= \mathrm{atan2}(Q, I)]$	rad	Eq. (3.2)
$\vec{V}_{\mathrm{ref}}$, $\vec{V}_{\mathrm{sum}}$	Station reference; vector sum of the cavity probes	V	Eq. (3.3)
$\vec{E}$	Error vector $[= \vec{V}_{\mathrm{ref}} - \vec{V}_{\mathrm{sum}}]$	V	Eq. (3.3)
$\vec{V}_{\mathrm{drive}}$	Drive signal to the RF modulator	V	Eq. (3.3)
G_m, θ_m	Gain and rotation angle of a baseband modulator	—, rad	Eq. (3.4)

B.9 Conventions

1. **Shunt impedance**: $R_s = V^2/(2P)$ throughout (circuit convention). Accelerator convention $R_a = 2R_s$.
2. **Synchronous phase**: ϕ_s measured from voltage crest (SLAC convention): $V_{\mathrm{RF}} \cos\phi_s = U_0$.
3. **Detuning**: $\Delta f = f_0 - f_{\mathrm{RF}}$. Negative = cavity below RF (normal above transition).
4. **Laplace variable**: s is the complex frequency of the baseband envelope; frequency responses are evaluated at $s = j\omega$.
5. **Discrete-time loops**: $x[m]$ is the value at the m-th sample or cycle.
6. **Arrow notation**: Baseband I/Q vectors (e.g. $\vec{V}_{\mathrm{ref}}$, $\vec{E}$).
7. **Reference tags**: [R*n*] = numbered reference (Appendix A), D*n* = disturbance class (§4.1).
8. **Equation numbering**: equations are numbered automatically as (section.sequence) and are always cited through a cross-reference, never by a literal number.

C Derivation of the Characteristic Equation

This appendix derives Eq. (8.4) for rigid point bunches in a uniform fill of M equally spaced bunches, with a narrowband impedance and a linear klystron, and then generalizes it to an external drive and to a klystron that treats amplitude and phase differently.

Motion of one bunch. Let $\Delta\phi$ be the phase of a bunch behind the synchronous phase, in RF radians (positive = late), and ΔE its energy deviation. Per turn the bunch gains $e[V_{\mathrm{RF}}\cos(\phi_s+\Delta\phi)+\delta V]-U_0$, where δV is the change of beam-induced voltage it meets; above transition a higher energy makes it arrive later, $d\Delta\phi/dt=\alpha_c\omega_{\mathrm{RF}}\,\Delta E/E_0$. To first order, with radiation damping added [R8],

$$\frac{d^2\Delta\phi}{dt^2}+2\alpha_{\mathrm{rad}}\frac{d\Delta\phi}{dt}+\omega_s^2\,\Delta\phi=\frac{\alpha_c\,\omega_{\mathrm{RF}}}{(E_0/e)\,T_{\mathrm{rev}}}\,\delta V, \tag{C.1}$$

with ω_s^2 and α_{rad} as in Eq. (8.5).

Beam current. Written as an envelope on the carrier $e^{j\omega_{\mathrm{RF}}t}$, like the signals of §3, the fundamental beam current of the unperturbed train is $I_{b1}e^{-j\phi_s}$, with $I_{b1}=2I_b$. A bunch that is late by $\Delta\phi$ multiplies its contribution by $e^{-j\Delta\phi}\approx 1-j\Delta\phi$, so the motion changes the current envelope by

$$\delta\vec{I}_b=-j\,I_{b1}\,e^{-j\phi_s}\,\Delta\phi. \tag{C.2}$$

Coupled-bunch mode n. In mode n the phase advances by $2\pi n/M$ from one bunch to the next, $\Delta\phi_k(t)=\mathrm{Re}\big[\Phi\,e^{st}e^{j2\pi nk/M}\big]$. Bunch k passes the cavities at $t_k=kT_{\mathrm{rev}}/M$ plus whole turns, where $e^{j2\pi nk/M}=e^{jn\omega_{\mathrm{rev}}t_k}$, so the bunches sample the single signal $\Delta\phi(t)=\mathrm{Re}\big[\Phi\,e^{s't}\big]$ with $s'=s+jn\omega_{\mathrm{rev}}$. Its aliases lie multiples of Mf_{rev} away, far outside the cavity resonance, and are dropped.

Induced voltage. The beam drives the cavities through the closed-loop impedance of Eq. (8.3), $\delta\vec{V}=-Z_b\,\delta\vec{I}_b$, and a bunch at the synchronous phase meets $\delta V=\mathrm{Re}\big[\delta\vec{V}e^{j\phi_s}\big]$. Because $\Delta\phi$ is real, $\Delta\phi=\frac{1}{2}\big(\Phi e^{s't}+\bar{\Phi}e^{\bar{s}'t}\big)$, and each exponential passes through Z_b at its own complex frequency. Collecting the terms in $e^{s't}$,

$$\delta V=\mathrm{Re}\Big[\tfrac{j}{2}\,I_{b1}\,\Phi\,\big(Z_b(s')-\overline{Z_b(\bar{s}')}\big)\,e^{s't}\Big]. \tag{C.3}$$

Result. Substituting Eq. (C.3) into Eq. (C.1) for bunch k and equating the coefficients of $e^{st}e^{j2\pi nk/M}$ gives $s^2+2\alpha_{\mathrm{rad}}s+\omega_s^2=j\kappa\,[Z_b(s')-\overline{Z_b(\bar{s}')}]$ with $\kappa=\alpha_c\omega_{\mathrm{RF}}I_{b1}/(2(E_0/e)T_{\mathrm{rev}})$, which is Eqs. (8.4) and (8.5). The two terms are the responses at the upper and the lower synchrotron sideband: for $s\approx j\omega_s$ and $n=0$ they are $Z_b(j\omega_s)$ and $\overline{Z_b(-j\omega_s)}$, which leads to Eq. (8.6) and to the Robinson criterion (4.1). Without feedback and with identical cavities, multiplying $D_0(s)$ by the cavity denominator $(s+\omega_{1/2})^2+\Delta\omega^2$ gives Pedersen's quartic [R35] with radiation damping added; at low current the first-order rate is that of Robinson [R14] and Wilson [R10].

External drive. An RF error that does not come from the motion adds to δV in Eq. (C.1) the voltage $\mathrm{Re}\big[\delta\vec{V}_{\mathrm{ext}}\,e^{j\phi_s}\big]$, where $\delta\vec{V}_{\mathrm{ext}}$ is the change of the total gap envelope that it causes with the bunches held fixed. An error common to all bunches drives only $n=0$, and the result becomes $D_0(s)\,\Delta\phi=\omega_s^2\,n_{\mathrm{eff}}$ with

$$n_{\mathrm{eff}}=\frac{\mathrm{Re}\big[\delta\vec{V}_{\mathrm{ext}}\,e^{j\phi_s}\big]}{V_{\mathrm{RF}}\sin\phi_s}, \tag{C.4}$$

which is $\varepsilon_A\cot\phi_s-\varepsilon_\phi$ for $\delta\vec{V}_{\mathrm{ext}}=V_{\mathrm{RF}}(\varepsilon_A+j\varepsilon_\phi)$ (Eq. (8.1)).

Two quadratures. A klystron near saturation responds differently to amplitude and phase (§8.10), and the bunch senses one combination of the two; neither fits a loop described by one complex gain. Write every

envelope as the real vector $(I, Q)^{\mathsf{T}}$ of its components and every complex-coefficient transfer function $F(s)$ as the real matrix

$$\mathbf{F}(s) = \begin{pmatrix} F_r & -F_i \\ F_i & F_r \end{pmatrix}, \qquad F_r = \tfrac{1}{2}\big[F(s) + \bar{F}(s)\big], \quad F_i = \tfrac{1}{2j}\big[F(s) - \bar{F}(s)\big], \quad \bar{F}(s) = \overline{F(\bar{s})}, \tag{C.5}$$

whose entries are the transforms of the real and the imaginary part of the impulse response. The klystron becomes $\mathbf{K} = \mathbf{R}(\theta_g)\left(\begin{smallmatrix} g_A & 0 \\ k_{\text{AP}} & 1 \end{smallmatrix}\right)\mathbf{R}(-\theta_g)$, with $\mathbf{R}(\theta)$ a rotation by θ and θ_g the phase of the generator current; $g_A = 1$, $k_{\text{AP}} = 0$ is a linear klystron. With $\mathbf{P}_\Sigma = \sum_k \mathbf{Z}_k$, $\mathbf{P}_{\text{m}} = \sum_k w_k \mathbf{Z}_k$ and the forward path $\mathbf{F}_{\text{fw}}(s) = \mathbf{K}\,\mathbf{C}_{\text{fw}}(s)/R_L$, where $C_{\text{fw}}(s) = C(s)\,\omega_a e^{-s\tau_d}/(s+\omega_a)$ is everything in Eq. (8.2) but the cavities, the closed-loop impedance seen by the beam is $\mathbf{Z}_b = \mathbf{P}_\Sigma\big(\mathbf{1} + \mathbf{F}_{\text{fw}}\mathbf{P}_{\text{m}}\big)^{-1}$. The motion changes the beam current along $\hat{b} = (-\sin\phi_s, -\cos\phi_s)^{\mathsf{T}}$ (C.2), and the bunch senses the voltage along $\hat{p} = (\cos\phi_s, -\sin\phi_s)^{\mathsf{T}}$ (C.4), so that

$$\begin{aligned} D_0(s) &= s^2 + 2\alpha_{\text{rad}}\, s + \omega_s^2 + 2\kappa\, \hat{p}^{\mathsf{T}}\mathbf{Z}_b(s)\,\hat{b}, \\ n_{\text{eff}}(s) &= \frac{\hat{p}^{\mathsf{T}}\mathbf{Z}_b(s)\big[\delta\vec{I}_{\text{kly}} - \mathbf{F}_{\text{fw}}(s)\,(\delta\vec{V}_{\text{m}} - \delta\vec{V}_{\text{ref}})\big]}{V_{\text{RF}}\sin\phi_s}, \end{aligned} \tag{C.6}$$

where $\delta\vec{I}_{\text{kly}}$ is noise at the klystron output (noise ahead of the klystron enters as $\mathbf{K}$ times its own vector) and $\delta\vec{V}_{\text{m}}$, $\delta\vec{V}_{\text{ref}}$ are as in Eq. (5.3). The coefficient of each source is its transfer function h_x of Eq. (8.12). For a linear klystron $\hat{p}^{\mathsf{T}}\mathbf{Z}_b\hat{b} = -\frac{j}{2}\big[Z_b(s) - \overline{Z_b(\bar{s})}\big]$, and the first line is Eq. (8.4) for $n = 0$ to rounding error; for a symmetric loop (cavities on tune, $\theta = 0$) each h_x is the open-loop path divided by $1 + G_{\text{OL}}$, or multiplied by T. With the klystron of §8.10 the roots agree with the time-domain simulation to 1%; with a linear klystron, white noise injected at the klystron output of that simulation reproduces the beam jitter of Figure 12 without feedback, at 0 dB and at 6 dB to within 11%. Two static checks are exact: without feedback a phase step of the klystron moves the beam by the same angle, and with an integral path a phase step of the reference does.

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